Homological Algebra

— *EYNTKA* Series —

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Contents

Part I

Homological Algebra

Throughout this book, the reader has encountered many functors, some notable examples being $\text{Hom}_R(-, N)$, $\text{Hom}_R(M, -)$ and $M \otimes_R -$ among many others. Each of these functors give some information about the objects and the morphisms passing through the functor. As we saw for the Hom and $M \otimes_R -$ functors, certain appropriate objects preserve exactness, while most preserve it only partially. Perhaps a crowning achievement of mid 20th century mathematics is to be able to quantify this lack of exactness, and rather spectacularly use the information given by this quantification to extract a (sometimes copious) amount of information about the objects of study. The great effectiveness with which homological algebra can produce interesting information may seem a little miraculous at first, for example, using homological algebra the notion of *dimension* as it is understood in geometry is readily translatable to algebra, or that we may measure how far away a module is from being a vector-space (details are in K-theory)[1] How this is the case is perhaps best motivated by going through learning the subject to see for oneself, but it may be helpful for some readers to have some of my (the authors) intuition presented to stand on some ground before proceeding.

If we have two objects M_1 and M_2, we may wish to consider a way of constructing a new object M given M_1 and M_2. Ideally, we may form a short exact sequence like so:

$$0 \to M_1 \to M \to M_2 \to 0$$

More generally, we may have a collection of objects $\{M_i\}_{i \in I}$ which we use to construct an object via a long exact sequence. However, it is often not the case that such an exact sequence is too strong, that is, it is not the case that $\text{im}(\varphi_i) = \ker(\varphi_{i+1})$. Even if such a construction exists, we often then want to "transform" this information via a functor which shall then lose us exactness (one can think of a functor as asking some sort of "question" whose output is the answer). In this case, it has been eerily common that the image of φ_i is contained in the kernel of φ_{i+1}:

$$\text{im}(\varphi_i) \subseteq \ker(\varphi_{i+1}) \tag{1}$$

Conceptually, this can be thought as the object "being obstructed" from fully being an extension. This new type of sequence is called a *cochain complex.* The reader with a good memory may recall **??** where it was given that another name for a long exact sequence is an exact cochain complex. In this way, the condition given by equation 1 is a generalization of the exact condition. The theoretical heart of homological algebra can be thought of how to extract "quantitatively" how a cochain complex *fails* to be an exact cochain complex. For example, in chapter **??**, it was have eluded that through the use of homology theory, the Ext functor gives us all the ways two modules M_1, M_2 may be extended. In this case, the failure of $\text{Hom}(M, -)$ or $\text{Hom}(-, N)$ preserving injectivity or surjectivity (equivalently, the "obstruction" to these functors being exact) gave rise to the Ext functor and the study of its homology.

The term "algebraic object" has been is a bit ambiguous. What type of algebraic objects can be used to in the theory of homological algebra? As it turns out, not all algebraic objects satisfy the right conditions for the theory of homological algebra (for example, **Grp** and **Fld** will be shown to be inappropriate for a theory of homology). However, there are non-algebraic objects for which a homological theory can be induced (famously, topological spaces and smooth manifolds admit a theory of homological algebra). This spurred the need to generalize the theory of homological algebra to the categorical level, where homological theory is well-defined on *abelian categories.* We thus shall start homological algebra with an overview of abelian categories. For those who are already familiar with some homological algebra, they may have studied homological algebra within the category of

[1] This is overviewed in chapter 3

R-**Mod**, which is a concrete abelian category. Doing so will in fact not loose any generality due to Freyd-Mitchell Theorem (theorem 1.1.15) which states that any [small] abelian category A admits a fully faithful exact functor to a category R-**Mod** for a suitable ring R. Working with the concrete category R-**Mod** brings it benefits as it may be easier to conceptualize the manipulation of elements, and so the reader may ask for the benefit in sticking to working with general abelian categories. As the author, I believe the benefits comes in being comfortable with more advanced category theory. Future algebraists are going to encounter more complicated problems involving category theory (higher order categories, topoi's, and stacks, are all examples the reader may see in the not too distant future). Hence, introducing a categorical perspective will familiarize the reader for more complicated categorical notions in their not too distant future[2].

The reader may wonder why a cochain complex is not called a chain complex. This is because we can work in the dual category, where all arrows are flipped. In the dual category, we shall get chain complexes. Homological algebra in the dual category is theoretically the same, however we shall see that cohomology is often computationally easier to work in, permitting more useful algebraic structures we can use to study. It is for this reason that most mathematicians work with cohomology rather than homology, even though every theorem we shall show for homology translates directly to cohomology[3]. In this book, we shall start with homology, and switch to cohomology when starting to work with more computational parts of the theory to get the reader accustomed to both modes of thinking.

Goal

Chapter 1 shall introduce the theory of abelian categories which shall be the types of categories in which homological algebra is well-defined.

The main result chapter 2 of the book is trying to convey is what information from R-modules give rise to certain properties in it's respective chain complex(es), and conversely what information from it's chain complex(es) gives us information about modules. We shall see that given an object M within an abelian category, there are an infinite number of ways we may associate a [co]chain complex to it. Thus, there will have to be a way to either associate a canonical [co]chain complex (in a functorial manner), or we assign [co]chain complexes to an object up to appropriate notion of isomorphism. The latter shall be the approach, where we shall see that we will want to consider [co]chain complexes up to something called *quasi-isomorphism* of [co]chain complexes. We shall then see what information we may deduce given different choices of [co]chain complex functors up to quasi-isomorphism.

After having developed the ground work for exacting cohomological information, in chapter 3 we shall put this knowledge to good use and apply cohomology theory to G-invariant functors and see what this this can tell us about groups. This shall also finally address what was foreshadowed in box:HERE where it was said the units of a Galois field form a trivial 1st homology group. We shall show how to quantify the obstruction of a module to being free via a bit of K-theory, and define the notion of homological dimension which shall line up with the results presented in the section on commutative algebra.

[2]assuming, naturally, that the reader holds great interest in algebra and shall keep pursuing branches within the field

[3]there is even an equivalent algebraic structure for the additional algebraic structure that is induced in cohomology, however it is less useful

The part will be concluded with the introduction chapter 4 which are the quintessential for any concrete computations in homological algebra and are an indispensable tool for any serious work in many disciplines of algebra and geometry (interesting examples being algebraic topology, algebraic geometry).

Connection to Singular Homology/Cohomology

For readers familiar with algebraic topology and the theory of (co)homology, the connection between the above abstraction and the homology theory of topological spaces comes from noticing that singular (co)homology on a manifold can be interpreted as taking the projective resolution $C_n(X) = \mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$. In particular, given M, we can apply $C_*(-)$ to it and get a projective (even free) resolution from which we extract homological information. This should also motivate the use of $\mathrm{Hom}_R(M, -)$ and $\mathrm{Hom}_R(-, N)$ as the "right" functors to be considering when generalizing classical algebraic topology to the algebraic setting. As the tensor product it adjoint (**??**) it is another natural candidate.

0.0.1 Foreshadowing: Yoneda's Lemma The Hom is very useful as it is "represented" by an element (either in the domain or codomain). Even more is true: it can be used to represent other functors. In section **??** we shall study *representable functors* which are functors where $F(-) \Rightarrow \mathrm{Hom}_C(A, -)$ for natural transformation $\Rightarrow$, category C, object $A \in C$, and $F : C \to \mathbf{Set}$. The Yoneda lemma will give a way of representing objects using Hom.

Historical Origins

While modern homological algebra is phrased in the language of categories, its machinery evolved from two distinct 19th-century sources which converged in the mid-20th century.

The algebraic lineage begins around 1890 with David Hilbert's Invariant Theory. To prove the Basis Theorem, Hilbert studied the relations ("syzygies") between generators of an ideal. He constructed a chain of free modules to resolve these relations:

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to M \to 0$$

Hilbert proved the *Syzygy Theorem*, showing that for polynomial rings, this chain of relations eventually terminates in an exact sequence of free modules. However, this theory encountered a significant obstruction when applied to rings with singularities (such as coordinate rings of singular curves). In these non-regular settings, Hilbert's resolutions do not terminate but continue infinitely. This failure shifted the mathematical focus from simply computing invariants to essentially classifying the severity of a singularity by the infinite length of its resolution.

Parallel to this, in the 1930s, number theorists like Emmy Noether, Richard Brauer, and Helmut Hasse were attempting to classify division algebras (generalizations of the quaternions) over number fields. They classified these algebras using "Crossed Products," which relied on complex systems of functions called "Factor Sets" ($c : G \times G \to K^\times$). While they successfully defined the *Brauer Group* of a field to organize these algebras, the calculations were notoriously opaque. The "Factor Sets" satisfied a messy algebraic identity that seemed arbitrary, acting as a computational blocker that obscured the underlying structure of the fields they were studying.

Simultaneously, from 1895 onwards, the topological roots of the subject were forming as Henri Poincaré classified manifolds. He decomposed spaces into simplices and represented their boundaries using incidence matrices. He defined Betti numbers as the ranks of these matrices. However, Poincaré's numerical approach had a blind spot: rank obliterates "torsion." A twisted shape like the projective plane appeared numerically identical to a trivial one (both have rank 0 in dimension 1).

In 1925, Emmy Noether revolutionized this view. She realized that the matrices topologists were diagonalizing were actually presentations of abelian groups. By shifting focus from the numerical rank to the *Homology Group* itself ($\ker \partial / \operatorname{im} \partial$), she captured the torsion information, effectively "algebraizing" the topology. She showed that the matrix was not just a calculator for a number, but a presentation of a structure.

The convergence of these fields occurred in the 1940s when algebraists attempted to classify *Group Extensions*. Saunders Mac Lane and Samuel Eilenberg attempted to classify these extensions and rediscovered the same "Factor Sets" used by Brauer and Hasse. They discovered that the algebraic identity these functions satisfied was *identical* to the boundary formula for a 2-simplex in topology; essentially they realized they were calculating $H^2(G, N)$ (see [1] for this computation in the EYNTKA series). This was the turning point: it proved that "cohomology" was not just a property of geometric shapes, but a fundamental algebraic structure that arose whenever one tried to "extend" one object by another, whether in topology, group theory, or number theory.

The grand unification arrived in 1956 when Cartan and Eilenberg realized that the topological concept of "boundaries" and the algebraic concept of "extensions" were instances of the same phenomenon: the *failure of exactness*. They observed that while Hilbert's resolution is exact, applying a functor (like the tensor product or Hom) usually destroys that exactness. They realized that the "homology" groups were simply the mathematical terms required to repair this broken exactness. This birth of the *Derived Functor* (Tor and Ext) transformed the field from the study of static objects to the study of how functors distort structure. To build this machinery, they faced a technical blocker: Hilbert's reliance on "Free Modules." While free modules worked for Homology (Tor), they failed for the dual theory of Cohomology (Ext), as the dual of a free module is rarely free. To achieve the necessary symmetry, they replaced the specific condition of "Freeness" with the categorical properties of *Projectivity* and *Injectivity*. This allowed them to define derived functors for any ring, regardless of whether a basis existed.

With this machinery established, the period from 1955 to 1980 saw these newly developed tools being used to solving the original obstructions in geometry and number theory. In geometry, Jean-Pierre Serre proved that a local ring is regular (non-singular) if and only if it has finite global homological dimension (theorem 3.2.8). This transformed the "problem" of infinite resolutions into a precise detection tool: the non-terminating resolution was effectively the algebraic signature of a geometric singularity. Alexander Grothendieck later introduced *Local Cohomology* ($H^i_{\mathfrak{m}}$) to quantify the depth and severity of these singularities, eventually leading to Intersection Homology which restored Poincaré Duality to singular spaces.

In number theory, the mysterious "Factor Sets" of the 1930s were identified as elements of the second Galois Cohomology group, $H^2(\mathrm{Gal}(L/K), L^\times)$. This realization allowed Emil Artin and John Tate to rewrite Class Field Theory entirely in the language of cohomology. The obscure identities of the past were revealed to be standard cohomological relations, and the *Tate Cohomology* groups ($\hat{H}^i$) were developed specifically to handle the arithmetic of finite groups, solving the classification problems that had stalled the previous generation (see [7] for the EYNTKA series exposition of modern class field theory).

1

Abelian Categories

This chapter focuses on the category in which we may define "cohomological theories" as not all categories are well-suited for this endeavor

1.1 Definitions and Properties

We have extensively worked with categories which have a multitudes of different properties. Some important terms to recall are the following:

- *zero-object*: an object 0 in a category C $0 \in \mathrm{Obj}(C)$ that is both the initial and final object. Such an object is also called a terminal object. Similarly, a zero morphism is a morphism $f : X \to Y$ such that for any other morphisms $g, h : X \to Y$, $fg = fh$ and $hf = gf$.
- *Products* and *Coproducts*: Given two objects $a, b \in \mathrm{Obj}(C)$, if there exists an object $c \in \mathrm{Obj}(C)$ that is the limit (resp. colimit) of the diagram:

  $$\overset{A}{\bullet} \qquad \overset{B}{\bullet}$$

  then c is the product (resp. coproduct) of a and b, and is usually written as $a \times b$ (resp. $a \amalg b$)
- *kernels* and *cokernels*: Given a morphism $f : a \to b$, the kernel (resp. cokernel) of f is an object k and a morphism $\iota : k \to a$ such that k is the limit (resp. colimit) of the diagarm:

  $$\overset{a}{\bullet} \xrightarrow{f} \overset{b}{\bullet}$$

  with the added restriction that the map $\varphi : k \to y$ must be the zero morphism (namely, by commutativity, $f\iota = 0_{kb}$ must be the zero morphism). In terms of diagrams, we get that that for any $\iota' : K \to A$ such that $f\pi\iota' = 0$, that there exists a morphism φ st the following diagram commutes:

$$\begin{tikzcd}
K' \arrow[r, "\iota'"'] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\varphi"'] & A \arrow[r, "\varphi"'] & B \\
& K \arrow[u, "\iota"'] &
\end{tikzcd}$$

and simmilarly for cokernels:

$$\begin{tikzcd}
& K' \arrow[d, "\pi'"] \arrow[dr, dashed, "\psi"] & \\
A \arrow[r, "\varphi"] \arrow[rr, bend right, "0"'] & B \arrow[r, "\pi"] & K
\end{tikzcd}$$

Kernels and cokernels do not necessarily exist in a category, or not all morphisms have kernels (ex. the category of finitely generated R-modules for any ring R need has morphisms without kernels). Furthermore, monomorphism need not be kernels (not all subgroups are normal) and epimorphisms need not be cokernels (recall $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism in **Ring**)

Not all categories we have worked with have these properties. For example, **Fld** doesn't have products (what would be the characteristic of the product of $\mathbb{F}_p$ and $\mathbb{F}_q$ for $p \neq q$?), some categories don't have zero morphisms (set being an example), some have monomorphisms/epimorphisms that do not correspond to kernels (as showed above). We first limit our attention to categories which have these nice properties:

Definition 1.1.1: Additive Category

Let **A** be a category. Then **A** is said to be *additive* if it has a zero-object, has both finite products and coproducts, and each $\mathrm{Hom}_A(X, Y)$ can be given an abelian group structure such that the composition of maps:

$$\mathrm{Hom}_A(X, Y) \times \mathrm{Hom}_A(Y, Z) \to \mathrm{Hom}_A(X, Z)$$

is bilinear. A functor between additive categories is *additive* if it is a group homomorphism on hom-sets.

A functor between two additive categories is called an *additive functor* if the maps between the hom-sets is a homomorphism.

The term "additive" comes froms the fact that we may interpret the additive category as a monoidal category (a category which we may add a functor $\otimes : C \times C \to C$ that is given the property of a monoid operation, for an additive category this would be the direct product), where the hom-sets are "enriched" to also be abelian groups. Another way of looking at additive categories is in which the algebra of matrices is well-defined.

Example 1.1.2: Additive Categories

1. The category of R-modules is an example of an additive category.

2. **Grp** is not additive, for $\mathrm{Hom}_{\mathbf{Grp}}(A, B)$ doesn't have an abelian group strucuture satisfying the above condition.

3. The category of matrices over a ring is an additive category.

4. The category of finitely generated R-modules is an additive category

A key realisation is that in the absence of kernels and cokernels as objects monomorphisms and epimorphisms take their place:

Lemma 1.1.3: Additive Categories: Monomorphisms and Kernels I

Let A be an additive category. Then kernels are monomorphisms and cokernels are epimorphisms

Proof:
Aluffi p.562-563

Note that kernels and cokernels need not exist in an additive category, hence the converse of each accession need not be the case (in a vacuous way). If a morphism φ has a kernel then we may identify exactly what this kernel is:

Lemma 1.1.4: Additive Categories: Monomorphisms and Kernels II

Let A be an additive category and $\varphi : X \to Y$ a morphism. Then:

1. If φ has a kernel, then φ is a monomorphism if and only if $0 \to X$ is it's kernel

2. If φ has a cokernel, then φ is a epimorphism if and only if $Y \to 0$ is it's kernel

Proof:
Aluffi p.563

As mentioned, it is not guaranteed that kernels and cokernels exist. We shall shortly axiomatically add them to our category by adding them into the definition of a new category that supersedes additive categories, but first we show a few more important results about abelian categories. Namely, in an additive category, products and coproducts are equivalent:

Lemma 1.1.5: Additive Categories: Products and Coproducts

Let A be an additive category and $X, Y \in \mathrm{Obj}(A)$. Then if $X \times Y$ exists, so does $X \amalg Y$, and if $X \amalg Y$ exists, so does $X \times Y$. Furthermore,

$$X \times Y \cong X \amalg Y$$

Proof:
Stacks Project (see this link)

Hence, additive categories behave very much like abelian groups, however as we pointed out kernels and cokernels need not exist. We thus define a new category that forces these conditions:

Definition 1.1.6: Abelian Category

[Abelian Category] Let A be an additive category. Then A is an *abelian* category if kernel and cokernels exist in A.

By the above lemmas, that is every monomorphisms is the kernel of some morphism, and every epimorphism is the cokernel of some morphism. Since each kernel is associated to a monomorphism, we may intuitively think of the association:

$$\text{kernel} \;\Leftrightarrow\; \text{submodule}$$

and similarly:

$$\text{cokernel} \;\Leftrightarrow\; \text{quotient module}$$

In many ways, an abelian category should be thought of as the minimum number of conditions necessary to define exact sequence[1]. To see this, we require the following:

Lemma 1.1.7: Abelian Category: Kernels And Cokernels

Let A be an abelian category. Then every kerne lis the kernel of its cokernel, and every cokernel is the cokernel of its kernel.

***Proof*:**
Aluffi p.565

Lemma 1.1.8: Abelian Categories: Isomorphisms

[Abelian Categories, Isomorphisms] Let A be an abelian category and $\varphi : A \to B$ a morphism in A. Then if φ is a monomorphism and an epimorphism, it is an isomorphism

Note that this does not hold in additive categories (see ref:HERE)

***Proof*:**
Aluffi p.566

Theorem 1.1.9: Abelian Category: Canonical Decomposition

[Abelian Category, Canonical Decomposition] Let A be an abelian category and $\varphi : A \to B$ a morphism. Then A can be decomposed in the following cannonical way:

$$\begin{tikzcd}
A \arrow[r, two heads] \arrow[rrr, bend left, "\varphi"] & C \arrow[r, "\tilde{\varphi}"'] & K \arrow[r, tail] & B
\end{tikzcd}$$

[1]Those familiar with some algebraic topology may find it insightful that an abelian category is the the category with ninimum number of condition so that the snake lemma (theorem 2.1.15) holds

***Proof*:**
Aluffi p.570

From the above, we may think of $\ker(\text{coker}) : K \to B$ as representing the "image" of φ:

Lemma 1.1.10: Abelian Category: Images

Let $\varphi : A \to B$ be a morphism in an abelian category, and let $\iota : K \to B$ be the kernel of the cokernel of φ. Then:

1. ι is a monomorphism
2. φ factors through ι
3. ι is initial with these properties

***Proof*:**
Aluffi p.572

Definition 1.1.11: Abelian Category: Image

Let $\varphi : A \to B$ be a morphism in an abelian category. Then the *iamge* of φ is $\ker(\text{coker}(\varphi))$ and is denoted $\text{im}(\varphi)$. Similarly, its co-image is $\text{coker}(\ker(\varphi))$ and is dentoed $\text{coim}(\varphi)$.

With our intuition from working with R-modules, given a morphism $\varphi : A \to B$ with image $\iota : K \iota B$, we may imagine there is an induced epimorphism $\overline{\varphi} : A \to K$. This is indeed the case:

Proposition 1.1.12: Abelian Category: sub object and quotient objects

Let $\varphi : A \to B$ be a morphism in an abelian category with image $\text{im}\varphi : K \to B$ and coimage $\text{coim}\,\varphi : AC \to A$. Then the induced morphisms $A \to K$ and $C \to A$ are (respectively) epimorphisms and monomorphisms.

***Proof*:**
Aluffi p.573

From the cannonical decomposition, we may say that the image and coimage are "isomorphic" (which is not strictly true since these are morphisms, however by the Freyd-Mitchell Theorem this is a safe way of looking at it).

Theorem 1.1.13: Abelian Category: Exactness

Consider a sequence of objects in an abelian cateogy

$$\cdots \to A \xrightarrow{\varphi} B \xrightarrow{\psi} C \to \cdots$$

Then this sequence is *exact* if:

1. $\psi \circ \varphi = 0$
2. $\mathrm{coker}\varphi \circ \ker \psi = 0$

Example 1.1.14: Categorical Exactness

1. Aluffi p.577

Freyd-Mitchell Theorem

As mentioned, R-mod is an abelian category, and as we shall develop a bit later it the categories R-mod "represent" all abelian categories.

Theorem 1.1.15: Freyd-Mitchell Theorem

Let A be a small abelian category. Then there is a fully faithful exact functor $A \to R$-Mod for a suitable ring R.

***Proof*:**
partial proof?

1.2 Category of Categories

(needed for derived categories proper definition)

2

Complexes and (co)Homology

Let $\mathbf{A}$ be a fixed abelian category, which by theorem 1.1.15 may without loss of generality it can be considered ${}_R\text{Mod}$. In this chapter, we shall focus on studying (co)chains of such modules, that is sequence of modules that respect equation (1):

$$\text{im}(d^n) \subseteq \ker(d^{n+1})$$

The collection of all chains shall be denoted $\text{Ch}(\mathbf{A})$. We shall see that $\text{Ch}(\mathbf{A})$ is an abelian category, the notion of extensions (and snakes lemma) is well defined, which will be explored. In the right categories, it is relatively easy to construct a concrete complex given the category has enough objects, for example $M \in \text{Obj}(R\text{-}\mathbf{Mod})$ always has a complex known as a *free resolution* (see **??**). There will be many different ways of creating complexes given an object, giving us different types of resolutions. We shall see that when a category has enough projective objects (resp. Injective objects), then "up to homotopy equivalence", projective resolutions shall be *initial* (resp. final) in the category of complexes $\text{Ch}(A)$. Thus, we shall take the time to develop the homology theory in categories with enough projective or injective objects. We shall see that under appropriate circumstances, taking (co)homology of a projective resolution of an object can give us some important information about said object.

Homology and cohomology are duals of each other, which means on the theoretical level they are the same. As cohomology is the more computationally useful tool, we shall be focusing on cohomology on this chapter, with the idea that the reader shall be comfortable being able to switch indices when necessary. Taking the cohomology of a complex shall be shown to be an additive functorial process, meaning for a given morphism of complexes $\alpha^\bullet : A^\bullet \to B^\bullet$, we shall get a morphism each cohomology $H^i(\alpha) : H^i(A^\bullet) \to H^i(B^\bullet)$ which will be shown can be to form a new complex $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$.

The goal of this chapter is to show how to extract information given a complexes (co)homology. Given this, it is important to know what information a morphism of complexes preserves in cohomology, that is given a morphism of complexes, $\alpha^\bullet : A^\bullet \to B^\bullet$, when is $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$ an isomorphism? This shall lead us to the notion of *quasi-isomorphisms*, and their "representation" through mapping cones. As it turns out, quasi-isomorphisms do not behave nicely in general. What we shall then seek out is a stricter concept known as a *homotopy equivalence* that will still be a quasi-isomorphism, but will be invertible in the appropriate category.

2.1 Basic Definition and Properties

Starting with defining the generalization of an exact sequence:

Definition 2.1.1: Chain Complex

A *chain complex* $(C_\bullet, d_\bullet)$ is a sequence of objects and morphisms:

$$\cdots \xleftarrow{d_i} M_i \xleftarrow{d_{i+1}} M_{i+1} \xleftarrow{d_{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d_i \circ d_{i+1} = 0$$

and each d_i is called a *boundary map*. Similarly, a *cochain complex* $(C^\bullet, d^\bullet)$ is a sequence of objects

$$\cdots \xrightarrow{d^i} M^i \xrightarrow{d_{i-1}} M^{i-1} \xrightarrow{d^{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d^i \circ d^{i-1} = 0$$

where $d^\bullet$ are is called the *differential*[a].

[a]Due to it's origin's in differential geometry and algebraic topology

Chains and cochains can be interchanged by taking

$$M_i = M^{-i}$$

Working with one or the other makes essentially no difference for homological algebra. In certain cases, some structures present themselves more evidently in the cohomological case (ex. a natural ring structure called the *cohomology ring* shall become important to us), and so we opt to work with the cohomology, even though at this stage this can be considered to be an arbitrary choice. The condition that $d^i \circ d^{i-1} = 0$ is the same as $\mathrm{im} d^{i-1} \subseteq \ker d^i$. If the complex is exact, then $\mathrm{im} d^{i-1} = \ker d^i$ for all i. We also often write $C^\bullet$ instead of $(C^\bullet, d^\bullet)$ if the mappings $d^\bullet$ are clear from context or implied. In some textbooks, even the bullet is dropped and M represents the complex. Sometime, for homology, the notation $\partial_\bullet$ shall be used instead of $d_\bullet$ due to it's connection to being the boundary map for topological spaces when creating complexes out of topological spaces[1].The condition that $d^i \circ d^{i-1} = 0$ (resp. $d_i \circ d_{i+1} = 0$) is often abbreviated to $d \circ d = 0$.

Due to historical origins in algebraic topology, the kernel and image of the differential

[1]Similarly, $d^\bullet$ stands for the *differential* due to it's link to the derivative, which would be explored in a class in algebraic topology, or see EYNTKA Algebraic Topology

Definition 2.1.2: Cycles and Boundaries

Let $C_\bullet$ (resp. $C^\bullet$) be a chain complex (resp. cochain complex). Then:

1. $\ker(d_n)$ is called the *n-cycles* of $C_\bullet$ and $\operatorname{im}(d_{n+1})$ is called the n-boundary of $C_\bullet$. The n-cycle is represented as $Z_n = Z_n(C_\bullet)$ and the *n-boundary* is represented as $B_n = B_n(C_\bullet)$

2. $\ker(d^n)$ is called the *n-cocycles* of $C^\bullet$ and $\operatorname{im}(d^{n-1})$ is called the *n-coboundary* of $C^\bullet$. The n-cocycle is represented as $Z^n = Z_n(C_\bullet)$ and the n-coboundary is represented as $B_n = B_n(C_\bullet)$

The name cycles and boundaries has it's origin in algebraic topology:

Example 2.1.3: Cycles and Boundaries

(Put that image from p.67 from alg top notes)

The collection of all complexes forms a Category:

Definition 2.1.4: Category Of Complexes

Let $\mathbf{A}$ be an abelian category. Then $\text{Ch}^\bullet(\mathbf{A})$ is a new category called the *category of complexes of A* where

1. $\text{Obj}(\text{Ch}^\bullet(\mathbf{A})) = \{\text{cochain complexes in } \mathbf{A}\}$

2. For two complexes $C^\bullet$ and $N^\bullet$, a morphism $\alpha^\bullet \in \text{Hom}_{\text{Ch}(\mathbf{A})}(C^\bullet, N^\bullet)$ consists of morphisms st the following diagram commute:

$$\begin{tikzcd}
\cdots \arrow[r] & M^{i-1} \arrow[r, "d^{i-1}_{C^\bullet}"] \arrow[d, "\alpha^{i-1}"] & M^i \arrow[r, "d^i_{C^\bullet}"] \arrow[d, "\alpha^{i-1}"] & M^{i+1} \arrow[r] \arrow[d, "\alpha^{i-1}"] & \cdots \\
\cdots \arrow[r] & N^{i-1} \arrow[r, "d^{i-1}_{N^\bullet}"] & N^i \arrow[r, "d^i_{N^\bullet}"] & N^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Similarly for $\text{Ch}_\bullet(\mathbf{A})$ with chain complexes.

If unambiguous, the bullet is dropped giving $\text{Ch}(\mathbf{A})$. It should be checked that $\text{Ch}(A)$ is indeed a category. There are many variations of $\text{Ch}(A)$ which will be important:

1. $C^+(A)$ is the full subcategory of $\text{Ch}(A)$ of complexes which are bounded from bellow, so that $L^i = 0$ for $i \ll 0$

2. $C^-(A)$ is the full subcategory of $\text{Ch}(A)$ of complexes which are bounded from above, so that $L^i = 0$ for $i \gg 0$

3. $C^{\geq 0}(A)$ (resp. $C^{\leq 0}$) for which $L^i = 0$ for $i \leq 0$ (resp. $i \geq 0$)

4. P, I are full subcategories of A consisting of projective and injective objects in A respectively, and hence we have $\text{Ch}(P)$, $\text{Ch}(I)$, and all its variants

There are also a few simple functors we may immediately define:

1. Take $\iota_r : \mathbf{A} \to \mathrm{Ch}(\mathbf{A})$ that sends an object A form $\mathbf{A}$ to a complex that is zero every and A in the rth degree. We shall commonly identify A with the complex given by ι_0.
2. Take $s_r : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$ which shifts over the complex, namely
$$C^\bullet \mapsto M[r]^\bullet$$
where $M[r]^i = M^{i+r}$ and $d^i_{M[r]^\bullet} = (-1)^r d^{ir+r}_{C^\bullet}$ (the $-1)^r$ shall be explained shortly).
3. the truncation functor shall truncate the complex in the natural way.

More importantly, $\mathrm{Ch}(A)$ is an abelian category, allowing the notion of exactness between complexes to be well-defined:

Lemma 2.1.5: Complex Category Is Abelian

Let A be an abelian category and $\mathrm{Ch}(A)$ the category of complexes of A. Then $\mathrm{Ch}(A)$ is an abelian category

***Proof*:**
Left as an exercise, check that:

1. morphisms between two complexes form an abelian group, namely since $\alpha^i + \beta^i$ will be well-defined
2. Finite products and coporducts exist in $\mathrm{Ch}(A)$ as a consequence of them existing in A
3. kernels and cokernls will come from the commutativity of the diagram defining morphisms of complexes

Since it is an abelian category, we shall be able to consider sequences of complexes, and in particular exact sequences of complexes:
$$\cdots \to L^\bullet \to C^\bullet \to N^\bullet \to \cdots$$
which give a commutative diagram like so:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+2} \arrow[r] \arrow[d, "\iota"] & A_{n+1} \arrow[r] \arrow[d, "\iota"] & A_n \arrow[r] \arrow[d, "\iota"] & A_{n-1} \arrow[r] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r] & B_{n+2} \arrow[r] \arrow[d, "\pi"] & B_{n+1} \arrow[r] \arrow[d, "\pi"] & B_n \arrow[r] \arrow[d, "\pi"] & B_{n-1} \arrow[r] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[r] \arrow[d] & C_{n+1} \arrow[r] \arrow[d] & C_n \arrow[r] \arrow[d] & C_{n-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd} \tag{2.1}$$

Example 2.1.6: Complexes

1. Given any object M, we may always create the trivial complex

$$\cdots 0 \to 0 \to M \to 0 \to 0 \to \cdots$$

   Verify that this is indeed a complex.

2. As mentioned, In R-**Mod**, the free resolution is an example of a complex.

3. Any short exact sequence can be thought of an (exact) complex if we add 0's that extend arbitrarily to the left and right.

4. Let $C^n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C^n = 0$ for $n < 0$. Let each d^n be defined as:

$$\overline{x} \mapsto \overline{4x}$$

   Show that this is a cochain complex.

5. Let $\{V_i, W_i\}_{i\in\mathbb{Z}}$ be a collection of vector spaces. Define:

$$A_n = V_n \oplus W_n \oplus V_{n-1}$$

   Combining the projection and inclusion maps $A_n \to V_{n-1} \to A_{n-1}$, show that $A_\bullet$ forms a chain complex.

6. Let $C^\bullet$ be a cochain complex, choose an object $A \in \mathbf{A}$, and apply $\mathrm{Hom}_A(-, A)$ to each object. Show that this is a cochain complex.

Since $\mathrm{Ch}(\mathbf{A})$ is an abelian category, kernels and cokernels exist and match with monomorphisms/epimorphisms, hence giving us the following:

Definition 2.1.7: Sub-complexes And Quotient Complexes

Let $C^\bullet$ be a (co)chain complex. Then a (co)chain complex $N^\bullet$ is called a *subcomplex* if each N_n is a sub-object of C_n and the differential on B_n is the restriction of the differential in C_n (categorically, the inclusion map $\iota : N^\bullet \to C^\bullet$ is a morphism of complexes).

Furthermore, C^n/B^n is a *quotient-complex*. Furthermore, by the canonical decomposition of morphisms in abelian categories, for any complex-morphism $\alpha^\bullet$, $\{\ker(\alpha^\bullet)\}$ and $\{\mathrm{coker}(\alpha^\bullet)\}$ is a sub-complex and quotient complex respectively.

2.1.1 Homology and Cohomology

Chain complexes need not be exact. The cohomology of a complex measures how far away a complex is from being exact, that is how far away a complex is form representing a sequence of extensions

Definition 2.1.8: Homology and Cohomology Of A Complex

Let $(C^\bullet, d^\bullet)$ be a complex. Then the ith homology and cohomology of the complex at M^i is, respectively:

$$H_i(C^\bullet) = \frac{\ker d_i}{\mathrm{im} d_{i+1}} \qquad H^i(C^\bullet) := \frac{\ker d^i}{\mathrm{im} d^{i-1}}$$

Though homology and cohomology are similar, it is not necessarily the case that $H^i(C^\bullet) = H_{n-i}(C^\bullet)$ for some appropriate n around which the chain can "pivot". We shall see such examples soon, and later also show under what conditions we may compare homology and cohomology[2]

Example 2.1.9: Cohomology Of Complex

Let's compute the cohomology of sequences in example 2.1.6.

1. The homology and cohomology at the 0th entry is:

$$\frac{\ker(d_0)}{\mathrm{im}(d_1)} = \frac{\ker(d^1)}{\mathrm{im}(d^0)} = M$$

   which shows that at this position, the chain is "off" by M in being exact.

2. As we have defined exact free resolutions, they are exact, meaning $\mathrm{im}(d^i) = \ker(d^{i-1})$, hence their homology and cohomology is always 0

3. Any exact sequence has trivial homology and cohomology

4. This sequence is not exact, and yet it's homoogy/cohomology is trivial (as the reader should verify). Such a sequence is called *acyclic*.

5. The reader should calculate this cohomology as an exercise

6. This cohomology shall be calculated later in this chapter.

Lemma 2.1.10: Cohomology Functor

For every $i \in \mathbb{Z}$, there exists a functor:

$$H^i : \mathrm{Ch}(A) \to A \qquad C^\bullet \mapsto H^i(C^\bullet)$$

which is additive and covariant.

This shows that taking cohomology is indeed a way of analyzing resolutions, for it preserves enough of the structures of modules and complexes. Notice it is not necessarily an exact functor, hence it will not in general preserve extensions.

[2]The universal coefficient theorem and Poincaré duality being two famous examples the reader may have heard of if they took algebraic topology

***Proof*:**
Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes and take $H^i(C^\bullet)$ and $H^i(N^\bullet)$. Let $x \in H^i(H^\bullet)$ so that $x = k + \mathrm{im}(d_M^{i-1})$ for d^{i-1}. Since $d^i(x) = 0$, by commutativity it must be that $\alpha^i(x) \in \ker(d_N^i)$. Furthermore, by commutativity we see that $\overline{\alpha^i}$ is zero on the image of $\mathrm{im}(d_M^{i-1})$, hence $H^i(\alpha^\bullet)$ is well-defined.

This functor is certainly covariant, and is additive since $\mathrm{Hom}_A(H^i(C^\bullet), H^i(N^\bullet))$ is an abelian group.

Notice that each H^i takes d^i to the zero map. Hence, we may consider the following functor:

$$H^\bullet : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$$

by taking $H^i(C^\bullet)$ for each i and connecting them by the zero-morphism. This is the cohomology of this complex then equals that of $C^\bullet$, namely:

$$H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$$

Lemma 2.1.11: Cohomology And Products

Let $\{A_\alpha\}$ be a collection of objects in $\mathbf{A}$. Then:

$$H^n(\oplus\alpha A_\alpha) = \oplus_\alpha H^n(A_\alpha) \qquad H^n(\prod_\alpha A_\alpha) = \prod_\alpha H^n(A_\alpha)$$

***Proof*:**
exercise.

Now, given a morphism of complexes $\alpha^\bullet : C^\bullet \to N^\bullet$, we may ask what (co)homological information such a morphism preserves. In other words, given a morphism $\alpha^\bullet$, when is $H^\bullet(\alpha^\bullet)$ an isomorphism? We give such morphisms a name:

Definition 2.1.12: Quasi-Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes. Then if $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ is an isomorphism, $\alpha^\bullet$ is called a *quasi-isomorphism*

One may wonder if the fact that the (co)homology is isomorphism is strong enough of a condition for the chains to be isomorphic. This turns out to be rather far from the case:

Example 2.1.13: Quasi-Isomorphism

1. The following counter example shall come up with many different minor variations, and so the reader should keep this in mind. Take the following morphism of complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

This is a quasi-isomorphism, as the reader should check, however it does not have an inverse. More generally, it is not even an equivalence relation, failing to be symmetric (show there is no morphism the other way around that is a quasi-isomorphism).

2. The natural maps $H^\bullet(C^\bullet) \to C^\bullet$ and $C^\bullet \to H^\bullet(C^\bullet)$ are quasi-isomorphisms, given the property of $H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$ discussed earlier.

3. Let $\iota(A) \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$ be the complex with A at the 0th position and zero's everywhere else. Say that $C^\bullet$ is a resolution such that $C^\bullet \to \iota(A)$ is a quasi-isomorphism. Show that this implies $C^\bullet$ is exact everywhere except possibly at $H^0(M^0)$

The lack of symmetry means that quasi-isomorphisms do not form an equivalence relation, but a *weak equivalence relation* (a relation that is reflective and transitive, but not symmetric). This is rather a pity, since if it did form an equivalence relation, then the equivalence class of morphisms up to weak isomorphism would hold all the necessary (co)homological information. Alas, this was not mean to be. Thus, in the pursuit of finding (co)homological invariants, it productive to ask how (co)homological information can be detected between (co)chain complexes. One idea is to map $\mathrm{Ch}(\mathbf{A})$ into another category $D(\mathbf{A})$ in which quasi-isomorphisms become isomorphisms. If quasi-isomorphisms formed equivalence relationships, we could have formed a congruence category which would satisfy this condition, however as just mentioned this cannot be the case.

Such a functor has already been encountered: the (co)homology functor is already one such example. It is more fruitful to as if this can be done in a *univerasl* way, that is if there is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd} \tag{2.2}$$

This category does indeed exist, and is called the *derived category* (hence the $D(-)$ notation). This category shall further be explored in the coming sections. In some nice categories $\mathbf{A}$, quasi-isomorphisms will line up with the nicer notion of *homotopy-equivalence*, in which case the derived category will be eminently understandable. We shall explore these all in greater detail section 2.2.

Though quasi-isomorphisms may not be ideal in computing (co)homological invariants, they are still important as the defining concept for (co)homological information. For example, the following shows the relationship between quasi-isomorphism and exactness:

Lemma 2.1.14: Quasi-isomorphisms And Exactness

Let $C^\bullet$ be a cochain complex. Then the following are equivalent:

1. $C^\bullet$ is exact
2. $C^\bullet$ is acyclic, that is $H^\bullet(C^\bullet) = 0$
3. The morphism of complexes $0 \to C^\bullet$ is a quasi-isomorphism

> ***Proof*:**
> Good exercise.

2.1.2 Snake Lemma

We often have multiple ways of constructions a resolution for an object M. We shall often extend resolutions using smaller resolutions. We then get cohomology of this new extended resolution by simply applying the cohomology functor. However, is is this functor exact? More generally, let's say we have a short exact sequence of complexes, then after applying cohomology do we get a short exact sequence of homology complexes? The answer is no: if

$$0 \to L^\bullet \to C^\bullet \to N^\bullet \to 0$$

is a short exact sequence, then we get the (not necessarily exact) sequence:

$$H^i(L^\bullet) \to H^i(C^\bullet) \to H^i(N^\bullet)$$

In terms of morphisms, if we have:

$$\alpha^\bullet : C^\bullet \to N^\bullet$$

Then the morphism after applying the cohomology functor $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ need not be exact, i.e. it doesn't preserve kernels and cokernels. Can we "fix" this in some way by recovering some exact sequence from this and thus allowing our analysis of building more complicated resolutions for M to be independent of cohomology. The answer is yes, Namely, if we take the cohomology of complexes, we may "snake" like in the following diagram:

$$\begin{tikzcd}
& 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & \\
\cdots \arrow[r, dotted] & H_{n+2}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_n(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n-1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\pi"] & H_n(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n-1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n+1}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_n(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n-1}(C) \arrow[r, dotted] \arrow[d, dotted] & \cdots \\
& 0 & 0 & 0 & 0 &
\end{tikzcd}$$

Theorem 2.1.15: Snake Lemma

> Given a short exact sequence of chain complexes, we may form a long exact sequence of modules using cohomology:
>
> $$\cdots \to H^n(A) \xrightarrow{i^*} H^n(B) \xrightarrow{\pi^*} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{i^*} H^{n+1}(B) \xrightarrow{\pi^*} H^{n+1}(C) \to \cdots$$

The following proof can be found (in the Homological variant) in EYNTKA Algebraic Topology

***Proof*:**
We first define the map $\partial : H_n(C) \to H_{n-1}(A)$. We should keep in mind the following diagram:

$$\begin{tikzcd}
 & A_{n-1} \arrow[d, "\iota"] \\
B_n \arrow[r, "\partial"] \arrow[d, "\pi"] & B_{n-1} \\
C_n &
\end{tikzcd}$$

Let $c \in C_n$ be a cycle so that $c \in \ker \partial_n$. Since π is surjective, $c = \pi(b)$ for some $b \in B_n$. Then by definition $\partial : B_n \to B_{n-1}$ so $\partial(b) \in B_{n-1}$. Then by commutativity $j(\partial(b)) = \partial(j(b)) = \partial(c) = 0$ (since c is a cycle), thus $\partial b \in \ker(\pi)$. Since $\ker j = \text{im} i$, $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ since i is injective, $\partial a = 0$. Thus, define $\partial : H_n(C) \to H_{n-1}(A)$ by $[c] \mapsto [a]$. This is well defined since:

1. The element a is uniquely determined by ∂b since i is injective
2. for differnet choice b' for b, we get $j(b') = j(b)$, os $b' - b \in \ker(j)$ which is equal to $\text{im}(i)$. Thus, $b' - b = i(a')$ for some a', ehcen $b' = b + i(a')$. But then this means we just chanage a to another representative (a homologous element) $a + \partial a'$, namely since
$$i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b = i(a'))$$
3. A difference choice of c within its homology class would have hte form $c = \partial c'$. To see this, since $c' = j(b')$ for some b',
$$c + \partial c' = c\partial j(b') = c + j(\partial b') = j(b + \partial b')$$
thus b is replaced by $b = \partial b'$, which gives ∂b and thus a is unchanged.

Finally, $\partial : H_n(C) \to H_{n-1}(A)$ is a homomorphism, for if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1, b_2, then
$$j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 = c_2$$
and
$$i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$$
hence
$$\partial([c_1] + [c_2]) = [a_1] + [a_2]$$
giving us the map ∂ is indeed a homomorphism. What's left to show is the following inclusions:

1. $\text{im} i_* \subseteq \ker j_*$. Since $ji = 0$, $j_* i_* = 0$
2. $\text{im} j_* \subseteq \ker \partial$. By definition of ∂, $\partial b = 0$ so $\partial j_* = 0$.
3. $\ker j_* \subseteq \text{im} i_*$. Let $b \in B_n$ with boundary represent a homology class in $\ker j_*$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjectiv,e $c' = j(b')$ for some $b' \in B_{n+1}$. Then since $\partial j(b') = \partial c' = j(b)$,
$$j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$$

Thus, $b - \partial b' = i(a)$ for some $a \in A - n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective, a is a cycle. Thus:

$$\iota_*[a] = [b - \partial b'] = [b]$$

showing that i_* maps onto $\ker j_*$

4. $\ker \partial \subseteq \mathrm{im} j_*$. By definition of ∂, if c represents a homology class in $\ker \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. Since

$$j(b - i(a')) = j(b) - ji(a') = j(b) = c$$

we get that $j_*[b - i(a')] = [c]$

5. $\ker i_* \subseteq \mathrm{im} \partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for seom $b \in B_n <$ then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, hence $\partial[j(b)] = [a]$

Doing this shall completing the proof.

Example 2.1.16: Uses Of Snake Lemma
(from alg top?)

There is another way of looking at chain complexes which shall be important in elucidating triangle categories in section ref:HERE.

Mapping Cones

Using a particular total complex, we can detect when a quasi-isomorphism is an isomorphism. This result requires the snake lemma to prove, hence the need to put this section after the snake lemma.

Definition 2.1.17: Mapping Cone

Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes. Then the *mapping cone* is the bi-complex denoted by $MC(\alpha^\bullet)$ defined by:

$$MC(\alpha)^i := L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

where the morphism $d^i_{MC(\alpha)} : MC(\alpha)^i \to MC(\alpha)^{i+1}$ are given by:

$$d^i_{MC(\alpha)}(l, m) := (-d_L i + 1(l), \alpha^{i+1}(l) + d^i_M(m))$$

The term mapping cone comes from the topological notion of a mapping cone of a function. Visually, the mapping cone can be seen as the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r] \arrow[drr] & L^{i+1} \arrow[r, "-d_L^{i+1}"] \arrow[drr, "\alpha^{i+1}"] & L^{i+2} \arrow[r] \arrow[drr] & \cdots & \\
 & \oplus & & \oplus & \\
\cdots \arrow[r] & M^i \arrow[r, "d_M^i"'] & M^{i+1} \arrow[r] & \cdots &
\end{tikzcd}$$

Given $MC(\alpha)^\bullet$, we may define two natural chain maps $C^\bullet \to MC(\alpha)^\bullet$ and $MC(\alpha)^\bullet \to L[1]^\bullet$ induced by the natural exact morphisms $M^i \to L^{i+1} \oplus M^i \to L^{i+1}$ which implies that:

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact.

Proposition 2.1.18: Mapping Cone And Exact Triangles

There is an exact triangle

$$\begin{tikzcd}
 & H^\bullet(C^\bullet) \arrow[dr] & \\
H^\bullet(L[1]^\bullet) \arrow[ur, "+1\ \delta"] & & H^\bullet(MC(\alpha)^\bullet) \arrow[ll]
\end{tikzcd}$$

where δ is the morphism induced by cohomology

***Proof*:**
Aluffi p.607

2.1.19 Remark Not all categories have this triangle property, categories which will have this will be called *triangulated categories*

Proposition 2.1.20: Quasi-isomorphism And Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$. Then $H^\bullet(\alpha^\bullet)$ is an isomorphism if and only if $MC(\alpha)^\bullet$ is an exact complex.

***Proof*:**
exercise

2.1.3 Chain Homotopies and Homotopic Categories

The notion of homotopies originated in topology as a weaker alternative to continuous functions[3]. The topological notion of homotopy settled on the level of continuous functions, where $f, g : X \to Y$ are homotopic (denoted $f \simeq g$) if there exists a continuous function $F : X \times [0,1] \to Y$ where $F(\cdot, t) = f_t$ is continuous, $f_0 = f$ and $f_1 = g$. If we represent X, Y as one dimensional lines, then we may visualize a homotopy as:

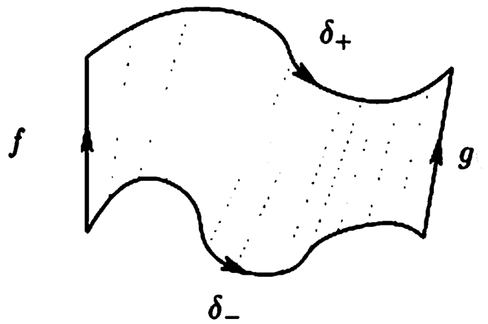


Figure 2.1: homotopy

In the development of homotopy theory, it was found that the (co)homology and homotopy of topological spaces (or their representations as chain complexes) is homology-invariant, hence homotopies are examples of quasi-isomorphisms. The proofs of these results turn out to be independent of the underlying topological structures and is purely algebraic, and hence the notion of homotopy has found a generalization as a purely algebraic notion between two (co)chain complexes.

Due to its origin in topology, the intuitions behind the following definitions will proceed more naturally with a more geometric build-up. Since homotopy may be interpreted purely algebraically, this geometrica build-up will be followed by a more algebraic interpretation. First, the reader may take for granted that given a topological space X, we may induce a chain $C_\bullet$, which is a topological object[4]. Given two continuous maps $f, g : X \to Y$, we may push forward the chain like so:

update image

[3] every homeomorphic is a homotopy equivalence, however not every homotopy equivalence is a homeomorphism, hence homotopy can be seen as weaker than continuity

[4] See EYTNKA algebraic topology or Hatcher as for more details

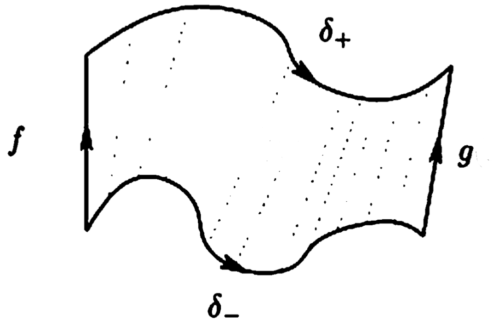


Figure 2.2: homotopy on chain

The space $h(C_\bullet)$ is a topological space, hence we may ask for it's border, $\partial h(C_\bullet)$. By inspecting the image, we see that:

$$\partial h(C_\bullet) = g(C_\bullet) - \delta_+ - f(C_\bullet) + \delta_- = g(C_\bullet) - f(C_\bullet) + h(\partial C_\bullet)$$

re-ordering, we get:

$$g(C_\bullet) - f(C_\bullet) = \partial h(C_\bullet) - h(\partial C_\bullet)$$

Abstracting notation, we get:

$$g - f = \partial h - h\partial \tag{2.3}$$

Now, recall that in a chain complex, the image of the function ∂ is called the boundary, and that by definition of homology the boundary becomes trivial. In a topological setting, this vocabulary reflects the concrete of the boundary map ∂: it maps a chain to it's boundary. What this means for us is that when taking homology on both sides, we get:

$$H^k(g|_{C_\bullet} - f|_{C_\bullet}) = \overline{0}$$

meaning the two maps are equivalent up to homology! In other words, this "geometric intuition" shows that a homotopy is an example of a quasi-isomorphism. Even better, homotopies shall form an equivalence relation, hence they are an example of a quasi-isomorphism that shall capture some (co)homological information (and in the case where all quasi-isomorphisms are homotopies, they could fully capture the (co)homological information!)

Though this is the origins of chain homotopies being (co)homology-invariant, since they may be interpreted in pure algebraic terms, it is worthwhile taking a moment to have an algebraic build-up as well. Algebraically, homotopies can be seen as forming equivalence classes of (co)chain complexes up to *obstruction from being split-exact.* Recall that a split exact sequence is a short exact sequence that is also split at each map, that is there exists a section morphism $s^n : C^n \to C^{n-1}$ such that $d^{n-1} \circ s^n \circ d^{n-1} = d$, or if we drop the indices and collect all the section maps:

$$d = dsd$$

In other words, it is possible to "inverse" the map d via the map s in such a way that it "cancels" the effect of d (for we may re-apply the map d after sd and get the same result). The simplest source of split-exact (co)chains are exact (co)chains of vector-spaces, for any vector-space maps split[5]. Given a

[5] More generally, any maps of projective modules split

cochain, we may write:

$$C^n = Z^n \oplus B' \qquad (B^n)' \cong C^n/Z^n = d(C^n) = B^{n+1}$$

$$Z^n = B^n \oplus (H^n)' \qquad (H^n)' = Z^n/B^n = H^n(C)$$

Hence, $C^n \cong Z^n \oplus B^{n+1}$. This allows for the definition of the section map via the composition:

$$C^n \hookrightarrow Z^n \hookrightarrow B^n \cong d(C^n) \cong B^{n+1} \subseteq C^{n+1}$$

We may thus define the maps $d \circ s$ and $s \circ d$ where $d \circ s(C^n) = B^n$ and $s \circ d(C^n)(B^n)' \cong B^{n+1}$. Adding these maps together, we get a map $ds + sd : C^n \to C^n$ whose kernel is the homology:

$$\ker(ds + sd)_n = H^n(C^\bullet)$$

The kernel (and cokernel) of the chain map $ds + sd$ is naturally $H^\bullet(C^\bullet)$. Working over more general R-modules (and more generally in an arbitrary abelian category), we loose the natural section maps, for example $0 \to \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$ does note have a section map for each map. Instead, we may define arbitrary maps $s^n : C^n \to D^{n-1}$ where we may put them in the following (*not* necessarily commutative) diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d"] \arrow[dl] & C^n \arrow[r, "d"] \arrow[dl, "s^n"] & C^{n+1} \arrow[r] \arrow[dl, "s^{n+1}"] & \cdots \arrow[dl] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Given any such collection of morphisms, we may define $f^n = s^{n+1} \circ d^n + d^{n-1} \circ s^n$ (or, if dropping indices and abbreviate, $f = sd + ds$). Notice that even though the maps $s^\bullet$ do not commute, $f^\bullet$ *will* create a commutative diagram, in particular it is a chain map. To see this, notice that (after dropping indices):

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

showing it is indeed a chain map:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d"] \arrow[dl, dotted] \arrow[d, "f^{n-1}"] & C^n \arrow[r, "d"] \arrow[dl, dotted, "s^n"] \arrow[d, "f^n"] & C^{n+1} \arrow[r] \arrow[dl, dotted, "s^{n+1}"] \arrow[d, "f^{n+1}"] & \cdots \arrow[dl, dotted] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Thus, given any collection of morphisms $s^\bullet$, we may naturally define a chain map using the differentials. Conversely, is every morphism of complexes representable as $ds + sd$. This is not in general the case; maps being representable like so are given a name:

Definition 2.1.21: Null-homotopic Maps

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then f is called *null-homotopic* if there exists a collection $\{s^i\}_{i \in \mathbb{Z}}$ of morphisms from C^i to C^{i-1} between two cochain complexes $C^\bullet$ and $D^\bullet$ such that

$$f^\bullet = ds + sd$$

The maps $s^\bullet$ are called the *chian contraction* of $f^b b$

To show that not all maps are null-homotopic, consider the following lemma:

Lemma 2.1.22: Null-homotopic And Split Exact Complexes

Let $C^\bullet$ be a complex. Then $C^\bullet$ is split exact if and only if the identity map is null-homotopic.

***Proof*:**
exercise

The name “null-homotopic” comes from how the (co)homology of such maps are trivial:

Proposition 2.1.23: Null-homotopic And Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be null-homotopic. Then $H^\bullet(f)$ is the zero map

***Proof*:**
Since f is null-homotopic, let $f = ds + sd$. Take a n-cycle representative x of an equivalence class of $H^n(C^\bullet)$. Then:

$$f(x) = d(s(x))$$

Hence, $f(x)$ is an n-boundary in D^n, hence $\overline{f(x)} = \overline{0}$ in $H^n(D^\bullet)$, as we sought to show. (this is an elementary result, but a similar proof is given in Aluffi p.612 if needed)

Thus, maps that are null-homotopic induce zero (co)homology maps![6] Thus, we may think of maps that are null-homotopic as being the “zero” maps. This leads us to the following definition:

Definition 2.1.24: Chain-homotopy

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be two morphisms of complexes. Then we say that these two maps are *chain homotopic* or just *homotopic* if there exists maps $h^i : C^i \to D^{i-1}$ such that

$$g^i - f^i = (d^{i-1}_{C^b}) \circ h^i + h^i \circ (d^i_{D^\bullet})$$

or, dropping indices:

$$g^\bullet - f^\bullet = dh - hd$$

If $f^\bullet, g^\bullet$ are homotopic, we write $f^\bullet \sim g^\bullet$

The s notation has been replaced with h notation to emphasize now the “homotopy” aspect of it rather than the “section” property that was evident in the case of vector spaces. If we were working with chains, then we would have

$$g_\bullet - f_\bullet = \partial h - h\partial$$

which, the reader may recognize as equation (2.3). If we were working with chains, then $h_i : C_i \to D_{i+1}$ would shift up a degree, which would be interpreted in the geometrical setting as moving *up* a dimension, in the same way that homotopies are a “dimension more” than the original spaces X, Y. Furthermore, the reader should verify the “up to homotopy” *is* an equivalence relation, as the reader should verify!

[6] TBD How much can we say about the converse, if a map induces the zero morphism, then is it null-homotopic?

Importantly, two chain-homotopic maps induce the same (co)homology:

Proposition 2.1.25: Chain-Homotopy and Cohomology

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be chain homotopic. Then

$$H^\bullet(f^\bullet) = H^\bullet(g^\bullet)$$

***Proof*:**
Since $f^\bullet$ and $g^\bullet$ are chain-homotopic, by proposition 2.1.23 for each element in $H^k(C^\bullet)$ (since it is an additive functor):

$$H^k(f^\bullet)(\overline{x}) - H^k(g^\bullet)(\overline{x}) = 0 \qquad \Rightarrow \qquad H^k(f^\bullet)(\overline{x}) = H^k(g^\bullet)(\overline{x})$$

as we sought to show.

The above result shows promise in using chain-homotopies to hold (co)homological information. The following definition is the important connection idea:

Definition 2.1.26: Homotopy Equivalent Chains

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then $f^\bullet$ is a *homotopy equivalence* if there exists a morphisms of complexes $g^\bullet : D^\bullet \to C^\bullet$ such that

$$f^\bullet \circ g^\bullet \sim 1_{D^\bullet} \qquad \text{and} \qquad g^\bullet \circ f^\bullet \sim 1_{C^\bullet}$$

If $f^\bullet : C^\bullet \to D^\bullet$ is a homotopy equivalence, then $C^\bullet$ and $D^\bullet$ are said to be *homotopy equivalent*

Proposition 2.1.27: Homotopy Equivalence Ans Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be a homotopy equivalence. Then $f^\bullet$ is a quasi-isomorphism, that is $H^\bullet(f^\bullet)$ is an isomorphism

***Proof*:**
Since $f^\bullet$ is a homotopy equivalence, we have $g^\bullet$ that is it's homotopic inverse. Then the maps $f^\bullet \circ g^\bullet$ and $g^\bullet \circ f^\bullet$ induce inverse morphisms of cohomology by proposition 2.1.25, and hence give an isomorphism, as we sought to show.

As demonstrated in example 2.1.13 not all quasi-isomorphisms are invertible even up to homotopy equivalence, hence quasi-isomorphisms are more general. As mentioned earlier, in nice enough categories these two concepts shall coincide.

Now that homotopy equivalences is established, can be asked if given an (additive) functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$, if there are two homotopic complexes in $\mathrm{Ch}(\mathbf{A})$, will the (co)homology be isomorphic in $\mathbf{B}$? In other words, when trying to define the derived category that was mentioned earlier, can this category be defined at least up to homotopy? The answer is yes! The proof of this result essentially follows from the following lemma:

Lemma 2.1.28: Homotopy Equivalence And Additive Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories, $C(\mathscr{F})$ the induced functor on chains. Then if $f^\bullet \sim g^\bullet$ in $\mathrm{Ch}(\mathbf{A})$, $C(\mathscr{F})(f^\bullet) \sim \mathrm{Ch}(\mathbf{B})(g^\bullet)$ in $\mathrm{Ch}(\mathbf{B})$. Furthermore, if $C^\bullet$ and $D^\bullet$ are homotopy equivalence, $C(\mathscr{F})(C^\bullet)$ and $C(\mathscr{F})(D^\bullet)$ are homotopy equivalent

***Proof*:**
Given the first assertion, the second is an immediate consequence, hence we prove the first. Let $f^\bullet \sim g^\bullet$ so that :

$$f^i - g^i = d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet}$$

Since $\mathscr{F}$ is an additive functor:

$$\mathscr{F}(f^i) - \mathscr{F}(g^i) = \mathscr{F}\left(d^{i-1}_{D^\bullet}\right) \circ \mathscr{F}(h^i) + \mathscr{F}\left(h^{i+1}\right) \circ \mathscr{F}\left(d^i_{C^\bullet}\right)$$

hence, the morphisms $\mathscr{F}(h^i)$ gives a homotopy between $\mathscr{F}(f^\bullet)$ and $\mathscr{F}(g^\bullet)$, as we sought to show.

Note this does *not* hold for quasi-isomorphisms (exercise 4.15 in Aluffi), needing the stronger condition of exact functors.

Theorem 2.1.29: Derived Category And Homotopy Equivalence

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then if $C^\bullet$ and $D^\bullet$ are homotopy-equivalent, then:

$$H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$$

that is, their cohomology is isomorphic.

***Proof*:**
exercise.

Hence, up to homotopy, the choice of cochain shall give the same cohomology information!

Homotopic Categories

With chain homotopies being a nicer sub-class of quasi-isomorphisms, their exploration is worth further pursuing. They are in fact almost the solution to the universal problem given in equation (2.2). Let us take for granted that a category $D(A)$ exists that satisfies the universal property. What necessary conditions can be deduced about $D(A)$?

Lemma 2.1.30: Derived Category And Additive Functors

Let $\mathbf{D}$ be an additive category, $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that takes quasi-isomorphisms to isomorphisms.

1. Let $C^\bullet$ be an exact complex in $\mathrm{Ch}(\mathbf{A})$. Then $\mathscr{F}(C^\bullet) = 0$, that is $\mathscr{F}(C^\bullet)$ is the zero object in $\mathbf{D}$.

2. Let $\alpha^\bullet : C^\bullet \to E^\bullet$ be a morphism of complexes that factors through an exact complex:

$$\begin{tikzcd} C^\bullet \arrow[rr, "\alpha^\bullet"] \arrow[dr] & & E^\bullet \\ & D^\bullet \arrow[ur] & \end{tikzcd}$$

   where $D^\bullet$ is exact. Then $\mathscr{F}(\alpha^\bullet)$ is the zero morphism.

***Proof*:**

1. Since $C^\bullet$ is exact, the zero morphism $0^\bullet : C^\bullet \to C^\bullet$ is a quasi-isomorphism, and so $\mathscr{F}(0^\bullet)$ maps to an isomorphism (i.e. an invertible morphism in $\mathbf{D}$). Thus:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)"'] \arrow[rr, bend left, "\mathrm{id}_{\mathscr{F}(C^\bullet)}"] & \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)^{-1}"'] & \mathscr{F}(C^\bullet) \end{tikzcd}$$

   Since $\mathscr{F}$ is additive functor, in particular a group homomorphism on hom-sets, $\mathscr{F}(0) = 0$. Thus:

$$\mathrm{id}_{\mathscr{F}(C^\bullet)} = 0$$

   It follows that $\mathscr{F}(C^\bullet)$ must be the zero object (as the reader should verify, see exercise ref:HERE)

2. Applying $\mathscr{F}$ to the diagram in the question, we get the diagram:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[rr] \arrow[dr] & & \mathscr{F}(E^\bullet) \\ & \mathscr{F}(D^\bullet) \arrow[ur] & \end{tikzcd}$$

   Since $\mathscr{F}(D^\bullet) = 0$ by the above result, we see that $\mathscr{F}(\alpha^\bullet)$ must be the zero morphisms, completing the proof.

Lemma 2.1.31: Derived Categories And Homotopies

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be homotopic morphisms in $\mathrm{Ch}(\mathbf{A})$. Then

$$\mathscr{F}(f^\bullet) = \mathscr{F}(g^\bullet)$$

Note that this was shown explicitly when $\mathscr{F} = H^i(-)$.

***Proof*:**
Since $\mathscr{F}$ is additive, it suffices to show that if $f^\bullet \sim 0$, then $\mathscr{F}(f^\bullet) = 0$. By lemma 2.1.30, it suffices to show $f^\bullet$ factors through an exact complex. The exact complex that shall be used is the mapping cone $MC(\mathrm{id}_{C^\bullet})$ for the identity map $\mathrm{id}_{C^\bullet} : C^\bullet \to C^\bullet$, namely

$$MC(\mathrm{id}_{C^\bullet})^i = C^{i+1} \oplus C^i$$

$$d^i_{MC(\mathrm{id}_{C^\bullet})}(a, b) = (-d^{i+1}_{C^\bullet}(a), a + d^i_{C^\bullet}(b))$$

By proposition 2.1.20, since the identity map $\mathrm{id}_{C^\bullet}$ is (trivially) a quasi-isomorphism, $MC(\mathrm{id}_{C^\bullet})$ is an exact complex.

Now, the goal is to find morphism of complexes where:

$$C^\bullet \to MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$$

the morphism of complexes $C^\bullet \to MC(\mathrm{id}_{C^\bullet})$ is readily available: for each i:

$$b \mapsto (0, b)$$

On the other hand, it is not immediately clear there is a morphism of complexes $MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$. To find such a morphism, we take advantage of the homotopy $f^\bullet \sim 0$, that is there exists maps:

$$h^i : C^i \to D^{i-1}$$

such that $f^\bullet = fh + hd$. With these, define the maps:

$$\pi^i : MC(\mathrm{id}_{C^\bullet})i \to D^i \qquad (a, b) \mapsto h^{i+1}(a) + f^i(b)$$

If this is indeed a morphism of complexes, then the map $f : C^\bullet \to D^\bullet$ factored through an exact complex and hence must be trivial. What must be verified is that:

$$d^{i-1}_{D^\bullet} \circ \pi^{i+1} = \pi^i \circ d^{i-1}_{MC(\mathrm{id}_{C^\bullet})} \tag{2.4}$$

The left hand side of equation (2.4) gives:

$$(a, b) \mapsto h^i(a) + f^{i-1}(b) \mapsto d^{i-1}_{D^\bullet}(h^i(a) + f^{i-1}(b))$$

while the right hand side gives:

$$(a, b) \mapsto (-d^i_{C^\bullet}(a), a + d^{i-1}_{L^\bullet}(b)) \mapsto -h^{i+1}(d^i_{C^\bullet}(a) + f^i(a + d^{i-1}_{C^\bullet}(b))$$

These two sides must be equal. Simplifying the equality, we get:

$$(d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet} - f^i)(a) = (f^i \circ d^{i-1}_{C^\bullet} - d^{i-1}_{D^\bullet} \circ f^{-1})(a)$$

for all $a \in C^{i+1}, b \in C^i$. But notice that both sides are zero: the right hand side b $f^\bullet$ is a morphism of complexes, while the left hand side because h is a homotopy and $f^\bullet \sim 0$. But then this gives us our desired result, completing the proof.

Thus, behave well for the desired universal property presented in equation 2.2. We formalize the category in question:

Definition 2.1.32: Homotopy Category

Let $\mathbf{A}$ be an abelian category. Then the *homotopy category* of $\mathbf{A}$, denoted $K(\mathbf{A})$ is the category whose objects ar ethe (co)chain complexes in $\mathbf{A}$ (i.e. $\text{Ch}(\mathbf{A})$), and whose morphisms are:

$$\text{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \text{Hom}_{\text{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$$

where $\sim$ is the homotopy relation, that is, the morphisms are morphisms of complexes up to homotopy.

These are indeed morphisms, namely $\sim$ is transitive, it respects composition (exercise). Naturally, the equivalent bounded variations $K^-(\mathbf{A})$ and $K^+(\mathbf{A})$ are obtained from $\text{Ch}^+(\mathbf{A})$ and $\text{Ch}^-(\mathbf{A})$.

Lemma 2.1.33: Homotopy Category Is Additive

Let $\mathbf{A}$ be an abelian category. Then $K(\mathbf{A})$ is an *additive category*

***Proof*:**
exercise

Note that in general homotopic categories are *not* abelian. Indeed, homotopic maps $f^\bullet \sim g^\bullet$ need not have the same kernel/cokernel, giving no natural definition of these concepts. The homotopic category is a type of category that is stronger than additive but weaker than abelian, known as a *triangulated category* (recall remark˜triangulatedCat).

In $K(\mathbf{A})$, homotopy equivalences become isomorphisms. This comes straight from the definition, for if $f^\bullet \circ g^\bullet \sim \text{id}$ in $\text{Ch}(\mathbf{A})$, then $f^\bullet \circ g^\bullet = \text{id}$ in $K(\mathbf{A})$, which can be thought of as formally inverting homotopy equivalences. Using this, we have the following almost universal property, or factorization

This really bugs me, this feels like the universal property, why isn't it

Proposition 2.1.34: Homotopy Category Factorization

Let $\mathscr{F} : \text{Ch}(\mathbf{A}) \to D$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Then ff factors uniquely through $K(\mathbf{A})$:

$$\begin{tikzcd}
\text{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\
K(\mathbf{A}) \arrow[ur, "\exists!"'] &
\end{tikzcd}$$

***Proof*:**
exercise

Hence, in the case where $K(\mathbf{A})$ is *not* the universal object satisfying equation (2.2), there must be a unique functor from $K(\mathbf{A})$ to $D(\mathbf{A})$.

2.2 Projective and Injective Resolutions

It has been alluded to a few times that for special categories, homotopy equivalence and quasi-isomorphism coincide. This shall now be covered in this section. In particular, this shall linked to the idea of an abelian category $\mathbf{A}$ being bounded from above or bellow, and having *enough projective objects* or *enough injective objects*. These categories shall furthermore be shown to be universal in some appropriate way, making their choice not arbitrary. Recall that in $\mathbf{A} = R\text{-}\mathbf{Mod}$, an R-module M Is called projective if $\mathrm{Hom}_R(M, -)$ is exact, and injective if $\mathrm{Hom}_R(-, M)$ is exact. Recall that projective/injective modules nicely lend themselves to a splitting condition on short exact sequences for which they are one of the modules forming them (recall **??** and **??**). This will be important, due to the link of chain homotopies with split exact sequences.

Definition 2.2.1: Projective And Injective Objects

Let $\mathbf{A}$ be an abelian category. Then

1. An object $P \in \mathbf{A}$ is said to be *projective* if the functor $\mathrm{Hom}_A(P, -)$ is exact.
2. An object $Q \in \mathbf{A}$ is said to be *injective* if the functor $\mathrm{Hom}_A(-, Q)$ is exact

?? and **??** hold for abelian categories more generally (in particular the splitting remarks), as the reader may venture to check. Since the dual of an abelian category is abelian (see exercise ref:HERE), the theory on projective objects is dual to that of injective object, and the functor $(-)^{\mathrm{op}}$ maps projective/injectives to injectives/projectives respectively. Thus, many proofs shall simply proof the result for projectives, the result for injectives coming automatically. Note that as we've seen in chapter **??**, in a fixed [abelian] category $\mathbf{A}$, projective and injective may have very different features.

In what follows, let $\mathbf{P}$ denote the full subcategory of $\mathbf{A}$ determined by the projective objects, and let $\mathbf{I}$ denote the full subcategory of $\mathbf{A}$ determined by the injective objects. Let $K^-(\mathbf{P})$ denote the full subcategory of $K(\mathbf{A})$ consisting of bounded-above complexes of projective objects of $\mathbf{A}$, and $K^+(\mathbf{I})$ be the full subcategory of $K(\mathbf{A})$ consisting of bounded-below complexes of injective objects of $\mathbf{A}$. Note that these are generally not abelian categories, and furthermore projective and injective objects need not exist:

Example 2.2.2: Category With And Without Projective/Injective Resolutions

1. The category of finite abelian groups has no projective or injective, hence no projective/injective resolution.
2. The category of finitely generated abelian groups has projectives, but no injectives, namely since $\mathbb{Z}^n$ for every n are objects this category. Thus, it is possible to create a free, and hence projective, resolution.
3. the dual of the above category has enough injectives but not enough projectives.
4. The category of R-modules has both projective and injective resolutions (projective resolutions comes from free resolutions, while injective resolutions comes from repeated application of **??**)

It is thus helpful to add the following condition to an abelian category

Definition 2.2.3: Enough Projective And Enough Injectives

Let $\mathbf{A}$ be an abelian category. Then:

1. we say that $\mathbf{A}$ as *enough projectives* if for every object A of $\mathbf{A}$, there exists a projective object P of $\mathbf{A}$ and an epimorphism $P \twoheadrightarrow A$.

2. we say that $\mathbf{A}$ as *enough injectives* if for every object A of $\mathbf{A}$, there exists a injective object Q of $\mathbf{A}$ and a monomorphism $A \hookrightarrow Q$.

With the important definitions established, the main objective of this section may start going under way. The goal is to show that if we have a quasi-isomorphism between two bounded-above complexes of projectives $f^\bullet : P_0^\bullet \to P_1^\bullet$ (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$), then $f^\bullet$ is in fact a homotopy-equivalence, from which it can be concluded that in $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$ (homotopy-classes of) quasi-isomorphisms are isomorphisms, which shall give us the desired derived category $D^-(\mathbf{A})$ and $D^+(\mathbf{A})$ in the case where A has enough projectives (resp. enough injectives).

Let us now revisit this splitting condition. Recall that if $C^\bullet$ in $\mathrm{Ch}(\mathbf{A})$ is exact, then the identity map $\mathrm{id}_{C^\bullet}$ and the trivial map induce the same morphisms in (co)homology. However, there do exist exact complexes that are *not* homotopic to 0. As discussed earlier, this directly links to the splitting condition, namely the identity is homotopic to 0 if the sequence is split exact. Being projective is exactly the condition that causes splitting behavior. With this in mind, let us prove the first lemma:

Lemma 2.2.4: Zero-morphism In Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$ (i.e. $P^i = 0$ for $i > 0$), and let $C^\bullet$ be a complex in $\mathrm{Ch}(\mathbf{A})$ such that $H^i(C^\bullet) = 0$ for $i < 0$ (i.e. has trivial cohomology in negative coefficients). Let $f^\bullet : P^\bullet \to C^\bullet$ be a morphism of complexes that induces the zero-morphism in cohomology: $H^\bullet(f^\bullet) = 0$. Then

$$f^\bullet \sim 0$$

i.e., $f^\bullet$ is null-homotopic.

The injective version of this statement is comes from dualizing the appropriate terms.

***Proof*:**
To show $f^\bullet$ is null-homotopic, we must show there exists maps $h^i : P^i \to C^{i-1}$ such that

$$f^i = d_{C^\bullet}^{i-1} \circ h^i + h^{i+1} \circ d_{P^\bullet}^i \tag{2.5}$$

as a diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{-2} \arrow[r] \arrow[dl, "h^{-2}"'] \arrow[d, "f^{-2}"] & P^{-1} \arrow[r] \arrow[dl, "h^{-1}"'] \arrow[d, "f^{-1}"] & P^0 \arrow[r] \arrow[dl, "h^0"'] \arrow[d, "f^0"] & 0 \arrow[r] \arrow[dl, "h^1=0"'] \arrow[d] & \cdots \\
\cdots \arrow[r] & C^{-2} \arrow[r] & C^{-1} \arrow[r] & C^0 \arrow[r] & C^1 \arrow[r] & \cdots
\end{tikzcd}$$

Naturally, $h^i = 0$ for $i > 0$. For the rest, we shall use induction. The base case is $i = 0$. Since the cohomology induced by $f^\bullet$ is zero, f^0 factors through the image of $d_{C^\bullet}^{-1}$:

$$\begin{tikzcd} & P^0 \arrow[d, "f^0"] & \\ C^{-1} \arrow[r, "d_{C^\bullet}^{-1}"] & \mathrm{im} d_{C^\bullet}^{-1} \arrow[r] & 0 \end{tikzcd}$$

Now, since P^0 is projective, there exists a morphism $h^0 : P^0 \to C^{-1}$ such that $f^0 = d_{C^\bullet}^{-1} \circ h^0$, as needed.

Next, assume h^i and h^{i+1} satisfies equation (2.5). The goal is to show h^{i-1} satisfies it too. By equation (2.5) and the fact that $P^\bullet$ is a complex:

$$d_{C^\bullet}^{i-2} \circ h^i \circ d_{P^\bullet}^{i-1} = (f^i - h^{i+1} \circ d_{P^\bullet}^{i-1}) = f^i \circ d_{P^\bullet}^{i-1} - h^{i+1} \circ 0 = f^i \circ d_{P^\bullet}^{i-1}$$

which can be seen by staring at the following diagram:

$$\begin{tikzcd} \cdots \arrow[r, dotted] & P^{i-1} \arrow[r, "d_{P^\bullet}^{i-1}"] \arrow[d, dotted] & P^i \arrow[r, "d_{P^\bullet}^{i}"] \arrow[dl, "h^i"'] \arrow[d, "f^i"'] & P^{i+1} \arrow[r, dotted] \arrow[dl, "h^{i+1}"'] \arrow[d, dotted] & {} \\ \cdots \arrow[r, dotted] & C^{i-1} \arrow[r, "d_{C^\bullet}^{i-1}"'] & C^i \arrow[r, dotted] & C^{i+1} \arrow[r, dotted] & {} \end{tikzcd}$$

Thus, since f^b is a morphism of cochain complexes:

$$d_{C^\bullet}^{i-1} \circ (f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}) = d_{C^\bullet}^{i-1} \circ f^{i-1} - f^i \circ d_{P^\bullet}^{i-1} = 0$$

Importantly, $(f^{i-1} - h^i \circ d_{P^\bullet}^{i-1})$ factors through $\ker d_{C^\bullet}^{i-1}$. Since $C^\bullet$ is exact at C^{i-1} for $i \leq 0$, $\ker d_{C^\bullet} = \mathrm{im}_{C^\bullet}^{i-2}$. Thus, we get:

$$\begin{tikzcd} & P^{i-1} \arrow[d, "f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}"] & \\ C^{i-2} \arrow[r, "d_{C^\bullet}^{i-2}"] & \mathrm{im}(d_{C^\bullet}^{i-2}) \arrow[r] & 0 \end{tikzcd}$$

Now, since P^{i-1} is projective, there exists a lift $h^{i-1} : P^{i-1} \to C^{i-2}$ such that

$$f^{i-1} - h^i \circ d_{P^\bullet}^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1}$$

which, shuffling around, gives us

$$f^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1} + h^i \circ d_{P^\bullet}^{i-1}$$

as we sought to show.

Corollary 2.2.5: Exactness Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{A})$, and $C^\bullet \in \mathrm{Ch}(\mathbf{S})$ an exact complex. Then every morphism $P^\bullet \to L^\bullet$ is homotopic to 0

***Proof*:**
Since every morphism to an exact complex induces zero-morphism in cohomology, this is a direct application of lemma 2.2.4.

Note the importance of the bounding from above/bellow (depending on whether working with projectives or injective) in the inductive argument, without it there wouldn't be an appropriate base case. Using the above results, the first quasi-isomorphism to homotopy equivalence can be established:

Corollary 2.2.6: Projectives With Quasi-isomorphism And Zero Morphisms

Let $P^\bullet$ (resp. $Q^\bullet$) be a bounded-above exact complex of projectives (resp. a bounded bellow exact complex of injectives). Then $P^\bullet$ (resp. $Q^b b$) is homotopy-equivalent to the zero-complex.

***Proof*:**
exercise (Aluffi made if 5.12 in his book, see p. 623)

Another important result that will be used in the next section is that quasi-isomorphisms are "non-zero-divisors" up to homotopy wrt morphisms from commplexes of projectives:

Lemma 2.2.7: Quasi-isomorphisms Of Projectives And Zero-divisor

Let $\mathbf{A}$ be an abelian category, $f^\bullet : C^\bullet \to D^\bullet$ be a quasi-isomorphism. Let $P^\bullet$ be a bounded-above complex of projectives, and let $\alpha^\bullet : P^\bullet \to C^\bullet$ be a morphism of cochian complexes st the composition:

$$P^\bullet \xrightarrow{\alpha^\bullet} C^\bullet \xrightarrow{f^\bullet} D^\bullet$$

is homotopic to the zero-morphism. Then $\alpha^\bullet \sim 0$

***Proof*:**
Let $h^i : P^i \to D^{i-1}$ define the homotopy between $f^\bullet \circ \alpha^\bullet$ and 0, namely:

$$-f^i \circ \alpha^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}$$

Take the mapping cone $MC(f)^\bullet$ of $f^\bullet$ and define the morphisms:

$$\beta^i = (\alpha' h^i) : P^i \to MC(f)^{i-1}$$

which may be visualized in the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[d, dotted, "\beta^{i-1}"] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[d, "\beta^i"'] & P^{i+1} \arrow[r] \arrow[d, "\beta^{i+1}"] & {} \\
\cdots \arrow[r] & C^{i-1} \oplus D^{i-2} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^i \oplus D^{i-1} \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+1} \oplus D^i \arrow[r] & {}
\end{tikzcd}$$

These give a morphism of cochain complexes $P^\bullet \to MC(f)[-1]^\bullet$. Indeed:

$$\beta^{i+1} \circ d^i_{P^\bullet} = (\alpha^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})$$
$$-d^{i-1}_{MC(f)} \circ \beta^i = (d^i_{C^\bullet} \circ \alpha^i, -f^i \circ \alpha^i - d^{i-1}_{M^\bullet} \circ h^i)$$

Then by definition of h^i and by the morphism property of $\alpha^\bullet$, these are equal. Now, by hypothesis, $f^\bullet$ is a quasi-isomorphism. Then by proposition 2.1.20 $MC(f)^\bullet$ is an exact complex. Then by corollary 2.2.6 $\beta^\bullet$ is homotopic to 0. Since $\alpha^\bullet$ is the composition:

$$P^\bullet \xrightarrow{\beta^\bullet} MC(f)[-1]^\bullet = L^\bullet \oplus M[-1]^\bullet \to L^\bullet$$

it follows that $\beta^\bullet \sim 0$ implies $\alpha^\bullet \sim 0$, as we sought to show.

Though $\alpha^\bullet$ must be homotopic to 0, it need not be the case that $\alpha^\bullet = 0$. Take for example:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathbb{Z} \arrow[r] \arrow[d, "\cdot 2"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

In this case, $f^\bullet$ is a quasi-isomorphism and $f^\bullet \circ \alpha^\bullet = 0$, and so $\alpha^\bullet \sim 0$, even though it is not equal to 0. With this, one more result is required before the desired goal:

Proposition 2.2.8: projectives, and one-sided inverses

Let $\mathbf{A}$ be an abelian category, $L^\bullet \in \mathrm{Ch}(\mathbf{A})$, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$, and $f^\bullet : C^\bullet \to P^\bullet$ a quasi-isomorphism. Then there exists a morphism of complexes $g^\bullet : P^\bullet \to C^\bullet$ such that:

$$f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$$

For injective, $Q^\bullet \in \mathrm{Ch}^+(\mathbf{I})$, $f^\bullet : Q^\bullet \to C^\bullet$ is a quasi-isomorphism, for which then there exists a morphism $g^\bullet : C^\bullet \to Q^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathrm{id}_{Q^\bullet}$. Notice that this is a one-sided inverse, hence is a slightly more general result than what is being built up to.

***Proof*:**
Since $f^\bullet : C^\bullet \to P^\bullet$ is a quasi-isomorphism, $MC(f)^\bullet = L[1]^\bullet \oplus P^\bullet$ is an exact complex. Define $\rho^\bullet = (0, \mathrm{id}_{P^\bullet})$ to be the morphism:

$$\rho^\bullet : P^\bullet \to MC(f)^\bullet$$

By corollary 2.2.5, since $MC(f)^\bullet$ is exact, $\rho^\bullet$ is homotopic to the zero morphism. Thus, there exists morphisms:

$$\widehat{h}^i : P^i \to MC(f)^{i-1} = L^i \oplus P^{i-1}$$

such that

$$\rho^i = d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet}$$

Visually, we may see this as:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[d, "\rho^{i-1}"'] \arrow[dl, "\widehat{h}^{i-1}"'] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[d, "\rho^i"'] \arrow[dl, "\widehat{h}^i"'] & P^{i+1} \arrow[r] \arrow[d, "\rho^{i+1}"] \arrow[dl, "\widehat{h}^{i+1}"'] & \cdots \\
\cdots \arrow[r] & C^i \oplus P^{i-1} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^{i+1} \oplus P^i \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+2} \oplus P^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Now, write out $\widehat{h}^i$ out in components so that $\widehat{h}^i = (g^i, h^i)$ where $g^i : P^i \to C^i$ and $h^i : P^i \to P^{i-1}$. We'll show that the collection $\{g^i\}_{i\in\mathbb{Z}}$ is a morphism of cochian complexes (i.e. they form commutative squares $g^{i+1} \circ d^i_{P^\bullet} = d^i_{C^\bullet} \circ g^i$) such that $f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$. Indeed, since

$$\begin{aligned}
(0, \mathrm{id}_{P^i}) &= \rho^i \\
&= d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet} \\
&= (d_{MC}(f)^\bullet)^{i-1} \circ (g^i, h^i) + (g^{i+1}, h^{i+1}) \circ d^i_{P^\bullet} \\
&= (-d^i_{C^\bullet} \circ g^i, f^i \circ g^i + d^{i-1}_{P^\bullet} \circ h^i) + (\beta^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})
\end{aligned}$$

It follows that:

$$\begin{aligned}
g^{i+1} \circ d^i_{P^\bullet} - d^i_{C^\bullet} \circ g^i &= 0 \\
\mathrm{id}_{P^i} - f^i \circ g^i &= d^{i-1}_{P^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}
\end{aligned}$$

which is exactly what we need, hence the morphism of complexes $g^\bullet : P^\bullet - > C^\bullet$ is the needed homotopy right-inverse of $f^\bullet$, as we sought to show.

With this, we can finally prove the desired result:

Theorem 2.2.9: Projective, Quasi-isomorphism Becomes Homotopy

Let $\mathbf{A}$ be an abelian category, and let $\alpha^\bullet : P_0^\bullet - > P_1^\bullet$ be a quasi-isomorphism between bounded-above complexes of projectives (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$). Then $\alpha^b b$ is a homotopy-equivalence.

***Proof*:**
The argument for projectives shall be given, the injective argument following by appropriate dualizing. By proposition 2.2.8, since $P_1^\bullet$ is in $C^-(\mathbf{P})$ and $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ is a quasi-isomorphism, there exists a morphism of complexes $\beta^\bullet : P_1^\bullet \to p_0^\bullet$ such that

$$\alpha^\bullet \circ \beta^\bullet \sim \mathrm{id}^\bullet_{P_1}$$

Then by proposition 2.1.25, $H^\bullet(\alpha^\bullet) \circ H^\bullet(\beta^\bullet) = \mathrm{id}$. Since $H^\bullet(\alpha^\bullet)$ is invertible, so is $H^\bullet(\beta^\bullet)$, and so $\beta^\bullet$ must be a quasi-isomorphism. Since $P_0^\bullet$ is also in $\mathrm{Ch}^-(\mathbf{P})$, we have that there exists a morphism $(\alpha^\bullet)' : P_0^\bullet \to P_1^\bullet$ such that $\beta^\bullet \circ (\alpha^\bullet)' \sim \mathrm{id}_{P_0^\bullet}$. Since:

$$(\alpha^\bullet)' \sim (\alpha^\bullet \circ \beta^\bullet) \circ (\alpha^\bullet)' = \alpha^\bullet(\beta^\bullet \circ (\alpha^\bullet)') \sim \alpha^\bullet$$

it follows that $\beta^\bullet \sim \alpha^\bullet \sim_{P_0^\bullet}$ (exercise 4.6 in Aluffi). Thus, $\alpha^\bullet$ is indeed a homotopy equivalence, as we sought to show.

Corollary 2.2.10: Projective In Homotopy Category

In $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$, (homotopy classes of) quasi-isomorphissm are isomorphisms.

***Proof*:**
follows immediately form theorem 2.2.9.

2.2.1 Resolutions

With the important equivalence between quasi-isomorphisms and homotopy-equivalence established in the case of complexes of projective (resp. projective) objects, we turn back our attention to the question of finding (co)homological invariants, focusing on categories with enough projective s(resp. Injectives). In particular, given an object A in an abelian category $\mathbf{A}$, it wouldb e desirable to associate to it a (co)chain of projective (resp. injective) objects in some universal way (up to homotopy) from which information can be extracted.

Let us first refine the definition of projective/injective resolutions of an object. Recall that there i a copy of $\mathbf{A}$ in $\mathrm{Ch}(\mathbf{A})$ given by mapping $A \mapsto \iota(A)$.

Definition 2.2.11: Projective And Injective Resolution

Let $A \in \mathrm{Obj}(\mathbf{A})$ be an object in an abelian category A. Then:

1. A *resolution* of A is a quasi-isomorphism $C^\bullet \to \iota(A)$ for some complex $C^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$.
2. A *projective resolution* of A is a quasi-isomorphism $P^\bullet \to \iota(A)$ where $P^\bullet$ is a complex in $\mathrm{Ch}^{\leq 0}(\mathbf{P})$
3. An *injective resolution* of A is a quasi-isomorphism $\iota(A) \to Q^\bullet$ where $Q^\bullet$ is a complex in $\mathrm{Ch}^{\geq 0}(\mathbf{I})$

Often, $P^\bullet$ and $Q^\bullet$ shall be referred to as the projective resolution and injective resolution respectively, omitting the quasi-isomorphism.

2.2.12 Remark There is a point of confusion that may arise. It may seem that the resolutions *themselves* are projective/injective objects in the abelian category $\mathrm{Ch}(\mathbf{A})$. However, this is not the case (exercise ref:HERE).

Recall further that any resolution $C^\bullet$ will be exact everywhere except possibly at C^0 (exercise, or double-check example 2.1.13). Hence, the non-trivial cohomological information lies in $H^0(C^\bullet)$. Let us now focus on resolutions for a particular object A: consider the category of homotopy classes of quasi-isomorphisms with *target* $\iota(A)$. Then we shall show that, if they exist, projective resolution are initial in this category.

$$\begin{tikzcd} P^\bullet \arrow[d, "\exists!"] \arrow[dr] & \\ C^\bullet \arrow[r, "\text{q-iso}"'] & \iota(A) \end{tikzcd}$$

Analogously, injective resolutions are final in homotopic category of quasi-isomorphisms with *source* $\iota(A)$. Given this universal property, this means that each projective resolution maps uniquely (in $K(\mathbf{A})$) to *every* resolution of A. The following lemma proves this:

Lemma 2.2.13: Projective Resolution As Initial Objects

Let $A \in \mathbf{A}$, $C^\bullet$ a resolution of A (note that $H^0(C^\bullet) = A$), and $P^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{P})$. Then any morphism

$$\varphi : H^0(P^\bullet) \to H^0(C^\bullet) \; [= A]$$

induces a morphism of complexes $\varphi^\bullet : P^\bullet \to C^\bullet$ that is unique up to homotopy equivalence.

***Proof*:**
The goal is to define $\varphi^i : P^i \to C^i$ for each $i \in \mathbb{Z}$. Since $\varphi^i = 0$ for $i > 0$, it is safe to replace $C^\bullet$ with its truncated version. From this, consider the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{-2} \arrow[r, "d^{-2}_{P^\bullet}"] \arrow[d, dashed, "\varphi^2"] & P^{-1} \arrow[r, "d^{-1}_{P^\bullet}"] \arrow[d, dashed, "\varphi^1"] & P^0 \arrow[r, "\pi"] \arrow[d, dashed, "\varphi^0"] & H^0(P^\bullet) \arrow[r] \arrow[d, "\varphi"] & 0 \\
\cdots \arrow[r] & C^{-2} \arrow[r, "d^{-2}_{C^\bullet}"] & C^{-1} \arrow[r, "d^{-1}_{C^\bullet}"] & \ker d^0_{C^\bullet} \arrow[r, "\mu"] & H^0(C^\bullet) \; [= A] \arrow[r] & 0
\end{tikzcd}$$

Then by the amendment to the resolution, the bottom complex is now exact.

Now, P^0 is projective, and maps to A via $\varphi \circ \pi$. Hence, by **??** there exists a lift, say φ^0

$$\begin{tikzcd}
P^0 \arrow[d, "\varphi^0"] \arrow[dr, "\varphi\circ\pi"] & \\
\ker(d^0_{C^\bullet}) \arrow[r, "\mu"] & A
\end{tikzcd}$$

Next, since $\ker(d^0_{C^\bullet}) \hookrightarrow M^0$, we have a map $\varphi^0 : P^0 : M^0$. Next, notice that

$$\mu \circ (\alpha^0 = d^{-1}_{P^\bullet} = \varphi \circ \pi \circ d^{-1}_{P^\bullet} = 0)$$

hence, $\alpha^0 \circ d^{-1}_{P^\bullet}$ factors through $\ker \mu$. Since the bottom is exact, we have $\mathrm{im} d^{-1}_{C^\bullet}$. Thus, we have the following diagram:

$$\begin{tikzcd}
 & P^{-1} \arrow[d, "\varphi^0 \circ d^{-1}_{P^\bullet}"] & \\
C^{-1} \arrow[r, "d^{-1}_{C^\bullet}"] & \mathrm{im}(d^{-1}_{C^\bullet} \arrow[r] & 0
\end{tikzcd}$$

hence, there exists a list φ^{-1}. From this, we may continue to inductively construct each φ^{-1} for $i > 1$, using exactness of the bottom. This then gives a morphism of complexes $\varphi^\bullet : P^\bullet \to M^\bullet$. Its uniqueness up to homotopy is given by lemma (ref;HERE), ctp.

Since these are initial (resp. final) objects in the category, we have a unique isomorphism beetween any two solutions to the universal problem. In this case, this may be stated as followed:

Proposition 2.2.14: Unique Projection Resolution

Let $\mathbf{A}$ be an abelian category. Then any two projective (resp. injective) resolutions of an object A in $\mathbf{A}$ are homotopy equivalent

***Proof*:**
Follows the above remark, but can also be directly deduced using the above lemma.

A final important observation is that this lift is functorial!

Proposition 2.2.15: Projective Resolution Functoriality

Let $\mathbf{A}$ be an abelian category, and let A_0, A_1 be objects of $\mathbf{A}$. Let $P_0^\bullet$ and $P_1^\bullet$ be projective resolutions for A_0 and A_1 respectively. Then every morphism $\varphi : A_0 \to A_1$ in $\mathbf{A}$ induces a morphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ that is unique up to homotopy

***Proof*:**
Notice that φ can be interpreted as a morphism $\varphi : H^0(P_0^\bullet) \to H^0(P_1^\bullet)$. The complex $P_0^\bullet$ consists of projectives, and $P1^\bullet$ is a resolution of A_1. Thus, by lemma 2.2.13, there exists a list α^b that is unique up to homotopy, as we sought to show.

Corollary 2.2.16: Full Subcategory Of Homotopic Category

Let $\mathbf{A}$ be an abelian category with enough projectives (resp. enough injectives). Then $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$).

***Proof*:**
important exercise!

2.2.2 Derived Category with Enough Projectives

Let us finally confirm that the solution to the universal problem in equation 2.2. Recall that the derived category is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd}$$

In the case where $\mathbf{A}$ has enough projectives (resp. enough injectives) and is bounded from above (resp. from bellow), then $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$) is the solution to the above universal problem. In particular, there is an equivalence of categories between $K^-(\mathbf{P})$ and $D^-(\mathbf{A})$. The formal definition of the derived category shall be given in section ref:HERE, hence this equivalence shall not be proven, however the universal property is readily provable, and once the derived category is formally defined it will be the equivalence essentially come to proper manipulation of universal properties.

To start, a general procedure for finding projective/injective resolutions for any (co)chain must be explored. The discussion so far has been for projective/injective resolution, that is quasi-isomorphisms $P^\bullet \to \iota(A)$ or $\iota(A) \to Q^\bullet$. Is it possible to replace $\iota(A)$ with any bounded complex? The answer is yes, though rather technical. The importance of this will be the existence of a functor p that maps $\mathrm{Ch}^-(\mathbf{P})$ into $K^-(\mathbf{P})$, which shall be important for solving the universal problem above, as well as finding functors between derived categories. Such functors shall also be key in solving the "extension problem" that was mentioned all the way back in chapter **??**.

Theorem 2.2.17: Resolution And Complexes

Let $\mathbf{A}$ be an abelian category with enough projectives and $C^\bullet$ a complex in $C^-(\mathbf{A})$. Then there exists a founded-above complex of projective $P^\bullet$ and a quasi-isomorphism $P^\bullet \to C^\bullet$, where $P^\bullet$ is uniquely defined up to homotopy equivalence. Furthermore, every morphism $\alpha^\bullet$ of complexes in $C^-(\mathbf{A})$ lifts to a morphism of the corresponding projective resolutions in $K^-(\mathbf{P})$, which is uniquely defined up to homotopy.

***Proof*:**
Aluffi p.632

Thus, given an abelian category $\mathbf{A}$ with enough projective, there is a functor

$$\mathscr{P} : C^-(\mathbf{A}) \to K^-(\mathbf{P})$$

associating each (bounded-above) complex $C^\bullet$ to a projective resolution $\mathscr{P}(C^\bullet)$ in a functorial manner (so that morphisms $f^\bullet$, $g^\bullet$ are lifted and $\mathscr{P}(g^\bullet \circ f^\bullet) = \mathscr{P}(g^\bullet) \circ \mathscr{P}(f^\bullet)$ since both sides are classes of homotopy lifts of $g^\bullet \circ f^\bullet$). The uniqueness of $\mathscr{P}$ is up to natural transformation, that is if another functor $\mathscr{P}'$ is chosen that assigns a different choice of projective resolution, then by theorem 2.2.17 there is a homotopy equivalence between $\mathscr{P}(C^\bullet)$ and $\mathscr{P}'(C^\bullet)$, hence an isomorphism $\varphi_{C^\bullet} : \mathscr{P}(C^\bullet) \to \mathscr{P}'(C^\bullet)$ in $K^-(\mathbf{P})$. Since the identity on $C^\bullet$ lifts to a unique map up to homotopy, the isomorphism $\varphi_{C^\bullet}$ is unique. Furthermore, given a morphism $f^\bullet : C^\bullet \to D^\bullet$, the diagram:

$$\begin{tikzcd}
\mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\sim"', "\varphi_{D^\bullet}"] \\
\mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet)
\end{tikzcd}$$

commutes (again by the uniqueness of lifts up to homotopy, can you see why)? Thus, we have the unique natural isomorphism for any two choices of functors giving a projective resolution.

Theorem 2.2.18: Universal Property Of Derived Category With Enough Projectives

Let $\mathbf{A}$ be an abelian category with enough projectives, and $\mathscr{P}$ a functor sending bounded-above cochains to projective resolutions. Then:

1. If $f^\bullet$ is a quasi-isomorphism in $C^-(\mathbf{A})$, then $\mathscr{P}(f^\bullet)$ is an isomorphism in $K^-(\mathbf{P})$
2. If $\mathscr{F} : C^-(\mathbf{A}) \to D$ is an additive functor that takes quasi-isomorphisms to isomorphisms, then there exists a unique functor $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ unique up to natural isomorphism such that the following diagram commutes up to natural isomorphism:

$$\begin{tikzcd} C^-(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d, "\mathscr{P}"'] & D \\ K^-(\mathbf{P}) \arrow[ur, "\widetilde{\mathscr{F}}"'] & \end{tikzcd}$$

   where commuting up to natural transformation means there exists a natural transformation $\nu : \widetilde{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$ such that:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}} \circ \mathscr{P}(C^\bullet) \xrightarrow{\sim} \mathscr{F}(C^\bullet)$$

   is an isomorphism for every complex $C^\bullet$ in $C^-(\mathbf{A})$.

The fact that the commutative diagram commutes up to natural isomorphism reflects that the homotopic category is categorically equivalent to the derived category. If $\mathbf{A}$ has enough injective, then we may instead define the functor $\mathscr{I} : C^+(\mathbf{A}) \to K^+(\mathbf{I})$ that shall satisfy the dual universal property.

***Proof*:**

1. Let $f^\bullet : C^\bullet \to D^\bullet$ be any morphism in $C^-(\mathbf{A})$. By theorem 2.2.17, $f^\bullet$ lifts to a morphism of resolutions which commutes up to homotopy in the following diagram:

$$\begin{tikzcd} \mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\mu_{C^\bullet}"'] & \mathscr{P}(C^\bullet) \arrow[d, "\mu_{D^\bullet}"] \\ C^\bullet \arrow[r, "f^\bullet"] & D^\bullet \end{tikzcd}$$

   where $\mu_{C^\bullet}, \mu_{D^\bullet}$ are the quasi-isomorphisms of the projective resolutions. Since the diagram commute up to homootpy, it commutes after taking cohomology, thus since $f^\bullet$ is a quasi-isomorphism, so is $\mathscr{P}(f^\bullet)$. Then by corollary 2.2.10, $\mathscr{P}(f^\bullet)$ is an isomorphism.

2. Define $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ via:

$$\widetilde{\mathscr{F}}(P^\bullet) := \mathscr{F}(P^\bullet) \qquad \widetilde{\mathscr{F}}(f^\bullet) := \mathscr{F}(f^\bullet)$$

   where $\widetilde{\mathscr{F}}(f^\bullet)$ can be any representative of $f^\bullet$. This map is well-defined, since homotopic morphisms have the same image in D (lemma 2.1.31). Naturally, $\widetilde{\mathscr{F}}$ is a functor. We need to check the commutativity up to morphism of the diagram and the uniqueness of $\widetilde{\mathscr{F}}$ up to natural isomorphism.

First, let $C^\bullet \in C^-(\mathbf{A})$. Then (by construction of $\mathscr{P}$, there exists a quasi-isomorphism

$$\mu_{C^\bullet} : \mathscr{P}(C^\bullet) \to C^\bullet$$

Since $\mathscr{F}$ maps quasi-isomorphisms to isomorphism, the induced morphism $\nu_{C^\bullet} := \mathscr{F}(\mu_{C^\bullet})$:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) = \mathscr{F}(\mathscr{P}(C^\bullet)) \to \mathscr{F}(C^\bullet)$$

is an isomorphism. What's left to check is that $\nu_{C^\bullet}$ defines a natural transformation. Let $f^\bullet : C^\bullet \to D^\bullet$ in $C^-(\mathbf{A})$. We want to show the following diagram commutes:

$$\begin{tikzcd}
\widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) \arrow[r, "\widetilde{\mathscr{F}}(\mathscr{P}(f^\bullet))"] \arrow[d, "\nu_{C^\bullet}"'] & \widetilde{\mathscr{F}}(\mathscr{P}(D^\bullet)) \arrow[d, "\nu_{D^\bullet}"] \\
\mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(f^\bullet)"] & \mathscr{F}(D^\bullet)
\end{tikzcd}$$

This diagram is obtained by applying $\mathscr{F}$ to the diagram discussed above the theorem:

$$\begin{tikzcd}
\mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\varphi_{D^\bullet}", "\sim"'] \\
\mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet)
\end{tikzcd}$$

This diagram commutes up to homotopy, hence the other diagram commutes directly (lemma 2.1.31).

For uniqueness up to isomorphism, say $\overline{\mathscr{F}} : K^-(\mathbf{P}) \to D$ is another functor satisfying the properties. Then there exists a natural isomorphism:

$$\overline{\nu} : \overline{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$$

Now, for any $P^\bullet \in K^-(\mathbf{P})$, take the composition:

$$\overline{ff}(P^\bullet) \overset{\overline{ff}(\mu_{P^\bullet})^{-1}}{\underset{\sim}{\to}} \overline{ff}(\mathscr{P}((P^\bullet)) \overset{\overline{\nu}_{P^\bullet}}{\underset{\sim}{\to}} = \widetilde{ff}(P^\bullet)$$

If φ is this composition, then it is easy to verify that $\varphi : \overline{\mathscr{F}} \to \widetilde{\mathscr{F}}$ is a natural isomorphism, completing the proof.

2.3 Derived Functors: Ext and Tor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Often, we study information about $\mathbf{A}$ by passing it through $\mathbf{B}$ (think of $- \otimes N$ or $\mathrm{Hom}_R(N, -)$). As alluded to in section **??**, such information may be the number of extensions two objects may have, the amount of torsion an object may have. In section **??**, it was claimed that this comes from "fixing" the lack of exactness of a left- or right-exact functor using homological algebra. We are finally in a position to make this precise. In this section, we shall show how to take a (not-necessarily left/right exact) additive functor between two abelian categories with $\mathbf{A}$ having enough projectives (resp. injectives) $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ and get a *derived functor*:

$$D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}) \qquad (\text{resp. } D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}))$$

This functor shall be used to develop a long-exact sequence in $\mathbf{B}$, given a short exact sequence of objects in $\mathbf{A}$.

2.3.1 Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then $\mathscr{F}$ extends to a functor:

$$\mathrm{Ch}(\mathscr{F}) : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{B})$$

namely, if $C^\bullet$ is a complex:

$$\mathscr{F}(d^{i+1}_{C^\bullet}) \circ \mathscr{F}(d^i_{C^\bullet}) = \mathscr{F}(d^{i+1}_{C^\bullet} \circ d^i_{C^\bullet}) = \mathscr{F}(0) = 0$$

hence, $\mathrm{Ch}(\mathscr{F})(C^\bullet)$ is indeed a complex. Then since homotopic morphisms are sent to homotopic morphisms (lemma 2.1.28), $\mathrm{Ch}(\mathscr{F})$ induces a morphism $K(\mathscr{F})$:

$$K(\mathscr{F}) : K(\mathbf{A}) \to K(\mathbf{B})$$

From the results we have proved, we have the following commutative diagram:

$$\begin{tikzcd} \mathbf{A} \arrow[r, "\mathscr{F}"] \arrow[d, hook, "\iota"'] & \mathbf{B} \arrow[d, hook, "\iota"] \\ K(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] & K(\mathbf{B}) \end{tikzcd}$$

Furthermore, if the complexes are bounded, then certainly $\mathrm{Ch}(\mathscr{F})$ sends bounded complexes to bounded complexes, and hence the same holds for $K(\mathscr{F})$. Now, if $\mathbf{A}$ has enough projectives so that we may consider $K^-(\mathbf{P})$, can we restrict $K(\mathbf{A})$ to $K(P(\mathbf{A}))$, and map to $K(P(\mathbf{B}))$ (where $P(\mathbf{A})$ and $P(\mathbf{B})$ are the projectives of $\mathbf{A}$ and $\mathbf{B}$ respectively)? This is not the case in general; for example we saw that $- \otimes N$ in R-**Mod** does not preserve projectives. If $\mathbf{B}$ has enough projectives, then we may choose a projective resolution functor $\mathscr{P}_B : \mathrm{Ch}^-(B) \to K^-(\mathbf{P}^{-1})$ mapping each complex to a projective resolution in a functorial manner. Since $\mathscr{P}_B$ sends homotopic morphisms to homotopic morphisms (lemma 2.2.7), $\mathscr{P}_B$ descends to a functor $K(\mathscr{P}_B)$. Combining these facts, we may define the following functor:

Definition 2.3.1: Left-Derived (right-Derived) Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Then the *left-derived functor* of $\mathscr{F}$ is the composition:

$$\begin{tikzcd} K^-(P(\mathbf{A})) \arrow[r, "\iota_A"'] \arrow[rrr, bend left, "\mathbf{L}\mathscr{F}"] & K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"'] & K^-(\mathbf{B}) \arrow[r, "K(\mathscr{P}_B)"'] & K^-(P(\mathbf{B})) \end{tikzcd}$$

where ι_A is the inclusion of $K^-(P(\mathbf{A}))$ as a full subcategory of $K^-(\mathbf{A})$. The resulting functor is denoted $\mathbf{L}\mathscr{F}$.

It may seem that the natural notation would be $D^-(\mathscr{F})$, however the $\mathbf{L}\mathscr{F}$ notation was established before the abstraction given above, and has stuck. Similarly, if $\mathbf{A}$, $\mathbf{B}$ has enough injectives, we may define the right-derived functor $\mathbf{R}\mathscr{F}$.

It is natural to want to characterize $\mathbf{L}\mathscr{F}$ and $\mathbf{R}\mathscr{F}$ in terms of some universal property. Naturally, this universal property would be defined up to natural isomorphism, for $\mathscr{P}_A$ and $\mathscr{P}_B$ are chosen up to natural isomorphism. This turns to be a bit subtle. For example, there is no natural functor which shall always make the following diagram commute:

$$\begin{tikzcd} \hat{\mathbf{A}} \arrow[r, dashed, "??"] \arrow[d, hook] & \hat{\mathbf{B}} \arrow[d, hook] \\ K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"] & K^-(P(\mathbf{B})) \end{tikzcd}$$

The reason is, it is not necessarily the case that a complex whose cohomology is concentrated at 0 would map via $\mathbf{L}\mathscr{F}$ to a complex with the same property. In fact, the reader may expect this, for the image of a functor $\mathscr{F}$ may present some non-trivial cohomological information, (think of $\mathscr{F} := \mathrm{Hom}_R(N, -)$ where N is non-projective). If the functor $\mathscr{F}$ is *exact*, then then there is essentially no "change in cohomology". This is captured in the following proposition:

Proposition 2.3.2: Derived Functors And Exactness

Let $\mathscr{F}$ be an additive *exact* functor. Then $\hat{\mathbf{A}}$ is in fact sent to $\hat{\mathbf{B}}$ via $\mathbf{L}\mathscr{F}$

***Proof*:**
Let $P^\bullet \in \hat{\mathbf{A}}$ so that $H^i(P^\bullet) = 0$ for all $i \neq 0$. Then $\mathbf{L}\mathscr{F}(P^\bullet)$ is given by choosing a projective resolution $P^\bullet_{\mathscr{F}(P^\bullet)}$ of $\mathscr{F}(P^\bullet)$. Since quasi-isomorphissm preserve cohomology and (key!) exact functors *commute* with cohomology (exercise ref:HERE):

$$H^i(\mathbf{L}\mathscr{F}(P^\bullet)) = H^i(P^\bullet_{\mathscr{F}(P^\bullet)}) \cong H^i(\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^i(P^\bullet)) = \mathscr{F}(0) \overset{i\neq 0}{=} 0$$

Thus, $\mathbf{L}\mathscr{F}(P^\bullet)$ does indeed map to an object of $\hat{\mathbf{B}}$.

Even more is true if $\mathscr{F}$ is exact: $H^i(\mathbf{L}\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^0(P^\bullet))$, saying that the restriction of $\mathbf{L}\mathscr{F}$ agrees with $\mathscr{F}$ (modulo equivalence of categories). Hence, in general we shall be interested in non-exact functors, usually functors that are only left or right exact.

Returning to finding an appropriate universal property for derived functors, another natural candidate would be given by taking a "step back" from the above diagram and consider instead if the following diagram commutes (up to natural isomorphism)

$$\begin{tikzcd} K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] \arrow[d, "\mathscr{P}_A"] & K^-(\mathbf{B}) \arrow[d, "\mathscr{P}_B"] \\ K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"'] & K^-(P(\mathbf{B})) \end{tikzcd}$$

in particular, that there may be a natural isomorphism:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

Unfortunately, this isn't in general the case

Example 2.3.3: Lacking Natural Isomorphism
(exercise 7.3 in Aluffi, should I put solution here?)

The following is the best we can hope for:

Proposition 2.3.4: Universal Property of (left-)Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then $\mathbf{L}\mathscr{F}$ satisfies the following universal property:

1. There is a natural transformation:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

2. for every functor $\mathscr{G} : K^-(P(\mathbf{A})) \to K^-(P(\mathbf{B}))$ and every natural tarnsformation $\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$, there is a unique (up to natural isomorphism) natural transformation $\mathscr{G} \rightsquigarrow \mathbf{L}\mathscr{F}$ inducign a factorization:

$$\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$$

A similar result holds for right-derived functors.

***Proof*:**
Aluffi p.644

(a word on composition)

2.3.2 Long Exact Sequences of Derived Functors

With the derived functor established and being the best candidate for holding (co)homology information, we proceed with extracting this informaiton: we may take $\mathbf{L}_i\mathscr{F} := H^{-i} \circ \mathbf{L}\mathscr{F}$. The reason for taking H^{-i} instead of i is since the indexing works out better. This produced a functor $\mathbf{L}_i\mathscr{F} : D^-(\mathbf{A}) \to B$. Notice that B doesn't need to have enough projectives for:

$$\mathbf{L}_i\mathscr{F}(P^\bullet) = H^{-i}(\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet))) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

where the last isomorphism comes from $\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet)))$ being quasi-isomorphic to $\mathrm{Ch}(\mathscr{F})(P^\bullet)$. Though this abstract approach is useful, the following more concrete version of $\mathbf{L}_i\mathscr{F}$ shall prove more useful for computations It feels like I should just skip to this, idk what the above really gives me here

Definition 2.3.5: ith Left-Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories where $\mathbf{A}$ has enough projectives. Then the *ith left-derived functor* $\mathbf{L}_i\mathscr{F} : A \to B$ is given by:

$$\mathbf{L}_i\mathscr{F} := H^{-1} \circ \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \circ \iota_A$$

The complex in $\mathrm{Ch}(\mathbf{B})$ with $\mathbf{L}_i\mathscr{F}(M)$ in the $-i$th degree with vanishing differentials is denoted $\mathbf{L}_\bullet\mathscr{F}(M)$.

To be precise, if $M \in \mathbf{A}$, then $\mathbf{L}_i(M)$ is found by

1. finding any projective resolution $P_M^\bullet$ of M
2. Applying $\mathrm{Ch}(\mathscr{F})$ to $P_M^\bullet$, $\mathrm{Ch}(\mathscr{F})(P_M^\bullet)$
3. take the $(-i)$th cohomology of this complex, $H^{-i}(\mathrm{Ch}(\mathscr{F})(P_M^\bullet))$

With all the build-up done so far, the resulting cohomology is independent of choice, for:

1. Any two projective resolutions are homotopy-equivalent
2. the image of homotopy-equivalent complexes via additive functors have isomorphic cohomology

Naturally, the *ith right-derived functor* $\mathbf{R}^i\mathscr{F}$ of an additive functor $\mathscr{F}$ and the assocaited complex $\mathbf{R}^\bullet\mathscr{F}$ is defined simplify (given $\mathbf{A}$ has enough injectives), namely

$$\mathbf{R}^i\mathscr{F} := H^i \circ \mathbf{R}\mathscr{F} \circ \mathscr{Q}_A \circ \iota_A$$

A note should be said about *contravriant functors*: since they can always be viewed as *covariant* functor $\mathbf{A}^{\mathrm{op}} \to B$, $\mathbf{A}$ would instead have to have the appropriate injectives or projectives swapped. The right-derived functor of an additive *contravariant functor* would need $\mathbf{A}$ to have enough *projectives*, since $\mathbf{A}^{\mathrm{op}}$ would have to have enough injectives. The $\mathbf{L}_i$ and $\mathbf{R}_i$ notation can be thought of as "looking" at the cohomology provided by $\mathscr{F}$. As mentioned earlier, if $\mathscr{F}$ is exact, then the derived functors are *trivial* in terms of cohomology, hence the derived functors are only interesting if $\mathscr{F}$ is not exact, at which point these functors give you interesting information.

Example 2.3.6: ith Derived Functors

It is finally time to bring pack section **??**

1. Let $\mathbf{A} = R\text{-}\mathbf{Mod}$ for a commutative ring R, pick an R-module N, and define

   $$- \otimes_R N$$

   The left-derived functor of $- \otimes_R N$ is labeled

   $$- \overset{\mathbf{L}}{\otimes}_R N : D^-(R\text{-Mod}) \to D^-(R\text{-Mod})$$

   The ith derived left-functor of $- \otimes_R N$ is denoted:

   $$\mathrm{Tor}_i^R(- \otimes_R N) = H^{-i}((-)_\bullet \otimes N)$$

where the right hand side of the inequality comes from the result elaborated in section **??**. It was mentioned that a flat resolution may be used instead of a projective resolution, however this was does not currently fit within the theory. This shall be addressed in the next section.

2. Similarly, $\mathrm{Hom}_R(-, N)$ and $\mathrm{Hom}_R(N, -)$ admits a right derived functors

$$\mathrm{Ext}^i_R(M, N) = H^i(\mathrm{Hom}_R(M, N^\bullet))$$

For $\mathrm{Hom}_R(-, N)$, don't forget that one must take the opposite category and a projective resolution.

The most important part of left-/right-derived functors is their use in fixing exactness after applying a not necessarily exact functor $\mathscr{F}$. On a high level, since the derived category there is not an abelian category (hence not necessarily containing kernels and cokernels, nor is it really feasible to investigate such categories), the notion of *triangulated categories* is useful for categories where given a "short exact sequence", we can create a long exact sequence (that is, a category in which snakes lemma can be applied). We shall not investigate the idea of triangulated categories, but the following is a concrete results needed for any rigorous exploration of triangulated categories.

Assuming the whole that that our category $\mathbf{A}$ has enough projectives, take a short exact sequence:

$$0 \to L \to M \to N \to 0$$

Since there is enough projectives, can the projective resolutions be arranged to create a short exact sequence? The answer is yes, and given the following apt name:

Lemma 2.3.7: Horseshoe Lemma

Let

$$0 \to L \to M \to N0 \to 0$$

be an exact sequence in an abelian category $\mathbf{A}$ with enough projectives, and let $P^\bullet_L, P^\bullet_N$ be projections resolutions of L, N respectively. Then there exists an exact sequence:

$$0 \to P^\bullet_L \to P^\bullet_M \to P^\bullet_N \to 0$$

where $P^\bullet_M$ is a projective resolution of M, inducing the original sequence in cohomology (i.e. applying H^0)

***Proof*:**
Visually, we want to find objects and maps to fill in this commutative diagram:

$$
\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}
$$

where the dotted lines and the P_M^i are the goal (the name of the lemma comes from this horseshoe shaped diagram). I claim that $P_M^i = P_L^i \oplus P_N^i$ (which the reader should recall or reprove is projective) with the natural inclusion and projection maps shall work. The argument is done via induction. Since P_N^0 is projective and $M \to N$ is an epimorphism, the morphism $P_N^0 \to N$ lifts to a isomorphism m $\beta : P_N^0 \to M$. On the other hand, P_L^0 maps to M via $\alpha : P_L^0 \to L \to M$. Thus, there is a morphism:

$$\pi_M := \alpha \oplus \beta : P_L^0 \oplus P_N^0 \to M$$

diving us the diagram

$$
\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^0 \arrow[r] \arrow[d] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d] & P_N^0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}
$$

The snake lemma and π_L being an epimorphism gives that the kernels of the vertical maps form an exact sequence, and hence we have an epimorphism from P_L^{-1} and P_N^{-1} to the respective kernels since the column of the original diagram are exact:

$$
\begin{tikzcd}
 & P_L^{-1} \arrow[d, "\pi_L^{-1}"] & & P_N^{-1} \arrow[d, "\pi_N^{-1}"] & \\
0 \arrow[r] & \ker \pi_L \arrow[r] & \ker \pi_M \arrow[r] & \ker \pi_N \arrow[r] & 0
\end{tikzcd}
$$

Now, repeat as before, defining $P_M^{-1} = P_L^{-1} \oplus P_N^{-1}$ and define $\pi_M^{-1} : P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M \to P_L^0 \oplus P_N^0$ giving the diagram:

$$\begin{tikzcd}
0 \arrow[r] & P_L^{-1} \arrow[r] \arrow[d, "\pi_L^{-1}"] & P_L^{-1} \oplus P_N^{-1} \arrow[r] \arrow[d, "\pi_M^{-1}"] & P_N^{-1} \arrow[r] \arrow[d, "\pi_N^{-1}"] & 0 \\
0 \arrow[r] & P_L^0 \arrow[r] \arrow[d, "\pi_L"] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d, "\pi_M"] & P_N^0 \arrow[r] \arrow[d, "\pi_N"] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

By then by the snake lemma, $P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M$ is an epimorphism, and hence we get exactness at $P_L^0 \oplus P_N^0$. Continuing inductively constructs the rest of the diagram.

It is tempting to say from this result that the mapping $\mathbf{A} \to K(\mathbf{A})$ is exact, however $K(\mathbf{A})$ is not abelian and hence this is simply undefined. The above is should instead be interpreted as follows: $P_M^\bullet$ is the mapping cone of a morphism: $\rho^\bullet : P_N[-1]^\bullet \to P_L^\bullet$, giving the distinguished triangle:

$$\begin{tikzcd}
& P_L^\bullet \arrow[dr] & \\
P_N^\bullet \arrow[ur, "+1"] & & MC(\rho)^\bullet[=P_M^\bullet] \arrow[ll]
\end{tikzcd}$$

This is as close as we can get to preserving exactness.

Corollary 2.3.8: Resolution Of Projective Resolutions

Let $M^\bullet$ be a complex bounded from above in a category $\mathbf{A}$ with enough projectives. Then there exists a complex $P_{M^\bullet}^b b$:

$$\cdots \to P_{M^{-3}}^\bullet \to P_{M^{-2}}^\bullet \to P_{M^{-1}}^\bullet \to\to P_{M^0}^\bullet \to 0$$

where each $P_{M^i}^\bullet$ is a projection resolution of M^i and inducing $M^\bullet$ in cohomology. If $M^\bullet$ is exact, then $P_{M^\bullet}^\bullet$ can be chosen to be exact.

***Proof*:**
Since $M^\bullet$ is a complex, it can be broken down into two short exact sequences:

$$0 \to K^i \to M^i \to I^{i+1} \to 0 \qquad 0 \to I^i \to K^i \to H^i \to 0$$

where K^i is the kernel of $M^i \to M^{i+1}$, I^{i+1} is its image, and H^i the cohomology at M^i. Chose arbitrary projective resolutions for each I^i and H^i for all i. Then by the horseshoe lemma, we get a projective resolution for K^i that fits within the short exact sequence:

$$0 \to P_{I^i}^\bullet \to P_{K^i}^\bullet \to P_{H^i}^\bullet \to 0$$

Applying the horseshoe lemma gain, we get a short exact sequence:

$$0 \to P_{K^i}^\bullet \to P_{M^i}^\bullet \to P_{I^{i+1}}^\bullet \to 0$$

Using these, the $P^\bullet_{M^i}$ can be used to construct a cochain complex, the differentials obtains by composition. If $M^\bullet$ is exact, then $I^i = K^i$, and so the construction gives $P^\bullet_{I^i} = P^\bullet_{K^i}$, so that $P^\bullet_{M^\bullet}$ is also exact.

Lemma 2.3.9: Projective Resolution and Functors

Let $\mathbf{A}$ be an abelian category and let

$$0 \to L^\bullet \to M^\bullet \to P^\bullet \to 0$$

be an exact sequence of complexes in $\mathbf{A}$, where P^i is projective for all i. Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then the sequence:

$$0 \to \mathscr{F}(L^b) \to \mathscr{F}(M^\bullet) \to \mathscr{F}(P^\bullet) \to 0$$

is exact

Note that there is nothing about $\mathscr{F}$ being an exact functor! This can be thought of as the generalization of the properties of projective modules which preserve exactness!!

***Proof*:**
Since P^i is projective, the sequence

$$0 \to L^i \to M^i \to P^i \to 0$$

splits. Thus

$$0 \to \mathscr{F}(L^i) \to \mathscr{F}(M^i) \to \mathscr{F}(P^i) \to 0$$

is split exact for all i (as the reader should check). Since exactness of a sequence of complexes is given by exactness at each degree, this gives the desired result.

Theorem 2.3.10: Long Exact Sequence From Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories, and assume $\mathbf{A}$ has enough projective. Then every short exact sequence in $\mathbf{A}$

$$0 \to L \to M \to N \to 0$$

[covariantly] functorially induces a long exact sequence in $\mathbf{B}$:

$$\begin{tikzcd}
0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] & \\
& L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
& L_0\mathscr{F}(L) \arrow[r] & L_0\mathscr{F}(M) \arrow[r] & L_0\mathscr{F}(N) \arrow[r] & 0
\end{tikzcd}$$

***Proof*:**
The functorial assignment will be left as an exercise. Given the short exact sequence, we get short exact sequence of projective resolutions, which by lemma 2.3.9 creates a short exact sequence:

$$0 \to \mathscr{F}(P_L^\bullet) \to \mathscr{F}(P_M^\bullet) \to \mathscr{F}(P_N^\bullet) \to 0$$

Since the cohomology of these complexes is precisely the left-derifevd functors of $\mathscr{F}$, the long exact cohomology sequence is given by Snakes lemma, completing the proof.

The change from short exact sequence to long exact sequence can be unified under the notion of triangles, namely this theorem can be seen as the triangle:

$$\begin{tikzcd}
 & \iota(L) \arrow[dr] & \\
\iota(N) \arrow[ur, "+1"] & & \iota(M) \arrow[ll]
\end{tikzcd}$$

inducing the triangle

$$\begin{tikzcd}
 & \mathbf{L}_\bullet(L) \arrow[dr] & \\
\mathbf{L}_\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{L}_\bullet(M) \arrow[ll]
\end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\leq 0}(\mathbf{B})$. Naturally, the dual situation happens for right derived functors for categories that have enough injectives, iin which case the above triangle induces a triangle:

$$\begin{tikzcd}
 & \mathbf{R}^\bullet(R) \arrow[dr] & \\
\mathbf{R}^\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{R}^\bullet(M) \arrow[ll]
\end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\geq 0}(\mathbf{B})$.

These consideration were for general additive functors. In most cases we saw, there is usually a left-exact or right-exact functor (indeed, the nature of the left-derived and right-derived series almost force us to choose “a side”). If a functor is given to be either left- or right-exact, then it can be directly recovered from their derived functors, as we shall now show.

Proposition 2.3.11: Right-Exactness And Derived Functors

L $\mathbf{A}$ be an abelian category with enough projective and $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact additive functor. Then

1. $\mathbf{L}_i\mathscr{F} = 0$ for $i < 0$
2. $L_0\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$

The dual result holds for the category with enough injectives. Do note that the *right*-exact functor is associated to the *left* derived functors, and vice-versa for left-exact functors which would have the associated *right* derived functors. This is because the derived functors are “fixing” the exactness on the other side.

***Proof*:**
Since projective resolution $P^\bullet$ of an object M in $\mathbf{A}$ is in $C^{\leq 0}(\mathbf{A})$, $C(\mathscr{F})(P^b b)$ is 0 in positive degree, and hence so is its cohomology. Since $H_i = H^{-i}$, the first claim is immediate.

For the second, an exact sequence:

$$P^{-1} \to P^0 \to M \to 0$$

will map to a exact sequence by $\mathscr{F}$ (due to right-exactness)

$$\mathscr{F}(P^{-1}) - > \mathscr{F}(P^0) - > \mathscr{F}(M) - > 0$$

This induces an isomorphism: $\nu_L : \mathbf{L}_0\mathscr{F}(M) = H^0(C(\mathscr{F})(P^\bullet)) \xrightarrow{\sim} \mathscr{F}(M)$. If $\varphi : M_0 - > M_1$ is a morphism, then there exists a morphism of projective resolutions $\alpha^\bullet : P_0^\bullet - > P_1^b b$:

$$\begin{tikzcd}
\cdots \arrow[r] & P_0^{-1} \arrow[r] \arrow[d, "\alpha^{-1}"] & P_1^0 \arrow[r] \arrow[d, "\alpha^0"] & M_0 \arrow[r] \arrow[d, "\varphi"] & 0 \\
\cdots \arrow[r] & P_1^{-1} \arrow[r] & P_1^0 \arrow[r] & M_1 \arrow[r] & 0
\end{tikzcd}$$

induced by φ where $H^0(\alpha^b b)$ agrees with φ. Applying $\mathscr{F}$ and truncating to $\alpha^\bullet$, we get the commutative diagram:

$$\begin{tikzcd}
\mathscr{F}(P_0^{-1}) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^{-1})"] & \mathscr{F}(P_1^0) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^0)"] & \mathscr{F}(M_0) \arrow[r] \arrow[d, "\mathscr{F}(\varphi)"] & 0 \\
\mathscr{F}(P_1^{-1}) \arrow[r] & \mathscr{F}(P_1^0) \arrow[r] & \mathscr{F}(M_1) \arrow[r] & 0
\end{tikzcd}$$

with exact rows since $\mathscr{F}$ is right-exact. But then the following diagram commutes:

$$\begin{tikzcd}
\mathbf{L}_0\mathscr{F}(M_0) \arrow[r, "\mathbf{L}_0\mathscr{F}(\varphi)"] \arrow[d, "\nu_{M_0}"] & \mathbf{L}_0\mathscr{F}(M_1) \arrow[d, "\nu_{M_1}"] \\
\mathscr{F}(M_0) \arrow[r, "\mathscr{F}(\varphi)"'] & \mathscr{F}(M_1)
\end{tikzcd}$$

completing the proof.

This finally gives the desired recovery of exactness promised way back. Given a short exact sequence

$$0 \to L \to M \to N \to 0$$

then applying a right-exact functor

$$\mathscr{F}(L) \to \mathscr{F}(M)\mathscr{F}(N) \to 0$$

will give an exact sequence, which can be stretched out to form a long exact sequence using the derived functors:

$$
\begin{tikzcd}
0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] & \\
 & L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
 & \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[r] & 0
\end{tikzcd}
$$

In this way, $\mathbf{L}_1\mathscr{F}$ measures the extent to which $\mathscr{F}$ fails to be left-exact, and each progressive $\mathbf{L}_i\mathscr{F}$ gives further measurement, further information, about this failure. When in a geometrical setting, this failure is often thought of as some form of obstruction (famously, in algebraic topology, this obstruction is interpreted as "'holes", in K-theory).

Similarly, $\mathbf{R}^-\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$ if $\mathscr{F}$ is left-exact and the right-derived functors fit into the diagram:

$$
\begin{tikzcd}
0 \arrow[r] & \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
 & R^1\mathscr{F}(R) \arrow[r] & R^1\mathscr{F}(M) \arrow[r] & R^1\mathscr{F}(N) \arrow[dll, "\delta_0"] & \\
 & R^2\mathscr{F}(R) \arrow[r] & R^2\mathscr{F}(M) \arrow[r] & R^2\mathscr{F}(N) \arrow[r] & \cdots
\end{tikzcd}
$$

Thus, we have successfully captured the cohomological information that a functor may provide!

Acyclic Functors

There still remain some questions to be answered, one that was mentioned was that instead of using a projective resolution for Tor, a flat resolution may be used instead and give the same answer.

Definition 2.3.12: Acyclic Functor

Let $\mathscr{F}$ be an additive functor. An object M of $\mathbf{A}$ is *$\mathscr{F}$-acyclic* (with respect to left-derived functors) if $\mathbf{L}_i\mathscr{F}(M) = 0$ for all $i \neq 0$.

Similarly, an object M is $\mathscr{F}$-acyclic with respect to right-derived functors if $\mathbf{R}^i\mathscr{F}(M) = 0$ for $i \neq 0$. Naturally, projectives are acyclic for every right-exact functor, and injective acyclic for every left-exact functor, however such a functor may admit other $\mathscr{F}$-acyclic objects.

Example 2.3.13: Functor-acyclic Objects

Let R be a commutative ring. Recall that an R-module M is *flat* if $- \otimes_R M$ is exact (equivalently, by symmetry of $\otimes$ that $M \otimes_R -$ is exact). Then flat modules are *acyclic* with respect to tensor products, namely if N is flat:

$$\mathrm{Tor}_R^i(M, N) = 0 \qquad \forall i \neq 0,\ \forall M$$

and symmetrically if M is flat. Thus, flat modules are $(- \otimes_R N)$-acyclic for all modules N.

What shall be built up to now is that $\mathscr{F}$-acyclic resolutions can be used to compute $\mathbf{L}_i\mathscr{F}$.

Theorem 2.3.14: Derived Functors And Acylcic Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact functor of abelian groups where $\mathbf{A}$ has enough projectives. Let $A^\bullet$ be a resolution of M where each A^i is $\mathscr{F}$-acyclic. Then for all i:

$$\mathbf{L}_i\mathscr{F}(M) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

Note that if each A^i is projective, the statement follows by definition.

***Proof*:**
The 0th and -1th position will be computed directly, and each successive position comes from induction. Take the exact sequence

$$A^{-1} \to A^0 \to M \to 0$$

which creates another exact sequence in $\mathbf{B}$ after applying the right-exact functor $\mathscr{F}$:

$$\mathscr{F}(A^{-1}) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Then:

$$H^0(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong \mathscr{F}(M) \cong \mathbf{L}_0\mathscr{F}(M)$$

Now, if K is the kernel of $A^0 \to M$, then we have a short exact sequence:

$$0 \to K \to A^0 \to M \to 0$$

Thus applying the left-derived functors we form the long exact sequence:

$$\mathbf{L}_1\mathscr{F}(A^0) \to \mathbf{L}_1\mathscr{F}(M) \to \mathscr{F}(K) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Since A^0 is acyclic, $L_1\mathscr{F}(A^0) = 0$, hence by exactness

$$\mathbf{L}_1\mathscr{F}(M) \cong \ker(\mathscr{F}(K) \to \mathscr{F}(A^0)) \cong H^{-1}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

where the last isomorphism is some manipulation of definition. For the rest, we do induction. Assume now that for all object of $\mathbf{A}$ and all indices $< i$. Truncating and shifting $\mathbf{A}^\bullet$ so that A^{-1} is in the 0 degree:

$$\cdots \to A^{-3} \to A^{-2} \to A^{-1} \to 0 \to \cdots$$

we get an $\mathscr{F}$-acyclic resolution of K, and so by the induction hypothesis $\mathbf{L}_{i-1}\mathscr{F}(K)$ can be computed by applying $\mathrm{Ch}(\mathscr{F})$ to the complex and taking cohomology, namely:

$$\mathbf{L}_{i-1}\mathscr{F}(K) \cong H^{-i}(A^\bullet)$$

On the other hand, the other terms in the long exact sequence obtained using the left derived functors gives:

$$\cdots \to \mathbf{L}_i\mathscr{F}(A^0) \to L_i\mathscr{F}(M) \to \mathbf{L}_{i-1}\mathscr{F}(K) \to \mathbf{L}_{i-1}\mathscr{F}(A^0) \to \cdots$$

since A^0 is $\mathscr{F}$-acyclic and $i > 1$, we get:

$$\mathbf{L}_i\mathscr{F}(M) \cong \mathbf{L}_{i-1}\mathscr{F}(K)$$

giving the desired result.

Thus, for the Tor functor, we may use a flat resolution to compute it the left-derived functors. Since $\mathscr{F}$-acyclic objects suffice in the computation, one can imagine that there is not need for $\mathbf{A}$ to have enough projectives to compute it. This is indeed the case, given that $\mathbf{A}$ has "enough $\mathscr{F}$-acyclic's". (this is an exercise in Aluffi, 8.14, with bifunctors).

2.4 Double Complexes

This final section shall be the cornerstone in being able to compute many complexes. It was mentioned in remark ref:HERE that $\mathrm{Ext}^i_R(M, N)$ may be computed by using either a projective resolution of M or an injective resolution on N, both give the same answer. More generally, we may compute the resolution of either M or N and get the same value. This section will now prove this result. This will make many computations much simpler, for usually one object in is readily understandable while the other is being analyzed.

To motivate this, switching from projective to $\mathscr{F}$-acyclic resolutions can be viewed in the following important alternative way. Consider the following diagram where the top row and right column are cochain:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^{-0}) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^0) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
& \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u]
\end{tikzcd}$$

in some way, there was a reason to go from the top chain to the right chain, with the intermediate values "interpolating" the result. The natural candidate for this would be a complex of complex. Needing to bound it in one direction, say we are looking at $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Such an object would be of the form:

$$\cdots \to M^{-3,\bullet} \xrightarrow{d_h^{-3,\bullet}} M^{-2,\bullet} \xrightarrow{d_h^{-2,\bullet}} M^{-1,\bullet} \xrightarrow{d_h^{-1,\bullet}} M^{0,\bullet} \to 0 \to \cdots$$

The h in the subscript is to represent the fact we are going *horizontally*, and the respective morphism of each $d_h^{-i,\bullet}$ will be going vertically and are denoted $d_v^{i,j}$, or simply d_v to simply show the function's direction:

$$\begin{tikzcd}
\cdots \arrow[r] & M^{-3,0} \arrow[r] & M^{-2,0} \arrow[r] & M^{-1,0} \arrow[r] & M^{0,0} \\
\cdots \arrow[r] & M^{-3,-1} \arrow[r] \arrow[u] & M^{-2,-1} \arrow[r] \arrow[u] & M^{-1,-1} \arrow[r] \arrow[u] & M^{0,-1} \arrow[u] \\
\cdots \arrow[r] & M^{-3,-2} \arrow[r] \arrow[u] & M^{-2,-2} \arrow[r] \arrow[u] & M^{-1,-2} \arrow[r] \arrow[u] & M^{0,-2} \arrow[u] \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Given a complex of complex, it would be nice to reduce this to being a complex, a more familiar setting. One strategy the reader may come up with is to take:

$$\bigoplus_{i+j=n} M^{i,j}$$

with the natural morphisms between them. However these morphisms *almost* complexes: instead of commutativity for each square, they are *anti-commutative*. To fix this, create a new type of object that will satisfy this condition:

Definition 2.4.1: Double Complex

Let $A = \{A_{p,q}\}$ be a family of objects from $\mathbf{A}$. Then A along with maps:

$$d^h : C_{p,q} \to C_{p-1,q} \qquad \delta^v : C_{p,q} \to C_{p,q-1}$$

with the condition of $d^h \circ d^h = 0$, $\delta^v \circ \delta^v = 0$ and $d^h \circ \delta^v + \delta^v \circ d^h = 0$ is called a *double complex* or *bicomplex*

We may visualize a double complex like so:

$$\begin{tikzcd}
& \cdots & \cdots & \cdots & \\
\cdots \arrow[r] & C^{p+1,q-1} \arrow[r] \arrow[u] & C^{p,q-1} \arrow[r] \arrow[u] & C^{p,q-1} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^{p+1,q} \arrow[r] \arrow[u] & C^{p,q} \arrow[r] \arrow[u] & C^{p,q} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^{p+1,q+1} \arrow[r] \arrow[u] & C^{p,q+1} \arrow[r] \arrow[u] & C^{p,q+1} \arrow[r] \arrow[u] & \cdots \\
& \cdots \arrow[u] & \cdots \arrow[u] & \cdots \arrow[u] &
\end{tikzcd} \tag{2.6}$$

Note this is not a commutative diagram, but an *anti-commutative* diagra. We can geet a complex of complex by introducing a sign:

$$d^{p,q} = (-1)^p \delta^v_{p,q} : C^{p,q} \to C^{p,q+1}$$

As mentioned, we can form a total complex from this:

Proposition 2.4.2: Total Complex

Let $A = \{A^{p,q}\}$ be a double complex. Then define the total complex $\mathrm{Tot}^{\Pi}(A)$ and $\mathrm{Tot}^{\oplus}(A)$ by:

$$\mathrm{Tot}^{\Pi}(A)_n = \prod_{p+q=n} A_n \qquad \mathrm{Tot}^{\oplus}(A)_n = \oplus_{p+q=n} A_n$$

with $d^{tot} = d^v + d^h$ so that $d \circ d = 0$

***Proof*:**
Computing:

$$d_{tot} \circ d_{tot} = (d_h + \delta_v) \circ (d_h + \delta_v) = d_h \circ d_h + d_h \circ \delta_v + \delta_v \circ d_h + \delta_v \circ \delta_v = 0$$

Note that the total complex does not exist in all categories, namely it may not exist in categories which are not closed under infinite products and coproducts (ex. in the category of finite abelian groups). A category which is closed for infinite products is called *complete* and *cocomplete* if it is closed for infinite sums. We can visualize a total complex by taking the diagram in equation (2.6) and moving it like so:

$$\begin{tikzcd}
 & & & M^{0,0} \\
 & & M^{-1,0} \arrow[ur, "d_h"] & M^{0,-1} \arrow[u, "\delta_v"'] \\
 & M^{-2,0} \arrow[ur, "d_h"] & M^{-1,-1} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
M^{-3,0} \arrow[ur, "d_h"] & M^{-2,-1} \arrow[u] \arrow[ur, "d_h"] & M^{-1,-2} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
\vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"']
\end{tikzcd}$$

It is then easy to visualize how we get a total complex form this.

Example 2.4.3: Total Complexs

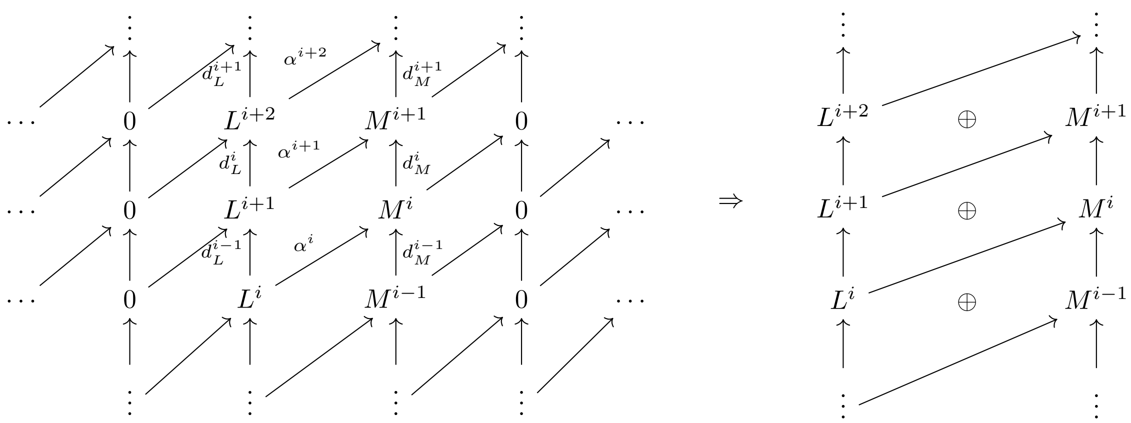
1. Consider a double complex:

$$\cdots \to 0 \to L^{\bullet} \xrightarrow{\alpha^{\bullet}} M^{\bullet} \to 0 \to \cdots$$

which we may represent as:

$$
\begin{tikzcd}
 & \vdots & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i+1} \arrow[r, "\alpha^{i+1}"] \arrow[u, "d_L^{i+1}"] & M^{i+1} \arrow[r] \arrow[u, "d_M^{i+1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i} \arrow[r, "\alpha^{i}"] \arrow[u, "d_L^{i}"] & M^{i} \arrow[r] \arrow[u, "d_M^{i}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i-1} \arrow[r, "\alpha^{i-1}"] \arrow[u, "d_L^{i-1}"] & M^{i-1} \arrow[r] \arrow[u, "d_M^{i-1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}
$$

we shall get the following:



which is exactly the definition of a mapping cone!

2. The operators $\mathrm{Hom}_A(-,-)$ and $-\otimes-$ both give rise to double complexes. Focusing on the $\mathrm{Hom}_A(-,-)$ example, consider a chain in $L^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{A})$ and $M^\bullet \in \mathrm{Ch}^{\geq 0}(\mathbf{A})$. These can be used to form a double complex, or isomorphically a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$:

$$\cdots \to \mathrm{Hom}_A(L^\bullet, M^0) \to \mathrm{Hom}_A(L^\bullet, M^1) \to \mathrm{Hom}_A(L^\bullet, M^2) \to \cdots$$

which creates the following upper-quadrant grid:

$$\begin{tikzcd}
\vdots & \vdots & \vdots & \\
\mathrm{Hom}_A(L^{-2}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-2}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-2}, M^2) \arrow[r] \arrow[u] & \cdots \\
\mathrm{Hom}_A(L^{-1}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-1}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-1}, M^2) \arrow[r] \arrow[u] & \cdots \\
\mathrm{Hom}_A(L^{0}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{0}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{0}, M^2) \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

Flipping the above diagram across the main diagonal, we would get the double complex:

$$\cdots \to \mathrm{Hom}_A(L^0, M^\bullet) \to \mathrm{Hom}_A(L^{-1}, M^\bullet) \to \mathrm{Hom}_A(L^{-2}, M^\bullet) \to \cdots$$

These two double complexes are isomorphic (as the reader should check). Due to this, by slight abuse of notation the total complex of both can be denoted $TC(\mathrm{Hom}_A(L, M))^\bullet$, where the degree i term is:

$$\mathrm{Hom}_A(L^{-i}, M^0) \oplus \mathrm{Hom}_A(L^{-i+1}, M^1) \oplus \cdots \oplus 0\mathrm{Hom}_A(L^{-1}, M^{i-1}) \oplus 0\mathrm{Hom}_A(L^0, M^i)$$

The reader should verify that:

(a) $\ker d^i_{tot}$ is equal to morphisms of cochains $L^\bullet \to M[i]^\bullet$
(b) The image $\mathrm{im} d^{i-1}_{tit}$ consists of morphisms that are homotopy equivalent to 0
(c) Thus, the cohomology of $TC(\mathrm{Hom}_A(M, N))^\bullet$ "paramaterizes" the morphisms in the homotopic category

To ground this, consider $TC(\mathrm{Hom}_A(M, N))^0 = \mathrm{Hom}_A(L^0, M^0)$. Then $d_{tot}(\alpha) = 0$ means that both $\alpha \circ d^{-1}_{L^\bullet} \in \mathrm{Hom}_A(L^{-1}, M^0)$ and $d^0_{M^\bullet} \in \mathrm{Hom}_A(L^0, M^1)$ both vanish. The reader should draw out this diagram, and notice that this will make α have the condition to be a morphism of cochain complexes $L^\bullet \to M^\bullet$!

$$H^0(TC(\mathrm{Hom}_A(M, N))^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(L^\bullet, M^\bullet) \overset{!}{=} \mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M^\bullet)$$

where the $\overset{!}{=}$ equality comes from the fact that there is no nontrivial homotopies in this case! For $H^i(TC(\mathrm{Hom}_A(M, N))^\bullet)$, with some care for signs, it can be identified with $\mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M[i]^\bullet)$

The cohomology of double complexes requires some book-keeping through the use of *spectral sequences*, which shall be explored in greater depth in chapter 4. For the current goal, it is enough to consider when a double-complex is exact:

Proposition 2.4.4: Exactness Of Complex Of Complex

Let $\mathbf{A}$ be an abelian category and let:

$$\cdots \to M^{-3,\bullet} \to M^{-2,\bullet} \to M^{-1,\bullet} \to M^{0,\bullet} - \cdots$$

be a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Then the corresponding total complex $^{\bullet}$ is exact

- The above chain is exact
- each complex $M^{i,\bullet}$ is exact

The same naturally holds for $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Note further that only one of the complexes need to be bounded, the boundedness being required for the proceeding inductive argument.

***Proof*:**

1. Flip the double complex corresponding to the above cochain, which the reader should check gives an exact complex $T^{\bullet}$
2. Aluffi p.671 (it's book keeping on induction)

Example 2.4.5: Exactness Of Double Complex

Let $P^{\bullet} \in C^{\leq 0}(\mathbf{A})$ be a complex and $L^{\bullet} \in C^{\geq 0}(\mathbf{A})$ an exact complex. Since each P^i is projective, $\mathrm{Hom}_{\mathbf{A}}(P^i, L^{\bullet})$ for each i. Hence, the complex:

$$0 \to \mathrm{Hom}_{\mathbf{A}}(P^0, L^{\bullet}) \to \mathrm{Hom}_{\mathbf{A}}(P^{-1}, L^{\bullet}) \to \mathrm{Hom}_{\mathbf{A}}(P^{-2}, L^{\bullet}) \to \cdots$$

is exact. Using example 2.4.3, we see that:

$$\mathrm{Hom}_{K(\mathbf{A})}(P^{\bullet}, L[i]^{\bullet}) = 0 \qquad \forall i \in \mathbb{Z}$$

This implies that every cochain morphism from a bounded-above projective complex to a bounded-bellow exact complex is homotopic to 0. This just proved corollary 2.2.5 (with an extra boundedness condition)! In some textbooks, due to the speed at which some results can be proved using double-complexes, they are introduced earlier. However, the more "down-to-earth" introduction has is favored in this chapter since it demonstrated key homological concepts.

Next, we look at resolutions of complexes.. As we saw in definition 2.2.11, a (projective) resolution is a quasi-isomorphism $M^{\bullet} \to \iota(A)$ for some complex $M^{\bullet} \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$, and more generally a projective resolution for a general complex is a quasi-isomorphism $P^{\bullet} \to C^{\bullet}$ for some projective complex $P^{\bullet}$. Generalizing the definition of resolution of the first kind, $\iota(A)$, and say that resolution of a complex $N^{\bullet} \in \mathrm{Ch}^{\bullet}(A) = A'$ is a complex of complex $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(A')$ and a quasi-isomorphism

$$\begin{tikzcd}
\cdots \arrow[r] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & N^{\bullet} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd} \tag{2.7}$$

where the diagram commutes. We shall show that this definition of a resolution induces a resolution as given in theorem 2.2.17. To see this, notice that in equation (2.7), $\alpha^\bullet \circ d_h = 0$, hence we can collapse the diagram like so:

$$\cdots \to M^{-2,\bullet} \xrightarrow{-d_h} M^{-1,\bullet} \xrightarrow{d_h} M^{0,\bullet} \xrightarrow{\alpha^\bullet} N^\bullet \to 0 \cdots$$

Call this complex $M_N^{\bullet;\bullet}$. Two important diagrams can be derived here. The first is by taking the total complex of $M^{\bullet,\bullet}$ and taking a morphism from this to $N^\bullet$:

$$\begin{tikzcd}
 & & & & 0 \arrow[r] & 0 \\
 & & & & M^{0,0} \arrow[u] \arrow[r] & N^0 \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[r] & N^{-1} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[r] & N^{-2} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] & {} \arrow[u]
\end{tikzcd}$$

This gives a morphism:

$$TC(\alpha^\bullet) : TC(M^{\bullet,\bullet}) \to N^\bullet$$

where the morphism is defined since $TC(-)$ is a functorial. Another diagram we can create is by considering the total complex of $M_N^{\bullet;\bullet}$:

$$\begin{tikzcd}
 & & & & & & N^0 \\
 & & & & M^{0,0} \arrow[urr] & \oplus & N^{-1} \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[urr] & \oplus & N^{-2} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[urr] & \oplus & N^{-3} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u]
\end{tikzcd}$$

These two are related in the following way:

Proposition 2.4.6: Total Complex Of Resolution

Let $M^{\bullet,\bullet}$ be a resolution of $N^\bullet$ and let $M_N^{\bullet;\bullet}$ be defined as above Then:

$$TC(M_N)^\bullet = MC(TC(\alpha))^\bullet$$

Proof:
exercise

Using this, we can see how the resolution in $\mathrm{Ch}^{\leq 0}(A')$ leads to a resolution as defined in theorem 2.2.17:

Theorem 2.4.7: Total Complexes And Resolution Of Complexes

Let $\mathbf{A}$ be an abelian category, $\mathbf{A}' = \mathrm{Ch}^{\leq 0}(\mathbf{A})$, and $N^\bullet \in A'$. Let $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(\mathbf{A}')$ with total complex $T^\bullet$.

1. Assume that $M^{\bullet,\bullet}$ is a resolution of $N^\bullet$ in $\mathrm{Ch}^{\leq 0}(\mathbf{A}')$. Then $T^\bullet$ is quasi-isomorphic to $N^\bullet$ in $\mathbf{A}'$.

2. Assuming the cohomology of $M^{i,\bullet}$ is concentrated at 0. Then $T^\bullet$ is quasi-isomorphic to the complex of cohomology objects induced by the complex $M^{\bullet,\bullet}$:

$$\cdots \to H^0(M^{-3,\bullet}) \to H^0(M^{-2,\bullet}) \to H^0(M^{-1,\bullet}) \to H^0(M^{0,\bullet}) \to 0 \cdots$$

Notice that this theorem is a direct extensions of proposition 2.4.4. This is also a special case of results we shall see in chapter 4

Proof:
Notice that (2) follows from (1) by flipping the corresponding double-complex across the diagonal, hence we prove the first. For the first, if $M_N^{\bullet,\bullet}$ is quasi-isomorphic, then $TC(M_N)^\bullet$ is exact by proposition 2.4.4, which by proposition 2.4.6 is the mapping cone of the morphism $T^\bullet \to N^\bullet$, which by proposition 2.1.20 shows that $\alpha^\bullet : T^\bullet \to N^\bullet$ is a quasi-isomorphism, as we sought to show.

Using this theorem, a simple alternative to resolving complexes (theorem 2.2.17). By the horseshoe lemma, every bounded-above complex $N^\bullet$ in an abelian category with enough projectives may be viewed as a complex induced by H^0 from a complex:

$$\cdots \to P^{-3,\bullet} \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to 0$$

where $P^{i,\bullet}$ are projective resolutions of N^i. Then by theorem 2.4.7(2) the total complex of $P^{\bullet,\bullet}$ is a projective resolution of $N^\bullet$. This then can give theorem 2.2.17. It is still worth presenting theorem 2.2.17 to firmly grasp the fundamentals of homological algebra.

2.4.1 Applications to Computations of Ext and Tor

Let us now show that we may compute $\mathrm{Ext}(M, N)$ or $\mathrm{Tor}(M, N)$ using a resolution of either term. Going back to the $\mathscr{F}$-acyclic resolution of an object M:

$$\cdots \to A^{-2} \to A^{-1} \to A^0 \to M \to 0 \to \cdots$$

We may create a complex of projective resolution by corollary 2.3.8:

$$\cdots \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to P_M^\bullet \to 0 \to \cdots$$

where $P^{i,\bullet}$ is a projective resolution of A^{-i}. Since the resolution by projective resolution's is exact, the rows of the double-complex corresponding to $P^{\bullet,\bullet}$ are projective resolutions of P^i_M. Now, applying the functor $\mathscr{F}$, we get the diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^{0}) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0}) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Since each row and column of this diagram (excluding the top row and right column) is exact, by theorem 2.4.7 the total complex of:

$$\cdots \mathscr{F}(P^{-2,\bullet}) \to \mathscr{F}(P^{-1,\bullet}) \to \mathscr{F}(P^{0,\bullet}) \to 0 \to \cdots$$

is quasi-isomorphic to both $\mathrm{Ch}(\mathscr{F})(A^\bullet)$ and $\mathrm{Ch}(\mathscr{F})(P^\bullet)$, that is:

$$TC(\mathscr{F}(P))^\bullet \sim \mathrm{Ch}(\mathscr{F})(A^\bullet) \sim \mathrm{Ch}(\mathscr{F})(P^\bullet)$$

Thus, by definition of quasi-isomorphism

$$\mathbf{L}_i\mathscr{F}(M) \cong H^i(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong H^i(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

Applying these results to Ext and Tor will finally give us the following results:

Theorem 2.4.8: Computation Of Tor and Ext

Let M, N be R-modules for a commutative ring R, and let $P_M^\bullet$ and $P_N^\bullet$ be projections resolutions of M and N respectively, and $Q_N^\bullet$ an injective resolution of N. Then:

$$\mathrm{Tor}_i^R(M,N) \cong H^i(P_M^\bullet \otimes_R N) \cong H^i(M \otimes_R P_N^\bullet)$$

and

$$\mathrm{Ext}_i^R(M,N) \cong H^i(P_M^\bullet \otimes_R N) \cong H^i(M \otimes_R Q_N^\bullet)$$

***Proof*:**
We shall show the result for Tor, the result following very similarly for Ext. Take the complex:

$$\cdots \to P_M^{-2} \otimes_R P_N^\bullet \to P_M^{-1} \otimes_R P_N^\bullet \to P_M^{0} \otimes_R P_N^\bullet \to 0 \cdots$$

Since each P_N^j is projective, the cohomology of the complex is concentrated at 0 and equals $M \otimes_R P_N^\bullet$. By theorem 2.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad M \otimes_R P_N^\bullet$$

Similarly, since each P_M^i is projective, the complex $P_M^i \otimes_R P_N^\bullet$ is a resolution of $P_M^i \otimes N$, i.e. the cohomology of each $P_M^i \otimes_R P_N^\bullet$ is concentrated at degree 0 and equals $P_M^i \otimes_R N$. By theorem 2.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad P_M^\bullet \otimes_R N$$

but then, the cohomologies are identical, as we sought to show.

2.5 *Further Topic in Homological Algebra

look at commented line here:

In this bonus section, we shall introduce formally give the definition of a derived category and introduce the notion of triangulated categories.

2.5.1 Derived Categories, Formally

As defined in this chapter, derived categories required enough projectives/injectives. However, it was mentioned that derived categories may be defined without this restriction, allowing for a category that that satisfies the universal property for abelian categories more generally. To define $\mathrm{D}(\mathbf{A})$ more generally, more theoretical build-up is necessary and shall not be carried out in this book, however presenting the idea of how to construct a general derived category will give the reader a chance to see some interesting applications of the concept if *localization* of a category. This section is heavily inspired by *Aluffi Chapter 0* which presented these idea's very concisely.

We shall have the objects still be $\mathrm{C}(\mathbf{A})$, however we need the quasi-isomorphisms to be invertible. Recall lemma 2.2.7 where it was shown that quasi-isomorphisms of projective resolutions have the property of being acting like non-zero divisors. We shall take full advantage of this idea, and essentially "formally invert" quasi-isomorphisms in the same vein as the construction of localization's for rings and modules. In fact, from an abstract perspective, the study of homological algebra can be thought of as the study of the localization of the category of (co)chain complexes (localized at the class of quasi-isomorphisms). When there is enough projectives, the localization of $\mathrm{C}(\mathbf{A})$ becomes $\mathrm{K}(\mathbf{a})$.

For the more general case, consider the following *roof diagram*:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A & & B \end{tikzcd}$$

Then we may formally add a morphism $\beta \circ \alpha^{-1} : A \to B$:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A \arrow[rr, dashed] & & B \end{tikzcd}$$

This can be thought of as the fraction. There are many technicalities being swept under the rug, like how do we insure these are well-defined morphisms (ex. how would we compose roofs?), while more fundamental problems like the amount of morphisms shouldn't exceed a *set*-amount (note that **A** and **B** are classes). Furthermore, we may ask if this is possible on any possible category, or should we restrict to abelian categories? Can any morphism be invertible, or do we need to restrict to a special subset (similar to a multiplicative set)? Giving this technical difficulties, we may imagine that from this the diagram

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\mathrm{id}_Z"] & \\ A \arrow[rr, dashed] & & Z \end{tikzcd}$$

would formally invert a morphism. For $\mathrm{D}(\mathbf{A})$, we would localize at the class of quasi-isomorphism of $\mathrm{K}(\mathbf{A})$. The category $\mathrm{D}(\mathbf{A})$ shall be the homotopic category of $\mathrm{C}(\mathbf{A})$ where a morphism $L^\bullet \to M^\bullet$ can be represented as a roof:

$$\begin{tikzcd} & N^\bullet \arrow[dl, "\text{quasi-iso}"'] \arrow[dr] & \\ L^\bullet & & M^\bullet \end{tikzcd}$$

(A word on how many abelian categories may induce the same derived categories. However, there is a special "triangulatedd structure", or t-structure which allows you to recover A)

2.5.2 Triangulated Categories

Homotopic categories are certainly almost never not abelian, for the kernels and cokernels of homotopic maps need not match (hence not being well-defined in the equivalence class). However, a derived functors $\mathrm{D}(\mathscr{F})$ always has a long exact sequence associated to it, which shows some weaker property these categories possess. These considerations are explored in this chapter. Take a short exact sequence of complexes:

$$0 \to L^\bullet \to M^\bullet \to N^\bullet \to 0$$

Then snakes lemma says that after taking homology, there is an interesting connection between $N^\bullet$ and $L^\bullet$. When working with the complex, there is no interesting connection between $N^\bullet$ and $L^\bullet$. We may connect these two complexes via the zero morphism: $N^\bullet \xrightarrow{0} L^\bullet$. From this, the short exact sequence of complexes may be interpreted as these three exact sequences:

$$N^\bullet \xrightarrow{0} L^\bullet \to M^\bullet$$
$$L^\bullet \to M^\bullet \to N^\bullet$$
$$M^\bullet \to N^\bullet \xrightarrow{0} L^\bullet$$

These three exact sequences are essentially can be combined if we fold any of them into the following triangle:

Definition 2.5.1: Exact Triangle

Let $L^\bullet, M^\bullet, N^\bullet \in \mathrm{Ch}(\mathbf{A})$ be three complexes. Then if there exists morphisms st the following diagram is exact (at each vertex):

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur] & & M^\bullet \arrow[ll] \end{tikzcd}$$

Then the diagram is called an *exact triangle* of $\mathrm{Ch}(\mathbf{A})$. This diagram is also sometimes denoted

$$L^\bullet \to M^\bullet \to N^\bullet \to$$

if it is easier to write it out flat.

Hence, the particular exact triangle given by folding any of the above exact sequences is:

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur, "+1"] & & M^\bullet \arrow[ll] \end{tikzcd}$$

where the $+1$ morphism represents the zero map. The $+1$ notation is to hint that this map will be a sort of shift from $N^\bullet$ to $L^\bullet[1]$. This can be stretched out into a single long sequence:

$$\cdots \to N^{i-1} \xrightarrow{0} L^i \to M^i \to N^i \xrightarrow{0} L^{i+1} \to \cdots$$

More explicitly, the triangle can be "unwrapped" to create the following nice diagram:

here, v good visual intuition

Applying the cohomology functor to the above long exact sequence and replacing the zero maps with the connecting morphism, we get:

$$\cdots \to H^{i-1}(N^\bullet) \xrightarrow{\delta^{i-1}} H^i(L^\bullet) \xrightarrow{0} H^i(M^\bullet) \xrightarrow{0} H^i(N^\bullet) \xrightarrow{\delta^i} H^{i+1}(L^\bullet) \to \cdots$$

Note that the maps δ^i did not come from the cohomology functor. In a sense, this is because $\mathrm{Ch}(\mathbf{A})$ is the wrong category to apply the cohomology functor, rather it is better to apply it in $\mathrm{D}(\mathbf{A})$. To make this connection, the notion of *distinguished triangles* would have to be developed in some level of detail. I do wish to do so since I'm interested in some of the connections they have, but for now I shall follow Aluffi's example and simply state the ingredients necessary on a shallow enough level to explain how the morphism δ would arise without needing any special considerations.

Let $\mathbf{A}$ be any Abelian category. We shall define the notion of a *distinguished triangle*. We first define a functor which is called the *translation functor*. Though it is called a translation functor, it shall be characterized by the axioms of a distinguished triangle. For the categories $\mathrm{K}(\mathbf{A})$ and $\mathrm{D}(\mathbf{A})$ this is the shift functor $L^\bullet \xrightarrow{+1} L^\bullet[1]$. Distinguished triangle are triangles that must satisfy 4 axioms:

1. The triangle:

$$\begin{tikzcd} & 0 \arrow[dr] & \\ A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"] \end{tikzcd}$$

has to be part of the collection of distinguished triangles

2. every morphism $\alpha : A \to B$ must be the side of a distinguished triangle.
3. Triangles that are isomorphic (as triangle diagrams) to distinguished triangles must also be distinguished
4. Distinguished triangles must be able to rotate, namely if:

$$\begin{tikzcd} & A \arrow[dr, "\alpha"] & \\ C \arrow[ur, "\gamma^{+1}"] & & B \arrow[ll, "\beta"] \end{tikzcd}$$

is a distinguished triangle if and only if

$$\begin{tikzcd} & A \arrow[dr, "\beta"] & \\ A[1] \arrow[ur, "-\alpha^{+1}"] & & B \arrow[ll, "\gamma"] \end{tikzcd}$$

is a distinguished triangle (think of visual ref:HERE, the 3D view, to intuit this axiom)

5. Finally, distinguished triangles must satisfy an axiom called the *octahedral axiom* which will not be necessary to cover for the remarks this section endeavors to explicate.

A category that contains a collection of distinguished triangles is called a *triangulated category*. The category $\mathrm{K}(\mathbf{A})$ is an example of such a category where the translation functor is the shift functor and all distinguished triangles are isomorphic to:

$$\begin{tikzcd} & L^\bullet \arrow[dr, "\alpha^\bullet"] & \\ MC(\alpha^\bullet) \arrow[ur, "+1"] & & M^\bullet \arrow[ll] \end{tikzcd}$$

Note how this diagram also fulfills the 3rd axiom: any map $\alpha^\bullet : L^\bullet \to M^\bullet$ is the edge of the above distinguished triangle where the other morphisms are the natural ones. Note also how the axioms prevent $\mathrm{Ch}(\mathbf{A})$ from being a triangulated category. Since

$$\begin{tikzcd} & 0 \arrow[dr] & \\ A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"] \end{tikzcd}$$

needs to be a distinguished triangle, then since $L^\bullet$ is the mapping cone of the zero-morphism $0 \to L^\bullet$ the rotation would axiom would require

$$\begin{tikzcd} & L^\bullet \arrow[dr, "\mathrm{id}_{L^\bullet}"] & \\ 0 \arrow[ur, "+1"] & & L^\bullet \arrow[ll] \end{tikzcd}$$

The mapping cone of the identity is not 0, and hence it cannot be a distinguished category. However, the mapping cone of the identity is *homotopically-equivalent* to 0, and hence the above triangle is indeed distinguished in $\mathrm{K}(\mathbf{A})$. Furthermore, a triangulated category, the cohomology functor will map distinguished triangles to exact triangles (in the case of $\mathrm{K}(\mathbf{A})$, this is because all distinguished triangles are isomorphic to the triangles with mapping cones), and hence give long exact sequences. Two notable examples of cohomological functors that map quasi-isomorphisms to isomorphisms for any triangulated category $\mathbf{A}$ are $\mathrm{Hom}(A,-)$ and $\mathrm{Hom}(=,A)$ for any object $A \in \mathbf{A}$.

The derived category $\mathrm{D}(\mathbf{A})$ is naturally a triangulated category, and the natural functor $\mathrm{K}(\mathbf{A}) \to \mathrm{D}(\mathbf{A})$ maps distinguished triangles to distinguished triangles and maps the translation functor to the translation functor. Derived categories are less restrictive than homotopic categories in the following important way: if

$$0 \to L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \xrightarrow{\beta^\bullet} N^\bullet \to 0$$

is a short exact sequence, then in general there is no distinguished triangle

$$\begin{tikzcd} & L^\bullet \arrow[dr, "\alpha^\bullet"] & \\ N^\bullet \arrow[ur, "\gamma^\bullet\ +1"] & & M^\bullet \arrow[ll, "\beta^\bullet"] \end{tikzcd}$$

In $\mathrm{K}(\mathbf{A})$ for any choice $\gamma^\bullet$ (even more is the case, $N^\bullet$ need not be homotopically equivalent to $MC(\alpha^\bullet)$). However, such a triangle *does* always exist in $\mathrm{D}(\mathbf{A})$), namely because there does exist a quasi-isomorphism $MC(\alpha)^\bullet \to N^\bullet$. Then applying the cohomology functor to this triangle shall give

$$\begin{tikzcd} & H^\bullet(L^\bullet) \arrow[dr] & \\ H^\bullet((N^\bullet) \arrow[ur, "+1"] & & H^\bullet((M^\bullet) \arrow[ll] \end{tikzcd}$$

where $+1$ is the connection homomorphism δ up to isomorphism. Furthermore, the 0 morphisms that connect the above in when applying the $H^\bullet(-)$ turned out to be particular choices in the more general case.

Finally, it should be noted that any functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ between Abelian categories shall induce a functor $\mathrm{K}(\mathscr{F}) : \mathrm{K}(\mathbf{A}) \to \mathrm{K}(\mathbf{B})$, and this functor is in fact a functor of triangulated categories. Hence, the bounded equivalents, as well as $\mathrm{L}(\mathscr{F}), \mathrm{R}(\mathscr{F}) : \mathrm{D}^\pm(\mathbf{A}) \to \mathrm{D}^\pm(\mathbf{B})$ are functor of triangulated categories. From this perspective, the results proven about derived functors is a consequence of this fact.

2.5.3 Stable Module Theory

I really want to talk about $f, g : M \to N$ being "homotopic" if $h = f - g$ can factor through a projective module $h : M \to P \to N$; and how this is equivalent to the homotopy equivalence we talked about earlier. This brings back the intuition from the geometric setting were we didn't need a resolution of modules to have two modules be homotopy equivalent! This also naturally leads to naturally define stably equivalent, namely M, N are stably equivalent if there exists a P such that:

$$M \oplus P \cong N \oplus P$$

This is further developed in [4], but it is definitely worth mentioning it here since it is an excellent prelude!

3

Applications of (co)Homology

With cohomology theory more securely under our belt, it is time to use it! All the work in this chapter can be learnt without the need for any algebraic topology background, as it is the goal of the book to minimize the need for external material. However, the proceeding ideas are abstractions of ideas in algebraic topology, and ignoring the source of inspiration for these ideas is itself a disservice to the pedagogy of the material. Therefore, the section will be written in such a way that a background in homology/cohomology will be advantageous. Namely, it is best if the reader is familiar with simplicial and singular homology, how cohomology is derived from $\mathrm{Hom}(C_\bullet(X), G) = C^\bullet(X; G)$ and the relation of homology to cohomology (ex. the universal coefficient theorem for cohomology). The reader may choose to consult [6] or [2] for the details.

This chapter does *not* present an in-depth coverage of any of the material, in particular these subjects are a large source of modern development in mathematics (as of the early 21st century). The material presented is to demonstrate to the reader the power of cohomological theories so that the reader has an easier time when reading the more dense books containing this information. For a book in the EYNTKA series covering group and galois cohomology, see [1], and for K-theory see [4].

In this chapter we:

1. Shall use the above theory to find cohomological invariants of groups
2. Shall show some ways of interpreting Ext and Tor
3. Shall show that the notion of dimension from the geometric setting can be brought over to the algebraic!
4. Some K-theory will certainly interest the reader

We shall not do sheaf cohomology, as (naturally) it required sheaf theory that was not covered, and it is an exploration for EYNTKA Algebraic Geometry [5].

3.1 Group Cohomology

We start with no link to algebraic topology, and shall include the relation to singular cohomology before the examples.

Given a (not necessarily abelian) group G, what would it mean to extract cohomological information from it? A (co)chain would have to naturally be formed, and a functor must be found to derive. Groups themselves don't form an abelian category, but it was shown that representations, that is $k[G]$-modules, do form them. This is perhaps a first place in which we may think we can study groups, however this category is too nice (think of (co)homology with $\mathbb{Z}$ vs. $\mathbb{R}/\mathbb{C}$ coefficients). What is more useful is it's generalization:

Definition 3.1.1: G-Modules

Let G be an arbitrary group, and M an abelian group. Then M is a G*-module* if there exists a [left] group-action on M. Equivalently, if there exists a $\mathbb{Z}[G]$-module structure on M.

The reader is free to parse through the technicalities of the group action. All $k[G]$-modules presented in chapter **??** are $\mathbb{Z}[G]$-modules with the action restricted. An example of a G-module that is not a representation is the action of the group $\mathrm{Gal}_K(L)$ on the module L/K (a galois extension of L over K). All finite groups G have a non-trivial G-module given by the action of G on the group-ring $k[G]$ (note the addition is abelian). From a geometric perspective, modules represent sheaves (or in special cases vector bundles) over some schemes,

As we have seen many times at this point, the object acts on an abelian group can give lots of information on the object acting on it and the object being acted on. Given a G-module M, we can define the following:

Definition 3.1.2: G-Invariant subgroup

Let M be a G-module. Then the G*-invariant* set of M is the set:

$$M^G := \{m \in M \mid g \cdot m = m, \ \forall g \in G\}$$

which is the largest trivial G-submodule of M.

It is easy to see that $(-)^G$ is a functor. The following will help in computing cohomology

Corollary 3.1.3: Hom Representation of G-invariant

$$M^G \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, M^G) \cong \mathrm{Hom}_G(\mathbb{Z}, M)$$

***Proof*:**
exercise (what G-action should be put on $\mathbb{Z}$?)

Similarly, we can define:

Definition 3.1.4: G-Coinvariant subgroup

Let M be a G-module. Then the G*-coinvariant* of M is the set:

$$M_G := \frac{M}{\langle (gm - m) \mid g \in G, m \in M \rangle}$$

and is the maximal invariant quotient module with the trivial action.

Just like $(-)^G$, $(-)_G$ is also a functor. The reader should further verify that

Proposition 3.1.5: Tensor Representation of Coinvariant Module

$$M_G \cong_G \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

Proposition 3.1.6: Invariants Is Left-exact

The functor $(-)^G$ is left-exact, the functor $(-)_G$ is right exact.

***Proof*:**
First, prove that $(-)^G$ is right-adjoint to the trivial-action functor.

$$\mathrm{Hom}(T(M), N) \cong \mathrm{Hom}(M, N^G)$$

And similarly for $(-)_G$; it is left-adjoint to the trivial-action functor. As $T(-)$ is an exact functor, $(-)^G$ and $(-)_G$ are left-exact and right-exact respectively, completing the proof.

Both of these functors are not exact:

Example 3.1.7: Lack Of Exactness's
Let $G = C_2$ be the cyclic group of order 2 and:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(1 \cdot a + \sigma \cdot b) = a + b$ and I_G is the kernel. This sequence is exact. Take $(-)^G$:

$$0 \to (I_G)^G \to (\mathbb{Z}[G])^G \xrightarrow{\epsilon^G} (\mathbb{Z})^G \to 0$$

i.e.:

$$0 \to 0 \to \Delta \xrightarrow{\epsilon^G} \mathbb{Z} \to 0$$

where $\Delta = \{(a, a) \mid a \in \mathbb{Z}\}$. Then ϵ^G is no longer surjective as $\epsilon^G(a, a) = 2a$. The above example can also be used to show $(-)_G$ is not left-exact. More generally, if $0 \to M_2^G \to M_2 \to M_3 \to 0$ is exact, taking $(-)^G$ doesn't change $M - 2^G$, but now we get an isomorphism between the 1st and second terms, so it cannot be surjective.

More generally, let G act trivially on a coset $[m]$ of a quotient M/L. Can this action always be lifted so that G acts trivially on a representative m?

Thus, it is fruitful to take the ith right-derived functor of $(-)^G$ (similarly the left-derived functors of $(-)_G$):

$$\mathbf{R}^i(-)^G := H^i(G,-)$$

which would give us the long exact sequence of abelian groups:

$$0 \to L^G \to M^G \to N^G \to H^1(G,L) \to H^1(G,M) \to \cdots$$

Since $M^G = \mathrm{Hom}_G(\mathbb{Z}, M)$ by taking an injective resolution of M, or a projective resolution of $\mathbb{Z}$, we can write:

$$H^i(G,M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

Note that $H^0(G,M) = M^G$, hence the higher order cohomology groups represent some failure for points to be fixed. Let us compute some examples to show the power of the machinery developed in chapter 2 before concretely demonstrating how the cohomology groups represent the obstruction of points being fixed.

Example 3.1.8: Group Cohomologies – Homological Algebra Computations

1. The trivial group has trivial cohomology
2. (Integers) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[G] \cong \mathbb{Z}[x, x^{-1}]$. Then since $\mathbb{Z}[x,x^{-1}]/(1-x) \cong_G \mathbb{Z}$ with the trivial action, define the following free (hence projective) resolution:

$$\cdots \to 0 \to \mathbb{Z}[x,x^{-1}] \xrightarrow{\cdot(1-x)} \mathbb{Z}[x,x^{-1}] \to 0 \to \cdots$$

   Applying the (contravariant) $\mathrm{Hom}_{\mathbb{Z}[x,x^{-1}]}(-,M)$, we can compute the cohomology of

$$\cdots \to 0 \to M \xrightarrow{\cdot(1-x)} M \to 0 \to \cdots$$

   Then we can conclude by direct computation:

$$H^0(\mathbb{Z}, M) \cong \ker(M \xrightarrow{\cdot(1-x)} M) = M^{\mathbb{Z}}$$

   and:

$$H^1(\mathbb{Z}, M) = \frac{M}{(1-x)M}$$

   and furthermore $H^i(\mathbb{Z}, M) = 0$ for $i \notin \{0,1\}$
3. Let $G = C_n$ be a finite cyclic group of over n with generator σ. Let us form a free resolution of $\mathbb{Z}[G]$ by taking advantage of the fact that $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ are zero divisors:

$$N \cdot (\sigma - 1) = (1 + \sigma + \cdots + \sigma^{n-1})(\sigma - 1) = \sigma^n - 1 = 1 - 1 = 0$$

   This gives a resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

   where

$$\epsilon\left(\sum_g a_g g\right) = \sum_g a_g$$

is the usual augmentation map. Then:

$$H^k(C_n, M) = \begin{cases} \frac{M^G}{NM} & k \text{ even, } k \geq 2 \\ \frac{{}_N M}{(\sigma-1)M} & k \text{ odd, } k \geq 1 \end{cases} \qquad \text{where} \qquad {}_N M = \{m \in M \mid N \cdot m = 0\}$$

For example, if $M = \mathbb{Z}$, then:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/m\mathbb{Z} & k \text{ even, } k \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Let $G = *_{s \in S}\mathbb{Z}$ be the free product of S copies of $\mathbb{Z}$. This has a simple free resolution, namely take:

$$\epsilon : \mathbb{Z}[G] \to \mathbb{Z}$$

where the kernel is:

$$I_S = \sum_{s \in S} \mathbb{Z}[G] \cdot (s-1)$$

It should be verified that I_S is a free G-module, hence we have a free-resoluion:

$$0 \to I_S \to \mathbb{Z}[G] \xrightarrow{\epsilon} \to \mathbb{Z} \to 0$$

and cohomology:

$$H^k(*_{s \in S}\mathbb{Z}, \mathbb{Z}^G) = \begin{cases} \mathbb{Z} & k = 0 \\ \bigoplus_{s \in S} \mathbb{Z} & k = 1 \\ 0 & k \geq 2 \end{cases}$$

5. (If you know some algebraic geometry) *Galois Cohomology*: Take the category of varieties W over k such that when extending scalars to $\overline{k}$ we have $V/\overline{k} \cong W/\overline{k}$. Let's label it $\mathbf{Var}_{k \to \overline{k}}$, and say that the objects are:

$$\mathrm{Obj}(\mathbf{Var}_{k \to \overline{k}}) = \left\{ W/k \;\middle|\; V/\overline{k} \cong W/\overline{k} \right\}$$

and the morphism are regular morphisms. Then I claim that:

$$\mathbf{Var}_{k \to \overline{k}} \cong H^1(\mathrm{Gal}_k(\overline{k}), \mathrm{Aut}(V/\overline{k}))$$

To see this, first note that an element $\sigma \in \mathrm{Gal}_k(\overline{k})$ acts on a $\overline{k}$-variety $W/\overline{k} = W_k \times_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(\overline{k})$ via:

$$\mathrm{id}_{W_k} \times \mathrm{Spec}\,(\sigma^{-1}) : W/\overline{k} \to W/\overline{k}$$

Or, for those less comfortable with fibered produced from algebraic geometry, they can think of this map as the "equivalent":

$$\mathrm{id}_{W_k} \otimes_k \sigma^{-1} : k[W] \otimes_k \overline{k}$$

With the action defined, let us look at the correspondence. Take $W_k \in \mathbf{Var}_{k \to \overline{k}}$ so that $\varphi : W_{\overline{k}} \xrightarrow{\sim} V_{\overline{k}}$ is an isomorphism. Then notice that this isomorphism induces a map $\varphi \circ \sigma^{-1} \circ \varphi^{-1}$, which is an automorphism on $W_{\overline{k}}$. The collection of all these maps as σ varies can uniquely define W_k (k is the fixed subfield over all σ). This is true up to a choice of $\mathrm{Aut}_k(V)$, and so we quotient out by it. Then we can see this creates the correspondence between these two categories, and shows how far away the category $\mathbf{Var}_{k \to \overline{k}}$ is from being invariant to the group-action of the galois group.

6. (if you know elliptic curves) An excellent example of where galois cohomology is immediately useful is in the theory of elliptic curves. On a high level, it is relatively easy to find the solution of an elliptic curve over $\mathbb{C}$, but it becomes much more difficult to solve it over finite extensions of $\mathbb{Q}$, or $\mathbb{Q}$ itself. If the reader has read (or is reading) [3], then the have encountered the short exact sequence of $\mathrm{Gal}_K(\overline{K})$-modules:

$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

where $E[m]$ are the m-torsion points of an elliptic curve ($m \geq 2$), K is a number field, $E(\overline{K})$ are the points of the elliptic curve over $\overline{K}$. Let $G = \mathrm{Gal}_K(\overline{K})$. Then when considering solutions over K, the short exact sequence sequence becomes a long exact sequences:

$$\begin{aligned} 0 \to E(K)[m] \to E(K) \xrightarrow{[m]} E(K) \xrightarrow{\delta} H^1(G, E[m]) \\ \to H^1(G, E(\overline{K})) \xrightarrow{[m]} H^1(G, E(\overline{K}) \to \cdots \end{aligned}$$

This is the foundation on which the torsion-free points of E/K will be computed!

7. If the reader knows about the classifying space BG, then:

$$H^n(BG, \mathbb{Z}) \cong H^n(G, \mathbb{Z})$$

where he left hand side is singular cohomology and the right hand side is group cohomology.

What obstruction information does this capture? Since there is a failure in surjectivity $\varphi : M \twoheadrightarrow N$ when applying $(-)^G$, this means that a fixed element $n \in N^G$ does *not* necessarily come form a fixed element $m \in M^G$. Thus, even though $\varphi(m) = n$ exists, it need not be the case that there exists an $m' \in M^G$ such that $\varphi^G(m') = n$. To measure this failure, take the short exact sequence

$$0 \to L \xrightarrow{\iota} M \xrightarrow{\varphi} N \to 0$$

and let $n \in N^G$. As $N^G \subseteq N$, there exists an $m \in M$ such that $\varphi(m) = n$. Consider any $g \in G$ and take $g \cdot m - m$. Then applying the G-homomorphism:

$$\varphi(g \cdot m - m) = g \cdot \varphi(m) - \varphi(m) = g \cdot n - n \overset{!}{=} n - n = 0$$

where $\overset{!}{=}$ comes from $n \in N^G$. By exactness $\ker \varphi = \mathrm{im} \iota$, so there must exist an element $\lambda_m(g) \in L$ such that

$$\iota(\lambda_m(g)) = g \cdot m - m$$

Now, $n \in N^G$ can be lifted if and only if $g \cdot m - m = 0$ for all $g \in G$, since then $m \in M^G$. In terms of elements of L:

$$\forall g \in G, \ \iota(\lambda_m(g)) = 0$$

by injectivity, it must be that $\lambda_m(g) = 0$ for all $g \in G$. Thus, the function $\lambda_m : G \to L$ is non-trivial if the lift does not exist. Let us explore the properties of λ_m. First, if $g_1, g_2 \in G$ then:

$$\begin{aligned}
\iota(\lambda_m(g_1g_2)) &= g_1g_2 \cdot m - m \\
&= g_1g_2 \cdot m - g_1 \cdot m + g_1 \cdot m - m \\
&= g_1 \cdot (g_2 \cdot m - m) + (g_1 \cdot m - m) \\
&= g_1\iota(\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1\lambda_m(g_2) + \lambda_m(g_1))
\end{aligned}$$

Thus, as f is injective (more specifically a monomorphism) we get:

$$\lambda_m(g_1g_2) = \lambda_m(g_1) + g_1\lambda_m(g_2)$$

A function with this property is given a name:

Definition 3.1.9: 1-Cocycle

A function $f : G \to M$ is a called a *1-cocycle*, *1-derivation*, or *crossed homomorphism* if:

$$f(gh) = f(g) + gf(h)$$

The set of all derivations is denoted $Z^1(G, M)$. If it is clear from context, the "1-" is dropped.

Now, the above λ_m was dependent on a choice of m. But $M \hookrightarrow N$ is surjective, and hence there may be another m' such that $\varphi(m') = n$. As φ is a G-homomorphism

$$\varphi(m - m') = c - c = 0$$

and so $m - m' \in \ker\varphi = \mathrm{im}\iota$. Thus, there is an $\ell \in L$ such that

$$m - m' = \iota(\ell) \qquad \text{or} \qquad m' = m + \iota(\ell)$$

Thus, let us relate $\lambda_{m'}$ to λ_m:

$$\begin{aligned}
\iota(\lambda_{m'}(g)) &= g \cdot m' - m' \\
&= g \cdot (m + \iota(\ell)) - m - \iota(\ell) \\
&= g \cdot m + g \cdot \iota(\ell)) - m - \iota(\ell) \\
&= (g \cdot m - m) + g \cdot \iota(\ell) - \iota(\ell) \\
&= \iota(\lambda_m(g)) + \iota(g \cdot \ell) - \iota(\ell) \\
&= \iota(\lambda_m(g) + g \cdot \ell - \iota(\ell))
\end{aligned}$$

Thus, as ι is a monomorphism:

$$\lambda_{m'}(g) = \lambda_m(g) + g \cdot \ell - \ell$$

Thus, the term $g \cdot \ell - \ell$ can be seen as the "trivial" part of the obstruction, as it does not change the difficulty in finding an obstruction. This is given a name:

Definition 3.1.10: 1-Coboundary

A function $f_m : G \to M$ is a called a *1-coboundary* or *principal derivation* if:

$$f_m(g) = g \cdot m - m$$

The set of all boundary maps is denoted $B^1(G, M)$.If it is clear from context, the "1-" is dropped.

Note that a coboundary is a cocycle:

$$f_m(gh) = gh \cdot m - m = gh \cdot m - h \cdot m + h \cdot m - m = f_m(g) + gf_m(h)$$

Hence $B^1(G, M) \subseteq Z^1(G, M)$. It shall be shown in a moment that:

$$H^1(G, M) \cong \frac{Z^1(G, M)}{B^1(G, M)}$$

First, let us show there is a map $L^G \to H^1(G, M)$. Indeed there is a well-defined G-module homomorphism:

$$\ell \mapsto [g \mapsto g \cdot m - m]$$

where m is in the fiber of ℓ.

Let us next motivate the 2nd order cohomology group. Let us assume the cocycle-representation and the derived functor representation are isomorphic for the next part. Let us start with the long exact sequence we know is given:

$$0 \to L^G \xrightarrow{\alpha} M^G \xrightarrow{\beta} N^G \xrightarrow{\delta^0} H^1(G, L) \xrightarrow{\alpha_*} H^1(G, M) \xrightarrow{\beta_*} H^1(G, N) \xrightarrow{\delta^1} H^2(G, L) \cdots$$

Consider $[\nu] \in H^1(G, N)$; let us call it a 1-obstruction. Then can it be lifted some $[\mu] \in H^1(G, M)$, namely does there exist $[\mu]$ such that $\beta_*([\mu]) = [\nu]$. To find out, first take some representative cocycle $\nu : G \to N$ (notice we are now working with a map, not an element). As $M \hookrightarrow N$ is surjective, we have a diagram:

$$\begin{tikzcd} & M \arrow[d, two heads, "\beta"] \\ G \arrow[ur, dashed, "\mu?"] \arrow[r, "\nu"'] & N \end{tikzcd}$$

We can certainly lift each point *locally*, namely given a point $\nu(g) = n \in N$ there is a point (or points) m such that $\beta(m) = n$[1]. But this lifting can fail *globally*, namely such a lift is not guaranteed to be a 1-cocycle. This mirrors the general pattern in cohomology where having local data of a function does not imply the existence of a global function that restricts to all the local data. Let us measure this failure of lifting. If the lifted μ is a cocycle, it would satisfy:

$$\mu(g_1g_2) - \mu(g_1) - g_1\mu(g_2) = 0$$

Thus, define the function:

$$\psi : G \times G \to M, \qquad \psi(g_1, g_2) = \mu(g_1g_2) - \mu(g_1) - g_1\mu(g_2)$$

[1]Technically, the axiom of choice is invoked here for making a potentially uncountable number of choices

Let us show that for any $(g_1, g_2) \in G \times G$, $\psi(g_1, g_2) \in \ker \beta$. Indeed:

$$\begin{aligned}
\beta(\psi(g_1, g_2)) &= \beta(\mu(g_1g_2) - \mu(g_1) - g_1\mu(g_2)) \\
&= \beta(\mu(g_1g_2)) - \beta(\mu(g_1)) - \beta(g_1\mu(g_2)) \\
&\stackrel{!}{=} \nu(g_1g_2)) - \nu(g_1) - g_1\nu(g_2)) \\
&= 0 && \nu \text{ is a cocycle}
\end{aligned}$$

where $\stackrel{!}{=}$ comes from μ being defined as a lift of ν and hence $\beta \circ \mu = \nu$. Thus, by exactness $\ker \beta = \mathrm{im}\alpha$ and so for each pair $(g_1, g_2) \in G \times G$, there exists a unique element in L that maps to $\psi(g_1, g_2)$, namely there exists a function $\Omega : G \times G \to L$ such that

$$\alpha(\Omega(g_1, g_2)) = \psi(g_1, g_2)$$

This Ω would now be our new 2-cocycle and are denoted $Z^2(G, M)$. Doing a similar but longer computations such as the one for the 1-cycle, we can find that:

$$g_1\Omega(g_2, g_3) - \Omega(g_1g_2, g_3) + \Omega(g_1, g_2g_3) - \Omega(g_1, g_2) = 0$$

Similarly, an explicit computation would fine that 2-boundaries (given a 1-chain ψ) are of the form

$$\Omega(g_1, g_2) = g_1 \cdot \psi(g_2) - \psi(g_1g_2) + \psi(g_1)$$

These are label $B^2(G, M)$ and:

$$H^2(G, M) \cong \frac{Z^2(G, M)}{B^2(G, M)}$$

The same process continues for higher order cohomology. For the reader who knows some algebraic topology, the following informal discussion grounds the higher order group-cohomology as the singular (or simplicial) cohomology of BG, the classifying space of G with the property that $\pi_1(BG) = G$ and $\pi_n(BG) = \{0\}$ for all $n > 1$ (see [6]). In particular:

$$H^n_{\mathrm{Sing}}(BG, M) \cong H^n(G, M)$$

Note that if M has a non-trivial G-module action, it corresponds to cohomology with local coefficients, but the geometric picture remains the same (the details are left for [1]). Using the nerve or bar construction on BG and considering the usual covering $EG \to BG$ we get the following correspondence:

Geometric Object in EG	Correspondence in BG	Algebraic equivalent
0-Simplices (Vertices)	A single 0-simplex, base point $*$.	elements of G
1-Simplices (Edges)	Directed edge for each element $g \in G$. An arrow from $*$ to $*$ labeled by g.	Functions on G
2-Simplices (Triangles)	Oriented triangle for each pair $(g_1, g_2) \in G \times G$. Edges are the 1-simplices for g_1, g_2, and the product g_1g_2.	Functions on $G \times G$
3-Simplices (Tetrahedral)	Oriented tetrahedron for each triple $(g_1, g_2, g_3) \in G \times G \times G$. Faces are the triangles for (g_2, g_3), (g_1, g_2g_3), etc.	Functions on $G \times G \times G$

n-Simplices	n-tuple $(g_1, \ldots, g_n) \in G^n$.	Functions on G^n

Where "functions" here means set-function (to take into account the "failed" lifting), and the transition maps d_i captures the G-module homomorphism information. Geometrically, the G-module homomorphisms correspond to G-equivariant maps $f : G^n \to M$.

Using this intuition, we can construct a standard way of defining a projective resolution of $\mathbb{Z}$ by G-modules. First, let $\mathbb{Z}[G^n]$ be a G-module with the diagonal action given by multiplying on the left, or more specifically:

$$g \cdot (g_1, ..., g_n) = (gg_1, ..., gg_n)$$

Note that this diagonal map corresponds to the natural action of left multiplication by g on EG. These modules are free G-modules; clearly $\mathbb{Z}[G]$ is a $\mathbb{Z}$-module with basis given by the elements G. Then for $n \geq 0$:

$$\mathbb{Z}[G^{n+1}] \cong \bigoplus_{(g_1,...,g_n)\in G^n} \mathbb{Z}[G] \cdot (1, g_1, ..., g_n)$$

which shows that these are free. From this, we can consider the following projective (even free) resolution:

Definition 3.1.11: Standard Resolution

Let G be a group. Then the *standard resolution* of $\mathbb{Z}$ by G-modules is the exact sequence of G-module homomorphism:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_{n+1}} \mathbb{Z}[G^n] \xrightarrow{d_n} \cdots \xrightarrow{d_2} Z[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where on the group G we define:

$$d_n(g_0, ..., g_n) = \sum_{i=1}^{n} (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$$

where the term $\hat{g}_i$ is omitted from the tuple. The map ϵ is also called the *augmentation map*.

Note that each $\mathbb{Z}[G^{n+1}]$ is a We claimed this is an exact sequence, however this ought to be verified.

Lemma 3.1.12: Standard Resolution is Exact

The standard resolution of $\mathbb{Z}$ by G-modules is exact

This shows that the nomenclature of cocycles and coboundaries above is used correctly.

***Proof*:**
This a proof that is about keeping track a lot of technical details, but it is all eminently doable, and so left to the willing reader.

Hence, after taking the contravariant functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], M) \to \cdots \to \mathrm{Hom}_{\mathbb{Z}[G^n]}(-, M) \xrightarrow{d_n^*} \mathrm{Hom}_{\mathbb{Z}[G^{n+1}]}(-, M) \to \cdots$$

where d_n^* maps $\varphi \mapsto \varphi \circ d_n$. Similarly, we can take the covariant functor

$$\cdots \to \mathbb{Z}[G^3] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{2*}} \mathbb{Z}[G^2] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{1*}} \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A \xrightarrow{\epsilon} 0$$

where d_{n*} is given by $(g_0, ..., g_n) \otimes a \mapsto d_n(g_0, ..., g_n) \otimes a$, the map ϵ is the augmentation map given by $\sum a_g g \mapsto \sum a_g$, and we view the modules as *right* $\mathbb{Z}[G]$-modules wit the action

$$(g_1, ..., g_n) \cdot g = (g_1 g, ..., g_n g)$$

Let us now better understand these hom-sets and relate to the above discussion. Let

Definition 3.1.13: coChain

Let

$$C^n(G, M) = (\mathrm{Hom}_{\mathbf{Set}}(G^n, M), +)$$

with group operation defined on the maps given by point-wise addition. This group is called the *n-cochain group*.

From this, we can define a boundary map:

$$\begin{aligned} d^n &: C^n(G, M) \to C^{n+1}(G, M) \\ d^n(f)(g_0, ..., g_n) &= g_0 f(g_1, ..., g_n) - f(g_0 g_1, g_2, ..., g_n) \\ &\quad + f(g_0, g_1 g_2, ..., g_n) - \cdots + (-1)^n f(g_0, ..., g_{n-2}, g_{n-1} g_n) + (-1)^{n+1} f(g_0, ..., g_{n-1}) \end{aligned}$$

Then $d^{n+1} \circ d^n = 0$, giving us a cochain complex. We can then have:

Definition 3.1.14: Cohomology Group – using Cocycles and Coboundaries

$$H^n(G, M) = \frac{\ker d^n}{\mathrm{im} d^{n-1}} = \frac{Z^n(G, M)}{B^n(G, M)}$$

We sum up many of the results that we've shown before using the new notation:

1. $C^0(G, M) = M$, and $H^0(G, M) \cong M^G$
2. $d^0 : C^0(G, M) \to C^1(G, M)$ is given by $d^0(a)(g) = ga - a$
3. $H^0(G, M) \cong \ker d^0 = M^G$
4. $\mathrm{im} d^0 = \{f : G \to M \mid \exists a \in M \text{ such that } f(g) = ga - a, \ \forall g \in G\}$ (principal crossed homomorphism)
5. $d^1 : C^1(G, M) \to C^2(G, M)$ given by $d^1(f)(g, h) = gf(h) - f(gh) - f(g)$
6. $\ker d^1 = \{f : G \to M \mid f(gh) = f(g) + gf(h)\}$ (crossed homomorphisms)
7. If $0 \to A \to B \to C \to 0$ is exact, we get an exact sequence of cochains for every n:

$$0 \to C^n(G, A) \to C^n(G, B) \to C^n(G, C) \to 0$$

We shall show that the collection of complexes given by the above is isomorphic to those given by the standard resolution

Proposition 3.1.15: Characterizing Hom of Resolution

Let M be a G-module. Then for every $n \geq 0$, we there is an isomorphism:

$$\Phi^n : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M) \xrightarrow{\cong} C^n(G, M)$$

where the map $\varphi : \mathbb{Z}[G^{n+1}] \to M$ goes to $f : G^n \to M$ given by:

$$f(g_1, ..., g_n) = \varphi(1, g_1, g_1g_2, g_1g_2g_3, ..., g_1g_2g_3 \cdots g_n)$$

The isomorphism Φ^n commutes with boundary maps, that is

$$\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$$

Hence we have an isomorphism from the category of cochain complexes with the standard resolution to the one with boundary-cycle maps.

> ***Proof*:**
> very doable, and worth doing at least once.

Due to the relevance of the standard representation, we have the following definition:

Definition 3.1.16: Induced and Coinduced G-module

Let G be a group and A an abelian group. Then:

$$\mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$$

with the G-action defined as $(g\varphi)(z) = \varphi(zg)$ is called the *coinduced G-module* associated to A. Similarly

$$\mathrm{Ind}^G(A) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$$

with G-action defined by $g(z \otimes a) = (gz) \otimes a$ is the is the induced G-module associated to A.

A result that is common in many Cohomologies theories is some version of the following:

Lemma 3.1.17: Induced and Coinduced Equivalence

Let G be a finite group and A an abelian group. Then

$$\mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$$

As a consequence, if $B = \mathrm{Ind}^G(A)$ or $B = \mathrm{CoInd}^G(A)$, then

$$H_0(G, B) \cong H^0(G, B) \cong A$$

and they are both zero for $n \geq 1$.

***Proof*:**
Define the map: $\alpha : \mathrm{Coind}^G(A) \to \mathrm{Ind}(G)(A)$ given by:

$$\varphi \mapsto \sum_{g \in G} g^{-1} \otimes \varphi(g)$$
$$(g^{-1} \mapsto a) \leftarrow\!\shortmid g \otimes a$$

with $(g^{-1} \mapsto a)(h) = 0$ for $h \in G \setminus \{g^{-1}\}$. These are certainly inverses, and easily verifiable to be G-module homomorphisms (this is a similar technique used in Maschkle's Theorem (see **??**).

The last part is left for exercise ref:HERE.

Proposition 3.1.18: Torsion of H^1

if G has order n and is finite, then every element of $H^1(G, M)$ has order divisible by n.

***Proof*:**
Let $\delta \in Z^1$ so that $\delta(gh) = \delta(g) + g\delta(h)$ for all $g \in G$. Sum both sides over $h \in G$ keeping g fixed. Then as multiplying by g is translation:

$$\sum_{h \in G} \delta(gh) = \sum_{h \in G} \delta(h) = S$$

As $\delta(g)$ is a constant, we can re-arrange and get:

$$(1 - g)S = |G|\delta(g)$$

Now, re-write it suggestively as:

$$g(-S) - (-S) = |G|\delta(g)$$

But now the left hand side is a coboundary, and hence the equation is 0 in $H^1(G, M)$ showing that the order of the group zero's any cocycle, completing the proof.

Proposition 3.1.19: M finitely generated, then H1 finite

Let M be a finitely generated G-module and G finite. Then $H^1(G, M)$ is finite.

***Proof*:**
Let $\langle m_1, ..., m_r \rangle = M$. Then C^1 is finitely generated as a $\mathbb{Z}$-module (note that G had to be finite). Then $Z^1(G, M)$ (as $\mathbb{Z}$-submodule) is finitely generated. But now by proposition 3.1.19, every element has order divisible by $n = |G|$. Thus, take the lcm of the order of the generators, giving a bound of the order of H^1, completing the proof.

Finally, let us prove the foreshadowing in box **??**. Let L/K be a cyclic extension so that $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$. The G acts on $L^\times$ in the natural way, and hence we find it's group cohomology. First:

Lemma 3.1.20: Cocycles using Norm and Trace

Let L/K be a cyclic extension $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ with generator σ, and $Z^1(G, L^\times)$ the 1-cocycles. Let $U_{L/K} = \left\{ l \in L \;\middle|\; \mathrm{Nm}_{L/K}(l) = 1 \right\}$ (sometimes this is denoted L^1). Then as groups:

$$\begin{aligned} Z^1(G, L^\times) &\cong U_{L/K} \\ c &\mapsto c(\sigma) \end{aligned}$$

Similarly:

$$Z^1(G, L) \cong T_{L/K}$$

where $T_{L/K}$ is the traceless elements[a]

[a] There is less common notation, sometimes it is denoted L^0

***Proof*:**
Let us first show the map is well-defined. Note first that a cocycle is determined by where it maps σ, namely for an arbitrary $c \in Z^1(G, L^\times)$:

$$\begin{aligned} c(\sigma^2) &= c(\sigma) \cdot \sigma(c(\sigma)) \\ c(\sigma^3) &= c(\sigma) \cdot \sigma(c(\sigma^2)) \\ &\vdots \end{aligned}$$

Next, note that $c(\sigma^m) = c(e) = 1$ since:

$$c(e) = c(e \cdot e) = c(e) \cdot e(\sigma(e)) = c(e) \cdot c(e)$$

as these map to $L^\times$, all elements are invertible, hence:

$$c(e) = 1 \in L^\times$$

Then:

$$1 = c(\sigma^n) = c(\sigma) \cdot \sigma(c(\sigma)) \cdot \sigma^2(c(\sigma)) \cdots \sigma^{n-1}(c(\sigma))$$

Letting $\beta = c(\sigma)$:

$$1 = c(\sigma^n) = \beta \cdot \sigma(\beta) \cdot \sigma^2(\beta) \cdots \sigma^{n-1}(\beta) = \prod_{g \in G} g(\beta) = \mathrm{Nm}_{L/K}(\beta)$$

showing the map is well-defined. As it is determined by where σ maps to, if $c_1, c_2 \in Z^1(G, L^\times)$ and $c_1(\sigma) = c_2(\sigma)$ then they agree on each σ^k and hence $c_1 = c_2$, showing injectivity. Surjectivity can be shown by defining $c(\sigma) = \beta$ for $\beta \in U_{L/K}$ and showing it satisfies the cocycle condition, which will be left as an exercise.

A similar argument can be done using the trace.

Theorem 3.1.21: Hilbert's 90th Theorem – Cohomology Upgrade

Using the above lemma's notation:

$$H^1(G, L^\times) \cong \{1\}$$

***Proof*:**
By Hilbert's 90th theorem (**??**), every norm 1 element is of the form:

$$\frac{\alpha}{\sigma^k(\alpha)}$$

But this is a coboundary element, hence all cocycles are coboundaries, completing the proof.

Note this theorem generalizes to finite galois extensions in general, see ref:HERE. This result should now make intuitive sense: the elements of K are *exactly* the $\mathrm{Gal}_K(L)$-invariant element. Note that if there were more complicated groups instead of $L^\times$, there can be more interesting cohomology groups, see [1] for more details.

Exercise 3.1.1

1. Let A, B be any G-modules. Show that $H^n(G, A \oplus B) \cong H^n(G, A) \oplus H^n(G, B)$
2. Show that $H^0(G, \mathrm{CoInd}^G(A)) \cong A$ and $H^n(G, \mathrm{CoInd}^G(A)) \cong 0$ for all $n \geq 1$ (hint: consider $\zeta : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^n], \mathrm{CoInd}^G(S)) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G^n], A)$ given by $\varphi \mapsto (z \mapsto \varphi(z)(1))$. What is the inverse of this map?)
3. Show that $H_0(G, \mathrm{Ind}^G(A)) \cong A$ and $H_n(G, \mathrm{Ind}^G(A)) \cong 0$ for all $n \geq 1$.

3.2 Homological Dimension

We now turn to measuring the "complexity" of modules and rings using the length of their resolutions. Before defining these measures rigorously, it is worth stepping back to 1890 to understand why mathematicians began counting the length of resolutions in the first place.

Historical Motivation: Invariant Theory

In the late 19th century, the landscape of algebra was dominated by *Invariant Theory*. Imagine a group G acting on a polynomial ring $S = \mathbb{C}[x_1, \ldots, x_n]$ (for example, by linear substitutions of variables). Mathematicians were looking for *invariant ring* S^G:

$$S^G = \{f \in S | g \cdot f = f \text{ for all } g \in G\}$$

The central question was if S^G finitely generated? Can we find a finite set of basic invariant polynomials $f_1, \ldots, f_m$ such that every other invariant is just a polynomial combination of these?

Hilbert showed we can map a polynomial ring onto the invariants:

$$\varphi : \mathbb{C}[y_1, \ldots, y_m] \twoheadrightarrow S^G \qquad y_i \mapsto f_i$$

and then the kernel $I = \ker(\varphi)$ represents the *relations* between the invariants. Hence, the study of invariants reduced to the study of the kernel. To find these invariants, 19th-century computationalists (like Paul Gordan) asked tried computing them and found the following problem:

1. **First Syzygies:** The ideal I describes the relations. By the Basis Theorem, I is finitely generated.
2. **Second Syzygies:** Are these relations independent? Or are there "relations between the relations"? If we have a relation $Ar_1 + Br_2 = 0$, that is a second syzygy.
3. **Higher Syzygies:** We can keep going. We can form a module of second syzygies and ask for its generators, then ask for the relations between *those* generators.

Mathematicians feared this chain of relations might continue forever, becoming infinitely complex. Hilbert's *Syzygy Theorem* proved that if you are working with a polynomial ring in n variables, this process *must stop*. Specifically, after at most n steps, the module of relations becomes free—there are no more "hidden" relations.

This was part of David Hilbert's seminal paper laying the foundation of commutative geometry. In particular, he proved three theorems: The *Basis Theorem*, the *Nullstellensatz*, and the *Syzygy Theorem*.

3.2.1 Etymology The word *Syzygy* comes from the Greek *syzygia*, meaning "yoked together". In astronomy, it refers to the alignment of three celestial bodies (like the Sun, Earth, and Moon). In algebra, it refers to a relation that "yokes" generators together to zero.

3.2.1 Projective and Global Dimension

We can now formalize Hilbert's discovery using the language of resolutions.

Definition 3.2.2: Projective Dimension

Let M be an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to M \to 0$$

If no finite resolution exists, we say $\mathrm{pd}_R(M) = \infty$.

Definition 3.2.3: Global Dimension

The *global dimension* of a ring R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in R\text{-}\mathbf{Mod}\}$$

Theorem 3.2.4: Global Dimension of Polynomial Rings

Let R be a commutative ring. Then

$$\operatorname{gldim}(R[x]) = \operatorname{gldim}(R) + 1$$

***Proof*:**
We will prove this by establishing both the upper bound $\operatorname{gldim}(R[x]) \leq \operatorname{gldim}(R) + 1$ and the lower bound $\operatorname{gldim}(R[x]) \geq \operatorname{gldim}(R) + 1$. If $\operatorname{gldim}(R) = \infty$, the equality trivially holds, so we may assume $\operatorname{gldim}(R) = n < \infty$.

Step 1: The Characteristic Exact Sequence Let M be any $R[x]$-module. We can view M as an R-module by restricting the scalars. We can then construct the extended $R[x]$-module $R[x] \otimes_R M$.

Define a sequence of $R[x]$-modules:

$$0 \to R[x] \otimes_R M \xrightarrow{\alpha} R[x] \otimes_R M \xrightarrow{\beta} M \to 0$$

where the maps are defined by:

- $\alpha(f(x) \otimes m) = xf(x) \otimes m - f(x) \otimes xm$
- $\beta(f(x) \otimes m) = f(x)m$

We must prove this sequence is exact.

- *Surjectivity of β:* For any $m \in M$, $\beta(1 \otimes m) = m$, so β is surjective.
- *Exactness at the middle:* Clearly $\beta(\alpha(1 \otimes m)) = \beta(x \otimes m - 1 \otimes xm) = xm - xm = 0$, so $\operatorname{im}(\alpha) \subseteq \ker(\beta)$. Conversely, any element in $R[x] \otimes_R M$ can be written as $\sum_{i=0}^{d} x^i \otimes m_i$. If it lies in $\ker(\beta)$, then $\sum_{i=0}^{d} x^i m_i = 0$. We can rewrite our original element as:

  $$\sum_{i=0}^{d} x^i \otimes m_i = \sum_{i=0}^{d} (x^i \otimes m_i - 1 \otimes x^i m_i)$$

  Notice that $x^i \otimes m - 1 \otimes x^i m = \alpha\left(\sum_{j=0}^{i-1} x^{i-1-j} \otimes x^j m\right)$. Thus, the element is in the image of α.
- *Injectivity of α:* Suppose $\sum_{i=0}^{d} x^i \otimes m_i \in \ker(\alpha)$. Then:

  $$\sum_{i=0}^{d} x^{i+1} \otimes m_i - \sum_{i=0}^{d} x^i \otimes xm_i = 0$$

  Because $R[x]$ is free over R, we can compare the coefficients of $x^k \otimes -$ degree by degree. For degree 0, we have $-1 \otimes xm_0 = 0 \implies xm_0 = 0$. For degree $k > 0$, we have $m_{k-1} - xm_k = 0 \implies m_{k-1} = xm_k$. For degree $d + 1$, we have $m_d = 0$. Working backward from $m_d = 0$, we get $m_{d-1} = xm_d = 0$, and by induction, $m_i = 0$ for all i. Hence, α is strictly injective.

Step 2: The Upper Bound
Let M be an $R[x]$-module. View M as an R-module, and take a projective resolution $P_\bullet \to M \to 0$ over R of length at most n (since $\mathrm{gldim}(R) = n$).

Because $R[x]$ is free (and thus flat) over R, tensoring the resolution with $R[x]$ preserves exactness. Therefore, $R[x] \otimes_R P_\bullet$ is a projective resolution of $R[x] \otimes_R M$ over $R[x]$. This proves that:

$$\mathrm{pd}_{R[x]}(R[x] \otimes_R M) \le \mathrm{pd}_R(M) \le n$$

Now, apply the functor $\mathrm{Ext}^*_{R[x]}(-, N)$ for an arbitrary $R[x]$-module N to our characteristic exact sequence. We obtain the long exact sequence:

$$\cdots \to \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, N) \to \mathrm{Ext}^{i+1}_{R[x]}(M, N) \to \mathrm{Ext}^{i+1}_{R[x]}(R[x] \otimes_R M, N) \to \cdots$$

Since $\mathrm{pd}_{R[x]}(R[x] \otimes_R M) \le n$, the Ext groups vanish for all indices $\ge n+1$. Therefore, setting $i = n+1$, we get:

$$0 \to \mathrm{Ext}^{n+2}_{R[x]}(M, N) \to 0$$

Since $\mathrm{Ext}^{n+2}_{R[x]}(M, N) = 0$ for all modules N, we conclude $\mathrm{pd}_{R[x]}(M) \le n+1$. Because this holds for all $R[x]$-modules, $\mathrm{gldim}(R[x]) \le n+1$.

Step 3: The Lower Bound
Since $\mathrm{gldim}(R) = n$, there exists an R-module M and an R-module K such that $\mathrm{Ext}^n_R(M, K) \neq 0$. We can elevate both M and K to the status of $R[x]$-modules by declaring that x acts trivially on them (i.e., $xM = 0$ and $xK = 0$).

Apply the characteristic exact sequence to M (viewed as this specific $R[x]$-module) and take the long exact sequence for $\mathrm{Hom}_{R[x]}(-, K)$. By the Tensor-Hom Adjunction, there is a natural isomorphism:

$$\mathrm{Hom}_{R[x]}(R[x] \otimes_R P_\bullet, K) \cong \mathrm{Hom}_R(P_\bullet, K)$$

Because $R[x] \otimes_R P_\bullet$ resolves $R[x] \otimes_R M$ over $R[x]$, taking the cohomology of both sides gives an isomorphism of the derived functors:

$$\mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \cong \mathrm{Ext}^i_R(M, K) \quad \text{for all } i \ge 0.$$

Let us examine the connecting map $\alpha^* : \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \to \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K)$ induced by α. On the level of the 0-th Hom group, for any $R[x]$-linear map $f : R[x] \otimes_R M \to K$:

$$(\alpha^* f)(1 \otimes m) = f(\alpha(1 \otimes m)) = f(x \otimes m - 1 \otimes xm)$$

Since x acts trivially on M, $1 \otimes xm = 0$. By $R[x]$-linearity, $f(x \otimes m) = x \cdot f(1 \otimes m)$. Because x also acts trivially on the codomain K, we get $x \cdot f(1 \otimes m) = 0$. Thus, the map α^* is identically the zero map on the entire cochain complex, and consequently, it is the zero map on the Ext groups.

Because the maps α^* are zero, our long exact sequence splits into short exact sequences for each i:

$$0 \to \mathrm{Ext}^i_R(M, K) \to \mathrm{Ext}^{i+1}_{R[x]}(M, K) \to \mathrm{Ext}^{i+1}_R(M, K) \to 0$$

Setting $i = n$, we get:

$$0 \to \mathrm{Ext}^n_R(M, K) \to \mathrm{Ext}^{n+1}_{R[x]}(M, K) \to \mathrm{Ext}^{n+1}_R(M, K) \to 0$$

Since $\mathrm{Ext}^n_R(M, K) \neq 0$, it directly follows that $\mathrm{Ext}^{n+1}_{R[x]}(M, K) \neq 0$. This proves that $\mathrm{pd}_{R[x]}(M) \ge n+1$.

Combining the upper and lower bounds gives $\mathrm{gldim}(R[x]) = \mathrm{gldim}(R) + 1$, completing the proof.

Hilbert's theorem is effectively the calculation of the global dimension of polynomial rings.

Theorem 3.2.5: Hilbert's Syzygy Theorem

Let k be a field and $R = k[x_1, \ldots, x_n]$ be the polynomial ring in n variables. Then every finitely generated R-module has finite projective dimension, and:

$$\operatorname{gldim}(R) = n$$

***Proof*:**
While Hilbert's original proof was highly computational, the modern approach is much cleaner. We must show two things: that the global dimension is at least n, and that it is at most n.

First, we show $\operatorname{gldim}(R) \geq n$ by finding a module with projective dimension exactly n. This module is the residue field $k \cong R/(x_1, \ldots, x_n)$. We can explicitly construct a free resolution for k called the *Koszul Complex*. Let V be a k-vector space of dimension n with basis $e_1, \ldots, e_n$. The resolution is given by the exterior algebra $\bigwedge^\bullet V \otimes_k R$:

$$0 \to \bigwedge^n V \otimes R \xrightarrow{d_n} \ldots \xrightarrow{d_2} \bigwedge^1 V \otimes R \xrightarrow{d_1} R \xrightarrow{\epsilon} k \to 0$$

where the differential is given by $d_1(e_i \otimes 1) = x_i$. Because the variables $x_1, \ldots, x_n$ form a regular sequence, this complex is exact. It has length exactly n, so $\operatorname{pd}_R(k) = n$.

Second, to show that *no* module has a resolution longer than n, we proceed by induction on n. For $n = 0$, $R = k$ is a field; every module is free, so the dimension is 0. For $n = 1$, $R = k[x]$ is a PID. Submodules of free modules are free, so any resolution stops at step 1. For the inductive step, we write $R = S[x_n]$ where $S = k[x_1, \ldots, x_{n-1}]$.

By theorem 3.2.4 ($\operatorname{gldim}(S[x]) = \operatorname{gldim}(S) + 1$) and our inductive hypothesis, $\operatorname{gldim}(S) = n - 1$, which forces $\operatorname{gldim}(R) \leq n$, completing the proof.

This theorem has implications beyond just counting relations:

1. *Degree of the Hilbert Polynomial:* Hilbert originally used this theorem to prove that the Hilbert Function $H_M(t) = \dim_k(M_t)$ of a graded module eventually becomes a polynomial. The finite resolution allows us to calculate $H_M(t)$ as an alternating sum of the Hilbert functions of free modules, which are essentially binomial coefficients $\binom{t+k}{k}$.

2. *Structure of Projective Modules:* A celebrated problem posed by Serre (related to this theorem) asked if every projective module over $k[x_1, \ldots, x_n]$ is free. This is the algebraic analogue of "vector bundles over affine space are trivial". This was solved affirmatively in 1976 (the *Quillen-Suslin Theorem*). This implies that over polynomial rings, "projective resolution" and "free resolution" are synonymous.

3.2.2 Dimension of Local Rings

Computing the global dimension for general rings seems daunting, as one must check *every* module. However, we can simplify this task using derived functors.

Proposition 3.2.6: Detecting Projective Dimension with Tor

Let R be a ring and M an R-module. The following are equivalent:

1. $\mathrm{pd}_R(M) \leq n$
2. $\mathrm{Tor}_i^R(M, N) = 0$ for all R-modules N and all $i > n$.
3. $\mathrm{Tor}_{n+1}^R(M, N) = 0$ for all R-modules N.

***Proof*:**
(1) $\Rightarrow$ (2): If $\mathrm{pd}_R(M) \leq n$, there is a resolution $P_\bullet$ of length n. Computing Tor using this resolution involves terms P_i which are zero for $i > n$, hence the homology vanishes. (2) $\Rightarrow$ (3): Trivial. (3) $\Rightarrow$ (1): Let $\cdots \to P_1 \to P_0 \to M \to 0$ be a projective resolution. Let $K_n = \ker(P_{n-1} \to P_{n-2})$. A standard dimension shifting argument (using the long exact sequence of Tor) shows that for any N:

$$\mathrm{Tor}_1^R(K_n, N) \cong \mathrm{Tor}_{n+1}^R(M, N)$$

By assumption (3), $\mathrm{Tor}_1^R(K_n, N) = 0$ for all N. This implies K_n is flat (and if R is Noetherian and M finitely generated, K_n is projective). Thus the resolution can stop at K_n, giving a length n projective resolution $0 \to K_n \to P_{n-1} \cdots \to M \to 0$.

It is more common that we consider rings that already have certain properties and verify how these conditions affect the global dimension. For example, given a local ring, the work is significantly reduced. In the local case, we can verify condition (3) by checking a *single* module: the residue field.

Lemma 3.2.7: Projective Dimension over Local Rings

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely generated R-module. Then:

$$\mathrm{pd}_R(M) \leq n \iff \mathrm{Tor}_{n+1}^R(M, k) = 0$$

Consequently, $\mathrm{pd}_R(M)$ is the smallest n such that $\mathrm{Tor}_i^R(M, k) = 0$ for all $i > n$.

***Proof*:**
The forward direction is immediate from proposition 3.2.6. For the converse, let $\cdots \to P_1 \to P_0 \to M \to 0$ be a *minimal* free resolution (meaning the matrices of the maps have entries in $\mathfrak{m}$). Then the differentials in the complex $P_\bullet \otimes_R k$ are all zero maps. Thus:

$$\mathrm{Tor}_i^R(M, k) \cong H_i(P_\bullet \otimes k) \cong P_i \otimes k \cong k^{\beta_i}$$

where β_i is the rank of the free module P_i. If $\mathrm{Tor}_{n+1}^R(M, k) = 0$, then $P_{n+1} \otimes k = 0$. By Nakayama's Lemma, this implies $P_{n+1} = 0$, so the resolution stops at n.

This leads to the fundamental equality attributed to Auslander, Buchsbaum, and Serre, which connects the homological algebra back to the geometry of the ring.

Theorem 3.2.8: Auslander-Buchsbaum-Serre Theorem

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring. Then the global dimension is determined by the residue field:

$$\operatorname{gldim}(R) = \operatorname{pd}_R(k)$$

Furthermore, $\operatorname{gldim}(R)$ is finite if and only if R is a *Regular Local Ring*.

This theorem is the bridge between algebra and geometry: "Regular" is a geometric condition (non-singular point), while Global Dimension is a homological condition (finite resolutions).

***Proof*:**
By definition, $\operatorname{gldim}(R) \geq \operatorname{pd}_R(k)$. We must show the reverse. Let $n = \operatorname{pd}_R(k)$. Let M be any finitely generated R-module. We check its projective dimension using lemma 3.2.7.

$$\operatorname{Tor}^R_{n+1}(M, k)$$

Since Tor is symmetric, $\operatorname{Tor}^R_{n+1}(M, k) \cong \operatorname{Tor}^R_{n+1}(k, M)$. Since $\operatorname{pd}_R(k) = n$, by proposition 3.2.6, $\operatorname{Tor}^R_{n+1}(k, -)$ is the zero functor. Thus $\operatorname{Tor}^R_{n+1}(M, k) = 0$. By lemma 3.2.7, this implies $\operatorname{pd}_R(M) \leq n$. Since this holds for all finitely generated M, $\operatorname{gldim}(R) = n$ (a standard limit argument extends this to non-finitely generated modules).

3.2.3 Beyond Regularity: The Hierarchy of Singularities

The Auslander-Buchsbaum-Serre theorem (theorem 3.2.8) presents a stark dichotomy: a local ring either has finite global dimension (and is geometrically smooth), or its global dimension is completely infinite.

For decades, algebraists sought to understand the "infinite" case. If a ring is singular, just *how* singular is it? How chaotic do the infinite free resolutions become? This inquiry led to a classification of local Noetherian rings—a hierarchy that measures the severity of singularities based on homological behavior.

To properly measure the complexity of rings that fail to be regular, we need to formalize the notion of a regular sequence, which we briefly encountered in the proof of Serre's Theorem (**??**):

Definition 3.2.9: Regular Sequence

Let R be a ring and M an R-module. A sequence of elements $x_1, \ldots, x_n \in R$ is called an *M-regular sequence* if:

1. $(x_1, \ldots, x_n)M \neq M$
2. For $i = 1, \ldots, n$, the element x_i is not a zero-divisor on the quotient module $M/(x_1, \ldots, x_{i-1})M$.

Intuitively, a regular sequence is a sequence of elements that "cut down" the dimension of the module as cleanly and efficiently as possible, without creating any hidden nilpotent. This allows us to define a new homological invariant.

Definition 3.2.10: Depth

Let $(R, \mathfrak{m})$ be a local Noetherian ring and M a finitely generated R-module. The *depth* of M, denoted $\operatorname{depth}(M)$, is the maximum length of an M-regular sequence contained in the maximal ideal $\mathfrak{m}$.

Depth is always bounded above by the Krull dimension ($\operatorname{depth}(R) \leq \operatorname{kdim}(R)$). The gap between a ring's depth and its dimension is one of the primary ways we measure how "bad" a singularity is.

Using this, we can make the Auslander-Buchsbaum-Serre theorem more refined by looking at modules over R:

Theorem 3.2.11: Auslander-Buchsbaum Formula

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a finitely generated R-module with finite projective dimension. Then:

$$\operatorname{pd}_R(M) + \operatorname{depth}(M) = \operatorname{depth}(R)$$

> ***Proof*:**
> see cite:HERE.

Using depth and other structural properties, local rings can be classified into a strict chain of implications:

$$\begin{aligned} &\text{Regular} \implies \text{Complete Intersection} \implies \text{Gorenstein} \\ &\qquad\qquad \implies \text{Cohen-Macaulay} \implies \text{Normal} \implies \text{Reduced} \end{aligned}$$

While these definitions are strictly for *local* rings, we say a general commutative ring has one of these properties if its localization at every prime ideal has it.

Complete Intersections: These are the "nicest" singularities. Geometrically, they are spaces cut out by exactly the correct number of equations (e.g., a 1D curve in $\mathbb{C}^3$ defined by exactly two equations). Algebraically, R is the quotient of a regular local ring by a regular sequence. While modules over these rings have infinite projective dimensions, their free resolutions are highly predictable, eventually becoming periodic.

Gorenstein Rings: These rings represent spaces with "symmetric" singularities. They are exactly the spaces where a generalized form of Poincaré Duality still holds. Algebraically, a local ring is Gorenstein if it has finite injective dimension over itself. The free resolutions over Gorenstein rings exhibit a mirror symmetry when the functor $\operatorname{Hom}_R(-, R)$ is applied.

Cohen-Macaulay Rings: This is perhaps the most important class of singular rings in algebraic geometry. A local ring is Cohen-Macaulay if its depth equals its Krull dimension ($\operatorname{depth}(R) = \dim R$). Geometrically, these spaces are "equidimensional" and lack hidden, embedded fuzzy components. Here, the Auslander-Buchsbaum formula perfectly aligns with the geometric dimension.

Normal Rings: A ring is normal if it is integrally closed in its total ring of fractions. Geometrically, this means the space is "smooth in codimension 1". For example, a 2D normal surface can have an isolated singular point (like the tip of a cone), but it cannot have a singular crease or folded line running through it.

Reduced Rings: A ring is reduced if it contains no non-zero nilpotents ($x^n = 0 \implies x = 0$). Geometrically, this simply means the space has no "thickness" or "fuzz"; it is a classical geometric space uniquely determined by the values of functions at its points.

3.3 The Local Criterion for Flatness

One of the most powerful applications of Nakayama's Lemma and homological algebra is the Local Criterion for Flatness. Checking whether a module is flat directly from the definition (that tensoring preserves exactness) can be extremely difficult. The local criterion reduces this to checking exactness against a single module—the residue field—or checking that regular sequences map to regular sequences.

Theorem 3.3.1: Local Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and let M be a finitely generated S-module. Then the following are equivalent:

1. M is flat over R.
2. $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$.
3. The natural multiplication map $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is an isomorphism.

***Proof*:**

The implication (1) ⇒ (2) is trivial since M is flat implies $\mathrm{Tor}_i^R(M, -) = 0$ for all $i > 0$. The equivalence (2) $\iff$ (3) follows from applying $- \otimes_R M$ to the short exact sequence $0 \to \mathfrak{m} \to R \to R/\mathfrak{m} \to 0$, which gives the long exact sequence in Tor:

$$0 = \mathrm{Tor}_1^R(R, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) \to \mathfrak{m} \otimes_R M \to R \otimes_R M \to R/\mathfrak{m} \otimes_R M \to 0$$

Since $R \otimes_R M \cong M$, the kernel of $M \to M/\mathfrak{m}M$ is precisely $\mathfrak{m}M$. Thus $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$ if and only if $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is injective (and it is always surjective).

For (2) ⇒ (1), we must show that $\mathrm{Tor}_1^R(M, N) = 0$ for all finitely generated R-modules N. We proceed by induction on the length of N. By Nakayama's Lemma (**??**), if $N \neq 0$, we can find a non-zero quotient $N \to R/\mathfrak{m}$. Let K be the kernel, so we have $0 \to K \to N \to R/\mathfrak{m} \to 0$. The long exact sequence in Tor gives:

$$\mathrm{Tor}_1^R(K, M) \to \mathrm{Tor}_1^R(N, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$$

Thus, $\mathrm{Tor}_1^R(K, M)$ surjects onto $\mathrm{Tor}_1^R(N, M)$. By iterating this process (using the Artin-Rees lemma and the $\mathfrak{m}$-adic filtration on N; **??**), one can show that $\mathrm{Tor}_1^R(N, M) \subseteq \mathfrak{m}\mathrm{Tor}_1^R(N, M)$. Since M is a finitely generated S-module and S is Noetherian, $\mathrm{Tor}_1^R(N, M)$ is a finitely generated S-module. By Nakayama's Lemma (applied over S, since $\mathfrak{m}S \subseteq \mathfrak{n}$), we conclude $\mathrm{Tor}_1^R(N, M) = 0$.

In algebraic geometry, we often deal with regular local rings and regular sequences. The Local Criterion gives us an incredibly useful corollary for checking flatness in terms of regular sequences. This is the algebraic engine behind an infamous result known as the "Miracle Flatness" theorem, an important "shortcut" for proving flatness in [5].

Corollary 3.3.2: Regular Sequence Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and M a finitely generated S-module. Let $x \in \mathfrak{m}$. Then M is flat over R if and only if:

1. x is a non-zero divisor on M, and
2. M/xM is flat over $R/(x)$.

By induction, if $x_1, \ldots, x_d \in \mathfrak{m}$ is a regular sequence in R, then M is flat over R if and only if $x_1, \ldots, x_d$ forms a regular sequence on M and $M/(x_1, \ldots, x_d)M$ is flat over $R/(x_1, \ldots, x_d)$.

***Proof*:**
($\Rightarrow$): If M is flat over R, applying $- \otimes_R M$ to the exact sequence $0 \to R \xrightarrow{\cdot x} R \to R/(x) \to 0$ yields the exact sequence $0 \to M \xrightarrow{\cdot x} M \to M/xM \to 0$. Thus x is a non-zero divisor on M. Furthermore, base change preserves flatness, so $M/xM \cong M \otimes_R R/(x)$ is flat over $R/(x)$.

($\Leftarrow$): Assume x is regular on M and M/xM is flat over $R/(x)$. We want to show M is flat. Since M/xM is flat over $R/(x)$, $\mathrm{Tor}_1^{R/(x)}(R/\mathfrak{m}, M/xM) = 0$. Combined with the fact that x is regular on M, a change-of-rings theorem for Tor implies $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$. By the Local Criterion for Flatness (theorem 3.3.1), M is flat over R.

Exercise 3.3.1

1. Let $R = k[x, y]$. Find a free resolution for the module $M = R/(x^2, xy, y^2)$. What is the length of this resolution?
2. Show that over $R = \mathbb{Z}$, the module $M = \mathbb{Q}$ has projective dimension 1, but does not have a finite *free* resolution if we require finitely generated free modules (which is impossible since $\mathbb{Q}$ isn't finitely generated).
3. Why does the Syzygy theorem fail for $R = k[x_1, x_2, \ldots]$ (infinite variables)? Find a module with infinite projective dimension.

4

Spectral Sequences

It is often the case that the (co)homology of an object is difficult to calculate, but the object can be broken down into simpler objects whose (co)homology is more readily computed. Perhaps the simplest type decomposition of an object is that of a filtration. The reader that read section **??** may recognize the notion of a filtration for an R-modules, which is a sequence:

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \supseteq \cdots$$

For the case of homology, the sequence usually is increasing. In the category *R-Mod*, each respective M_i will have an associated $P_i^\bullet$ associated to it creating a natural grid

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots \arrow[r] & P_{i,j} \arrow[r] \arrow[d] & P_{i,j+1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & P_{i+1,j} \arrow[r] \arrow[d] & P_{i+1,j+1} \arrow[r] \arrow[d] & \cdots \\
 & \vdots & \vdots &
\end{tikzcd}$$

From this grid, we will be able to extract (co)homological information connecting (co)homology of M_i and M_{i+1}. The spectral sequence then will be a way of combining this information to give cohomological information about M. The goal of this chapter is to go about the formalization of these concepts and give some interesting examples. As it turns out, there are many ways of creating grid of information, and so the definition of spectral sequences takes this into account and may at first seem a little separated from this original intuition, however examples will be given to make the connection.

4.1 Definition

On a high-level, a spectral sequence arises from a special type of exact triangle:

Definition 4.1.1: Exact Couple

Let A, E be two object in an abelian category $\mathcal{A}$. Then the exact triangle

$$\begin{tikzcd} & A \arrow[dr, "\alpha"] & \\ E \arrow[ur, "\gamma"] & & A \arrow[ll, "\beta"] \end{tikzcd}$$

is called an *exact couple*. Notice that the object A is repeated twice.

No assumption are done on the function γ. Define the function $d : E \to E$ where $d = \beta \circ \gamma$. Since an exact couple is an exact triangle, $\gamma \circ \beta = 0$, and so

$$\delta \circ \delta = \beta \circ \gamma \circ \beta \circ \gamma = \beta \circ 0 \circ \gamma = 0$$

Hence the (co)homology of E with respect to d is well-defined and is

$$E = \frac{\ker d}{\mathrm{im} d}$$

We may furthermore define:

- $A' = \mathrm{im}\alpha$
- $\alpha' : A' \to A'$ by restricting α to A'
- $\beta' : A' \to E'$ by defining $\beta(\alpha(a)) = [\beta(\alpha)]$
- $\gamma' : E' \to A'$ by defining $\gamma([e]) = \gamma(e)$ (which by exactness is well-defined)

What is interesting is the following

Lemma 4.1.2: Derived Couple

Given an exact couple, then the E' and A' along with the above map give rise to another exact couple, namely

$$\begin{tikzcd} & A' \arrow[dr, "\alpha'"] & \\ E' \arrow[ur, "\gamma'"] & & A' \arrow[ll, "\beta'"] \end{tikzcd}$$

is an exact triangle

***Proof*:**
This is diagram chasing (p.523 of Hatcher chap 5 if you want to verify)

The resulting exact couple is called a derived couple, where the nomenclature arose separately from the nomenclature for derived categories (it arose in originally in algebraic topology). Thus, since the result is another exact triangle, the process can repeat itself indefinitely, giving a sequence:

Definition 4.1.3: Spectral Sequence

Let $\mathbf{A}$ be an abelian category. Then a *spectral sequence* is a sequence of tuples $\{(E_i, d_i)\}_i$ where $d_i : E_i \to E_i$ are maps with the property that $d_i \circ d_i = 0$ and $E_{i+1} = \ker d_i/\text{im} d_i$

4.2 Important Examples

(I would like to at least put the Künneth Spectral Sequence and Serre-Spectral Sequence, as well as any accessible examples I may find in Aluffi. Another two would be cool to add is the Hodge to de Rham spectral sequence and the Mayer-Vietoris spectral sequence, but I think that should be put in EYTNKA alg geo or diff geo)